

Redundant Systems SIMATIC S7-1500R/H

Technical Details / V21



General Hints

Unless otherwise specified, all features described in the following slides apply to both S7-1500H and S7-1500R systems.

Examples:

“Failure scenarios for S7-1500R/H” → Valid for S7-1500R and S7-1500H
No information about R or H in a slide



“Failure scenarios for S7-1500R” → Valid for S7-1500R only



“Failure scenarios for S7-1500H” → Valid for S7-1500H and S7-1500HF only



Icons for failure scenarios



Cause

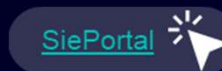


No effect



Subsequent fault

Links to SiePortal (former Mall/SIOS)



Overview of the documentation for the S7-1500R/H System (I)

These slides offer an overview of the functions and applications of the S7-1500R/H system. For detailed information, please refer to the official product documentation.

Type	Document	Link in SiePortal
System Manual	S7-1500R/H Redundant System	109754833
Equipment Manual	CPU 1513R-1 PN	109754834
	CPU 1515R-2 PN	109754835
	CPU 1517H-4 PN	109754836
	CPU 1518HF-4 PN	109784207
Getting Started	SIMATIC S7-1500R/H Getting Started	109757712
Function Manual	Communication	59192925
Function Manual	Cycle and response times	59193558
Function Manual	Diagnostics	59192926
Function Manual	Structure and Use of the CPU Memory	59193101
Function Manual	S7-1500 PID control	108210036
Function Manual	Using the trace and logic analyzer function	64897128
Function Manual	Designing interference-free controllers	59193566



Overview of the documentation for the S7-1500R/H System (II)

These slides offer an overview of the functions and applications of the S7-1500R/H system. For detailed information, please refer to the official product documentation.

Type	Document	Link in SiePortal
Function Manual	Webserver	109977246
Programming and Operating Manual	SIMATIC Safety - Configuring and Programming	54110126
Programming Manual	Programming Style guide for S7-1200/1500	109478084
System Manual	Security with SIMATIC S7-Controller	90885010
Product Information	Syslog Messages	109823696
Product Information	S7-1500/ET 200MP, S7-1500R/H	68052815
Equipment Manual	SIMATIC S7-1500 and ET 200MP active backplane bus	109778694



SIMATIC S7-1500 Redundant Systems

Table of contents

What's new in
Firmware V4.1 (V21)

Motivation and
Product Strategy

System Overview

PROFINET
System Redundancy

Network
Configurations

Failure Scenarios

Communication via
System IP and Device IP

OPC UA Communication
and Web-API

Safety with
Redundant Systems

Security

Loading a Project

Programming
Recommendations

Specific Functions for
R/H Controller

Feature Comparison /
Restrictions

Add Ons

What's new in Firmware V4.1 (V21)

Important Information

Using **Firmware V4.1.0** for redundant S7-1500R/H controllers in combination with TIA Portal V21 and the “CiR” feature **requires** installing **TIA Portal V21.0 Hotfix 1**, which can be downloaded via the following link: 

Firmware V4.1 are available through the following links:

CPU Type	MLFB	Firmware
CPU 1513R-1 PN	6ES7513-1RM03-0AB0	
CPU 1515R-2 PN	6ES7515-2RN03-0AB0	
CPU 1517H-4 PN	6ES7517-4HQ10-0AB0	
CPU 1518HF-4 PN	6ES7518-4JT10-0AB0	

Firmware V4.0 and lower of R/H controller can be used without restrictions in TIA Portal V21.

SIMATIC S7-1500 Redundant Systems

What's new with Firmware Version 4.1 / TIA Portal V21

V21

Configuration in RUN (CiR)

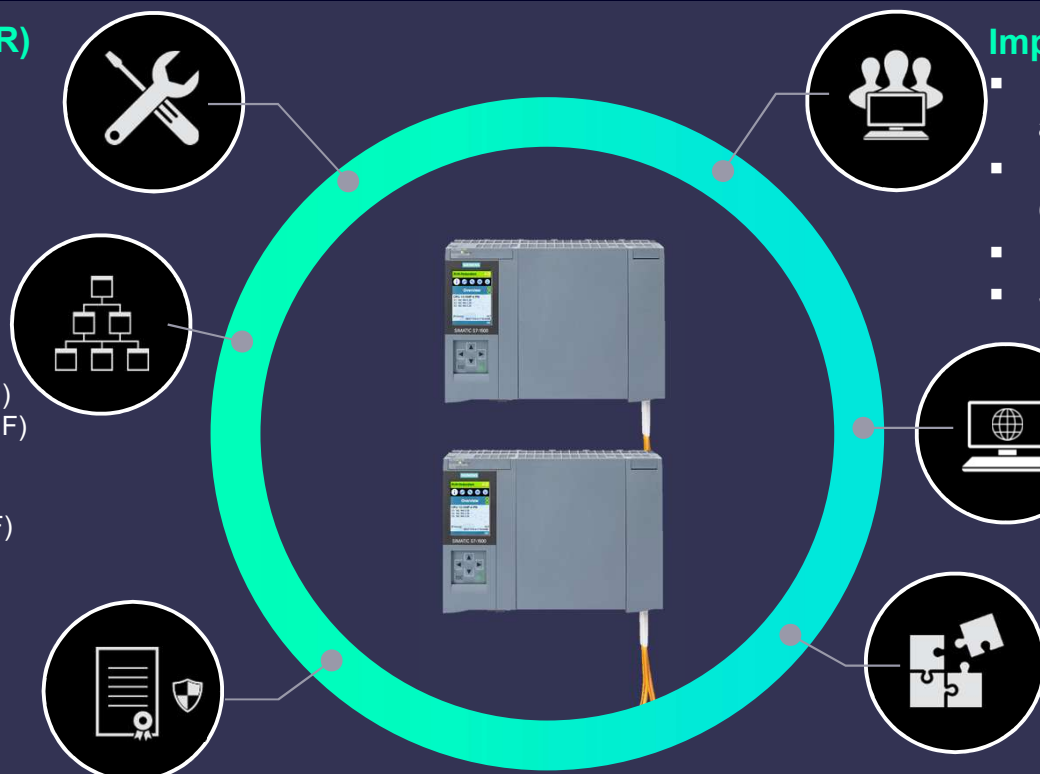
- Change hardware configuration during operation [Details](#)

Quantity structures for H-CPU

- 512 PN IO-Devices
- OPC UA Nodes:
30.000 → 100.000 (CPU 1517H)
50.000 → 200.000 (CPU 1518HF)
- OPC UA Monitored Items:
25.000 → 50.000 (CPU 1517H)
25.000 → 60.000 (CPU 1518HF)

Dynamic Certificate Management

- Renew Certificates during Operation using OPC UA GDS Push



Improved Usability

- FW-Update via list of accessible devices [Details](#)
- Move System-IP also for OPC UA Server [Details](#)
- Improved MRP-Ring Status
- SINAMICS Startdrive support

Web Server

- System Web Pages [Details](#)
- Full Web-API support

Additional functions

- TO Ident support
- Activate/Deactivate IP-forwarding in user program

CiR Configuration in RUN

What's new in Firmware V4.1 (V21)

What's new with Firmware Version 4.1 / TIA Portal V21

Configuration in RUN – 1st Step

V21

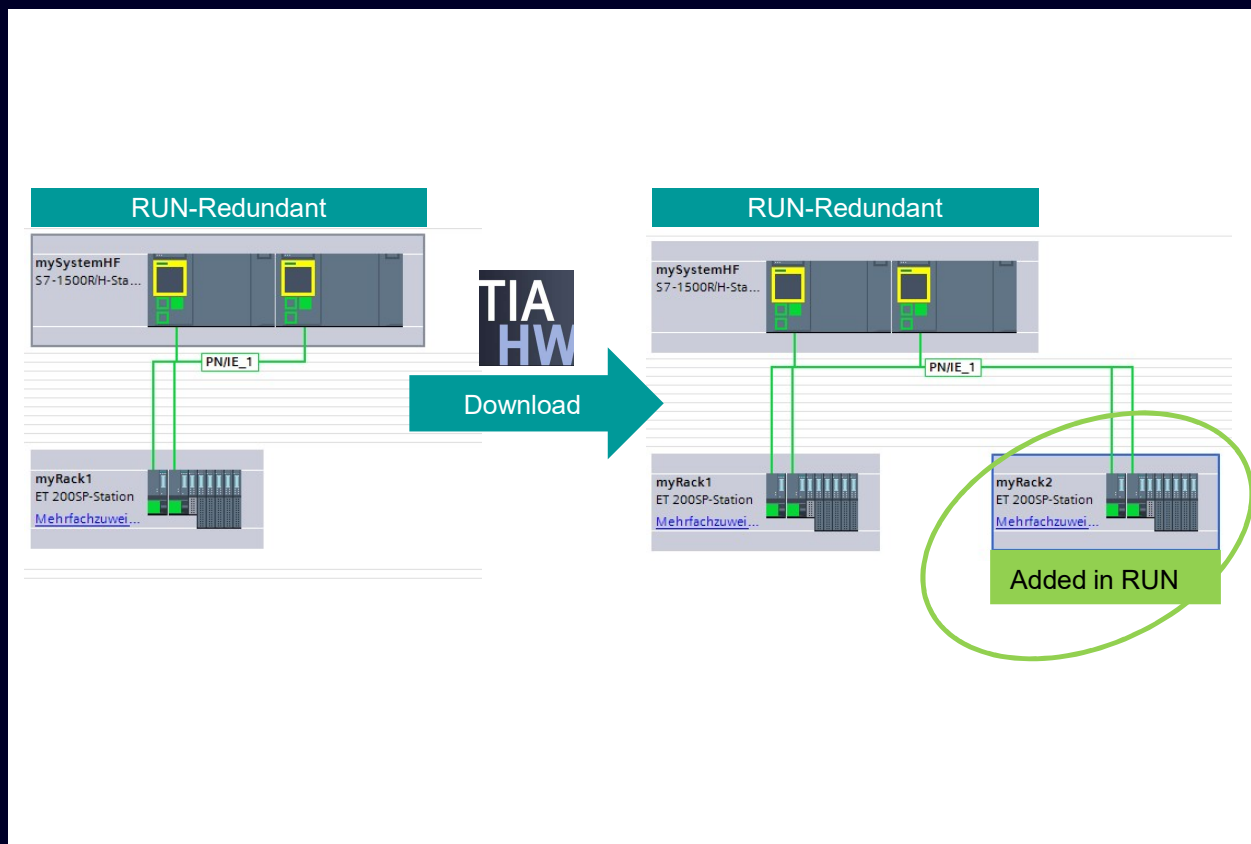
Configuration in RUN (CiR) allows to **change the hardware configuration** and load it **during Run-Redundant** operation.

Following scenarios are supported:

1. Add new PROFINET IO-Device,
2. Change settings of IO-modules (e.g., enable wire break detection).

Additional CiR-functions will follow in next versions.

CiR is supported by
S7-1500R and **S7-1500H**
redundant system



What's new with Firmware Version 4.1 / TIA Portal V21

Configuration in RUN – Use Cases

V21

Configuration in RUN (CiR) enables the following use cases:

Operation Phase

- Expand an existing plant without production downtime
- Adjust hardware properties (e.g., for a different temperature sensor) during operation
- Avoid communication interruptions to the SCADA system, as hardware changes no longer require SYNCUP

Commissioning Phase

- Reduction of engineering times because the system does not have to be stopped and restarted when adjusting IO-module parameters.

Engineering Phase

- Possible expansions do not have to be planned for from the beginning.



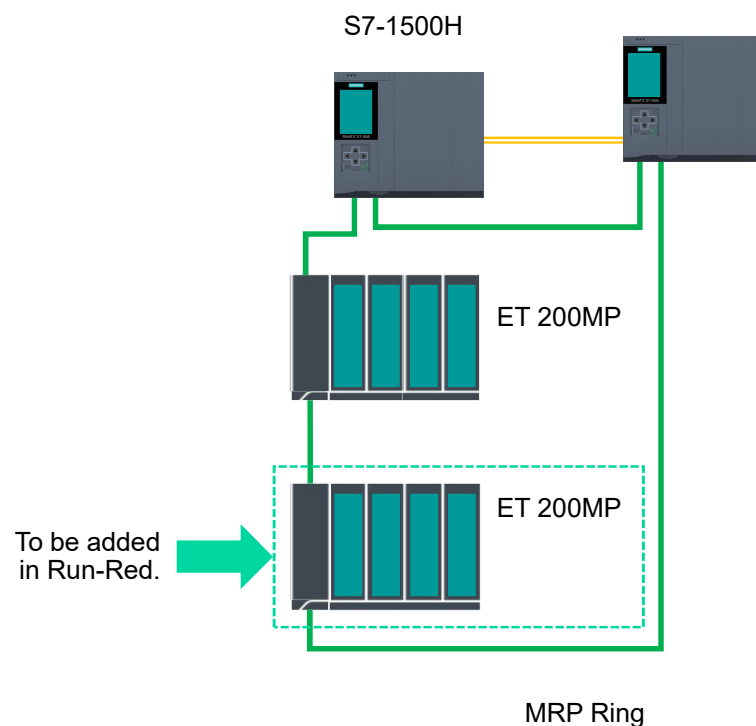
What's new with Firmware Version 4.1 / TIA Portal V21

Configuration in RUN

V21

Requirements from target group “End Customer”

As an end customer, I need to **add new PROFINET IO devices** during operation - without stopping the redundant system or interrupting the process - to meet evolving process requirements.



What's new with Firmware Version 4.1 / TIA Portal V21

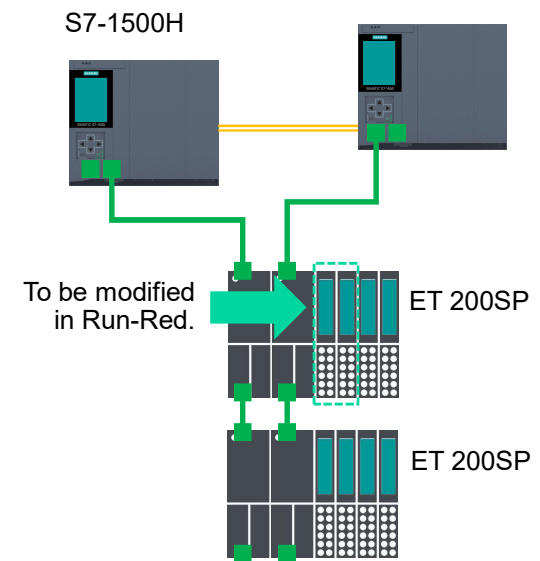
Configuration in RUN

V21

Requirements from target group “Commissioning engineers”

As a commissioning engineer, I need to **adjust IO module parameters** during operation - without impacting other parts of the process.

This allows my colleagues to continue their commissioning work undisturbed while I make necessary hardware changes.



What's new with Firmware Version 4.1 / TIA Portal V21

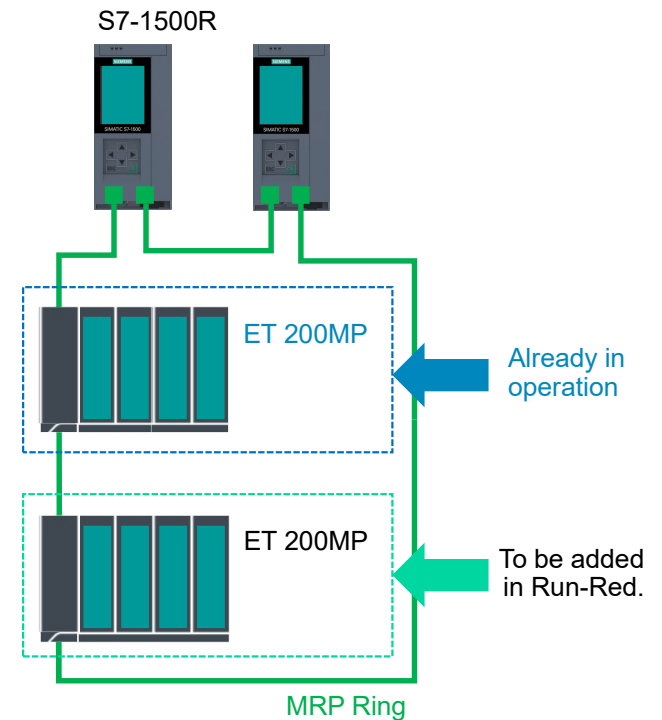
Configuration in RUN

V21

Requirements from target group “System Integrator”

As a system integrator, I need to migrate an existing installation from S7-400H / ET 200M to S7-1500H / ET 200SP step by step, and during operation.

Adding new ET 200SP stations must not interrupt the operation of already migrated stations, ensuring continuous process availability.



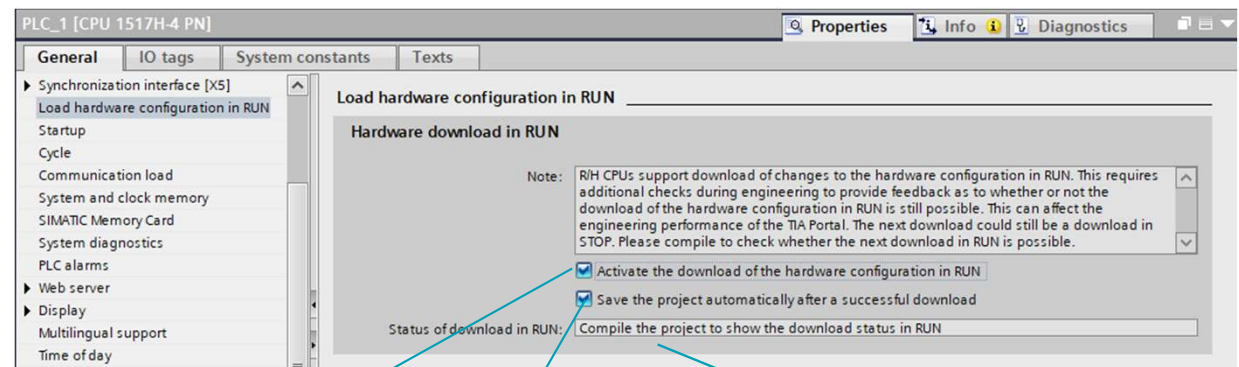
What's new with Firmware Version 4.1 / TIA Portal V21

Configuration in RUN

V21

General preconditions for CiR

- TIA Portal V21 HF1
- Firmware version $\geq$ V4.1
- CiR-Feature is activated
- A first download was completed with at least one PROFINET device.
- As the network topology cannot be modified in run with this TIA Portal version, it is advisable to avoid configuring it.



Activation of
CiR

Ensure consistence between
offline project and online project

Status information if CiR
is possible or not

Note: CiR is not supported with **Failsafe** components, as F-Program changes require a controller stop.

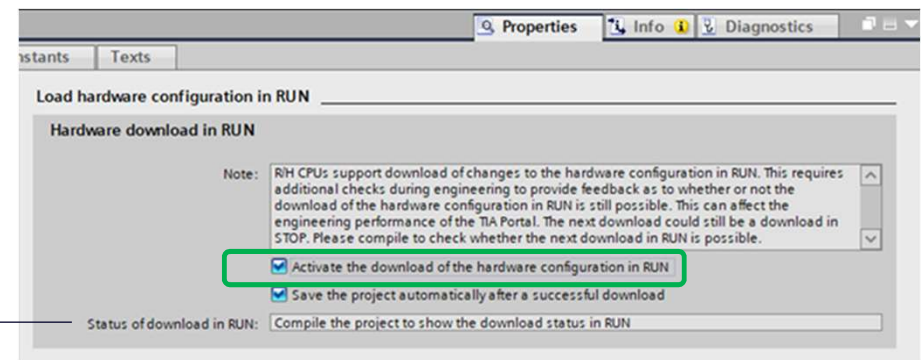
What's new with Firmware Version 4.1 / TIA Portal V21

Configuration in RUN

V21

How it works in “operation”

- TIA Portal checks during compilation whether the configuration can be loaded during operation.
- Status information is provided after compile.
- Only the changes (delta) in the hardware configuration are loaded in CiR mode; therefore, the offline and online projects must match.
- For possible status messages, see the table.



Status of download in RUN	Cause / Result
Configuration in RUN is possible	Download will work without any bump.
Configuration in RUN is possible with restart of devices	The settings of at least one device that does not support CiR have been changed / The affected devices are restarted during the download.
Configuration in RUN is not possible	Changes have been made that prevent downloading.

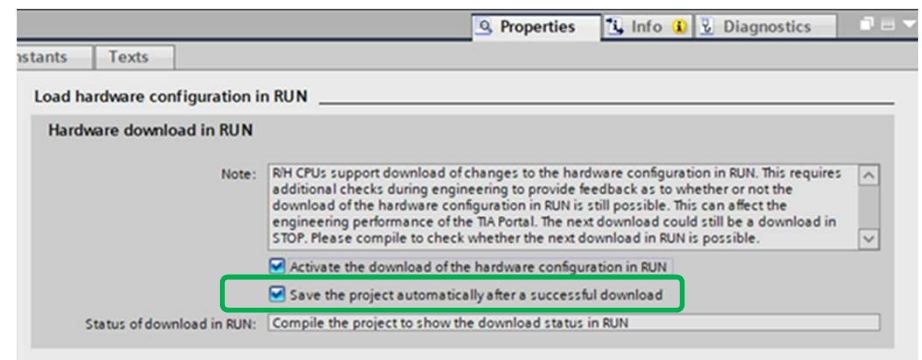
What's new with Firmware Version 4.1 / TIA Portal V21

Configuration in RUN

V21

Consistency between offline and online project

- CiR operates only if the CPU's online project matches the offline project.
- TIA Portal verifies synchronization before downloading.
- To maintain consistency, it is recommended to automatically save the project after a successful download



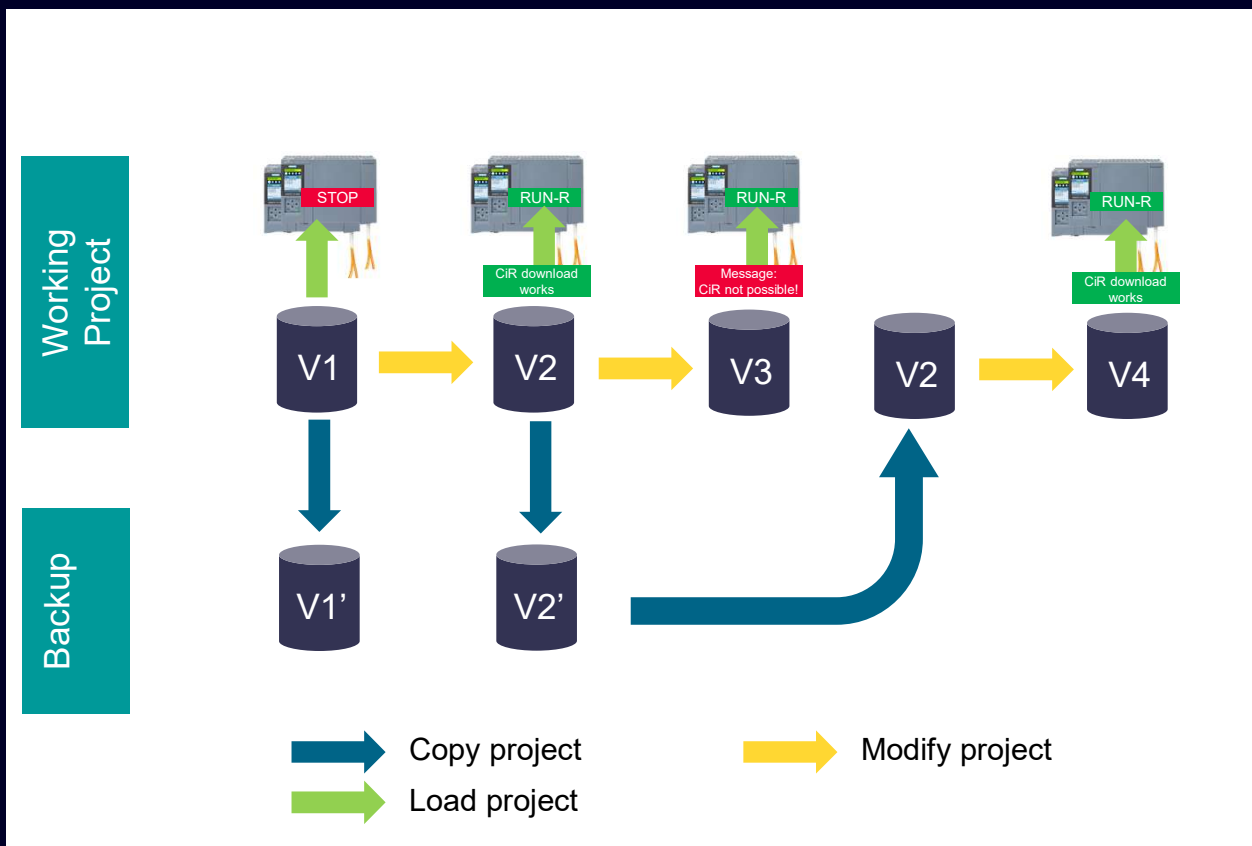
What's new with Firmware Version 4.1 / TIA Portal V21 Configuration in RUN

V21

Backup strategy for project data

Recommendation for project handling:

- Because some changes may not be loadable during operation, always save the last working project version.
- The diagram illustrates a recommended backup strategy.



What's new with Firmware Version 4.1 / TIA Portal V21

Configuration in RUN

V21

Following CiR-functions are supported:

- Use case “Add new device” → works with any PN-IO device (even S1)
- Use case “Change settings of parameters” → see following list

IO-Device Family	Product	PN System Red.	MLFB	Firmware Version
ET 200SP	IM 155-6 PN R1	R1	6ES7155-6AU00-0HM0	From V6.1
ET 200MP	IM 155-5 MF HF	S2	6ES7155-5MU00-0CN0	From V5.3.1
ET 200SP HA	IM 155-6 PN HA	R1, S2	6DL1155-6AU00-0PM0	From V1.3
ET 200iSP	IM 152-1 PN	R1, S2	6ES7152-1BA00-0AB0	From V1.0

For these devices, parameter changes can be made during operation without bump.

Other PROFINET devices can also be reparametrized in the “Run-Redundant” state, but the IO device will experience a temporary failure during the update →

Status of download in RUN: Configuration in Run is possible with restart of devices

| Other new features

What's new in Firmware V4.1 (V21)

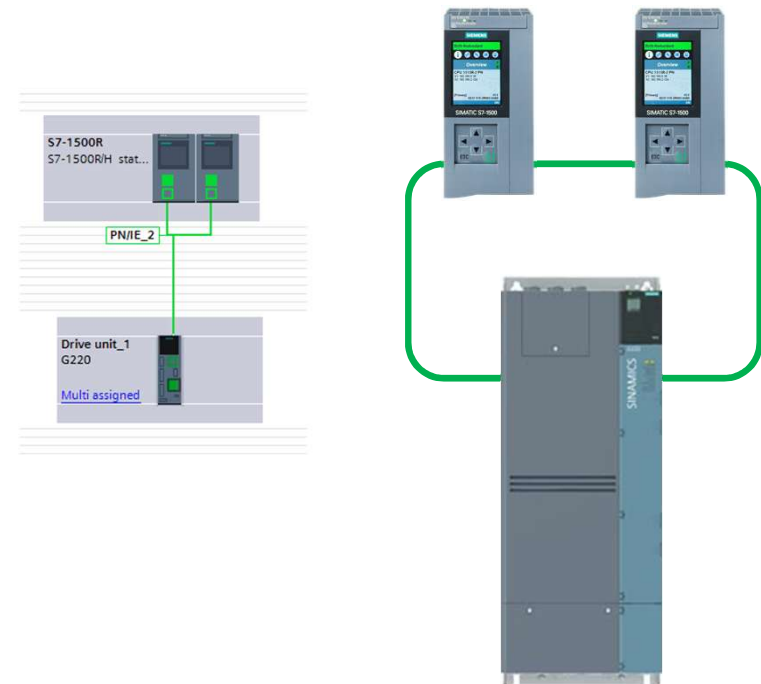
What's new with Firmware Version 4.1 / TIA Portal V21

SINAMICS Startdrive Support

V21

Easy configuration of SINAMICS drives

- SINAMICS drives can be assigned to both controllers in an S7-1500R/H redundant system.
- This enables drive configuration in **SINAMICS StartDrive** without workarounds (such as dummy projects or GSD-only configuration).
- TIA Openness is therefore available for all devices.



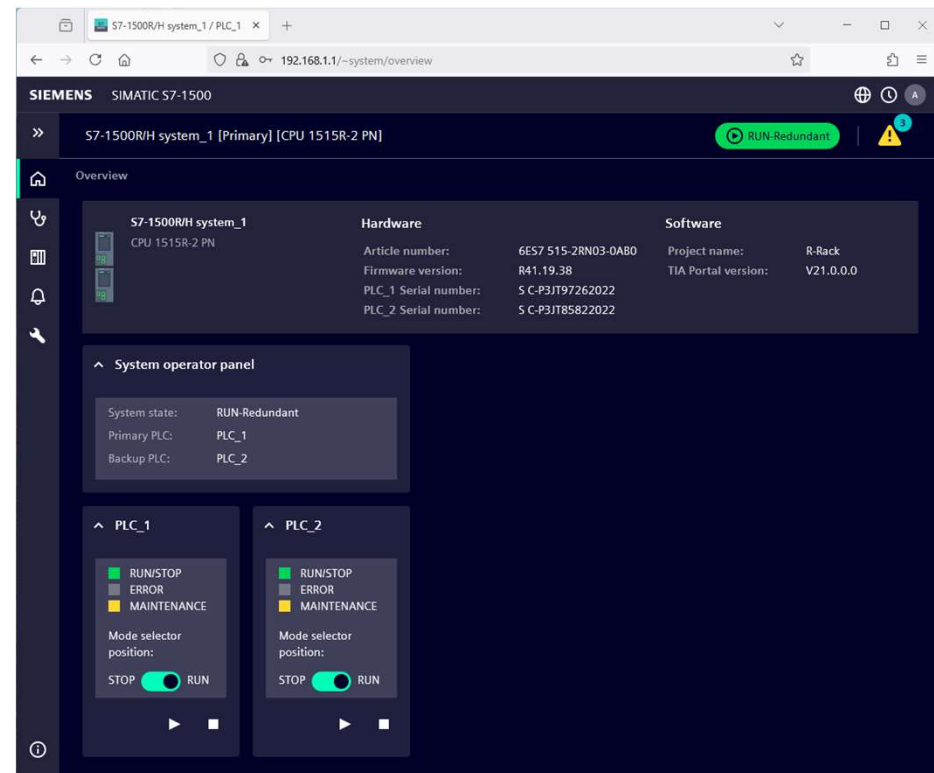
What's new with Firmware Version 4.1 / TIA Portal V21

System Webpages for S7-1500R/H

V21

Web-Browser support

- System Web Pages are now available for R/H controller.
- Both, System-IP and Device-IP Addresses can be used to connect the webserver.



What's new with Firmware Version 4.1 / TIA Portal V21

OPC UA GDS Push

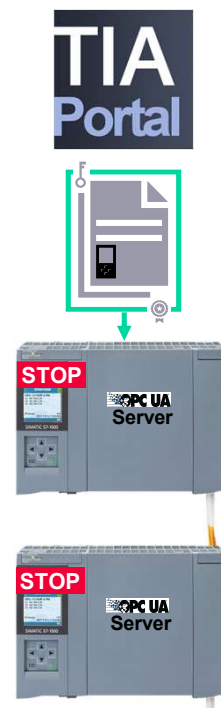
V21

Dynamic Certificate Handling

- Certificate Update in RUN-Redundant via OPC UA, using server methods.
- For services: OPC UA Server, Web server.
- Self-signed or CA-based certificates.
- So far only for S7-1500, **with FW 4.1 also for S7-1500R/H controller**

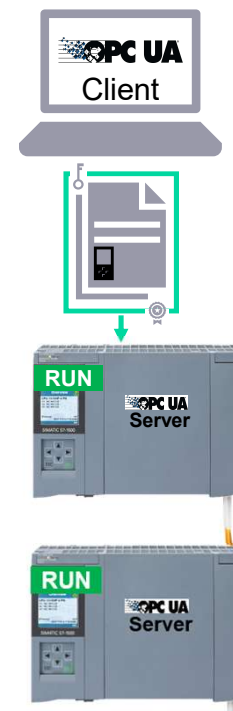
Up to Firmware V4.0

Hardware configuration



New in V4.1

OPC UA – GDS Push



What's new with Firmware Version 4.1 / TIA Portal V21

FW-Update via list of accessible devices

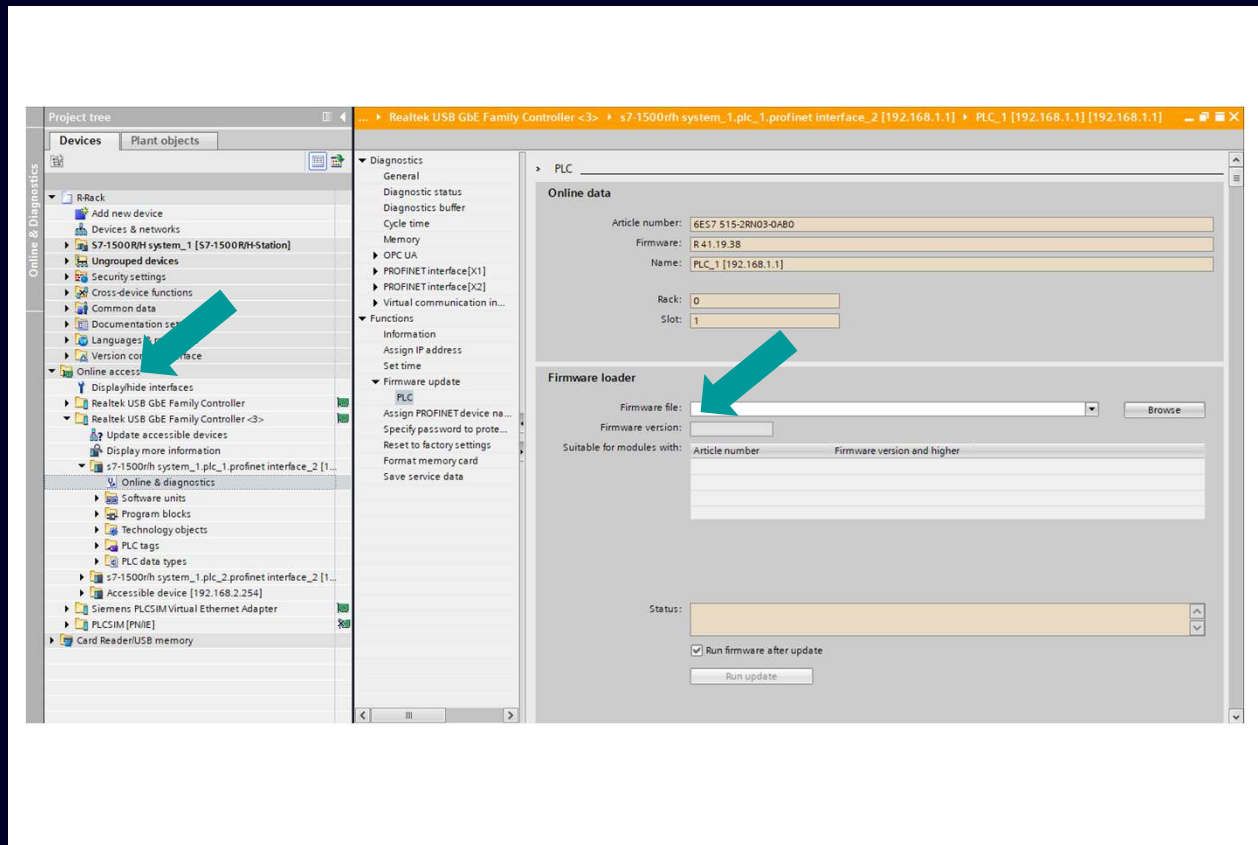
V21

Previous situation

Previously, a matching TIA Portal project was required to update the firmware of R/H CPUs

New feature in TIA Portal V21

Now, firmware updates can be started from the list of accessible devices. This makes it easier to use.



What's new with Firmware Version 4.1 / TIA Portal V21

Improved MRP-Ring status detection

FW V4.1

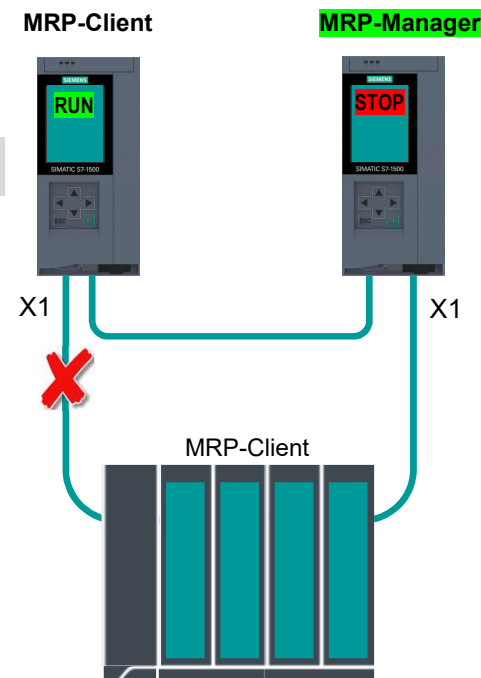
With firmware V4.1, the MRP ring status is available even when the MRP Manager is in STOP mode.

The PROFINET data record library can be used for this purpose.

See 

Note: TIA Portal V21 is not required to use this function.

LPNDR_ReadMRPState
mrpRingStateDomain=**Open ring**



Motivation and Product Strategy

SIMATIC S7-1500 Redundant Systems

Motivation

Avoiding plant downtimes

High availability during operation, avoidance of production downtimes



Prevention of data loss

The data is retained and long recovery times after a failure are eliminated

Prevention of damage

Avoidance of unplanned production stops, where the product to be processed would be permanently damaged.



Operation without persons on site

Trips for maintenance are easier to plan

Save maintenance effort

Applicative solutions are usually complicated and difficult to maintain

Redundant Systems reduce costs

SIMATIC S7-1500 Redundant Systems

Product Strategy SIMATIC S7-1500R/H

Based on standard S7-1500 CPUs and PROFINET

- Based on hardware of standard and failsafe CPUs
- Based on the PROFINET communication standard



Transparent programming (like standard)

- Standard engineering tools incl. all programming languages
- Complete Integration in TIA Portal
- No special redundancy know-how required
- Easy scaling:
Standard CPU → S7-1500R → S7-1500H
Standard F-CPU → S7-1500HF



Scale extensively

- Scaling the switching time
- Scaling the redundancy architecture
- Scaling performance from CPU 1513 to 1518



SIMATIC S7-1500 Redundant Systems

Continuous innovations since market launch

FW V2.8

V16 – Extended Programming 11/2019

- Program Download in Run-Redundant
- PID control loop functions
- Support of S7-Graph and ProDiag
- X1 to X2 Routing
- Additional alarming and diagnosis

FW V2.6

V15.1 – Market Launch 11/2018

- 3 new controller:
1513R / 1515R / 1517H
- ET 200SP and ET 200MP with S2 redundancy

FW V3.0

V18 – Full Redundancy 11/2022

- System Redundancy R1 for S7-1500H
- New ET 200SP redundant IM
- Flexible Network Structures
- Long Distance (40km) H-Sync
- HW-Update for R-CPU

FW V2.9

V17 - Safety 05/2021

- new high availability and safety controller:
CPU 1518HF
- PLCSIM Advanced supports R/H
- MRP Interconnection
- Recipe Functions

FW V4.0

V20 – HW Update 11/2024

- HW-Update for H-CPU
- Full Retentive Memory
- Move System IP to Backup CPU
- Single Mode for R-CPU
- Central User Management
- Profiling

FW V3.1

V19 – Communication 01/2024

- Multiple CP cards with Hot-Swap
- OPC UA Server
- Support of IE/PB LINK HA
- Webserver API

FW V4.1

V21 – Continuous Operation 11/2025

- Configuration in RUN
(1st step)
- GDS Push
- R/H Webpages

System Overview

SIMATIC S7-1500 Redundant Systems

System Overview

CPU Type

Synchronization

Hot-Standby

Max distance between CPUs

PROFINET System Redundancy

Structure of the PROFINET network

Redundant S7-1500R



CPU 1513R / CPU 1515R

via **PROFINET Ring (MRP)**

Yes fail-over time ca. 300 ms

100 m *)

S2 and switched S1

MRP Ring

High Available S7-1500H



CPU 1517H / CPU 1518HF

via **Sync-Module / FO**

Yes fail-over time ca. 50 ms

40 km

R1, S2 and switched S1

Any

*) Can be extended with media converter

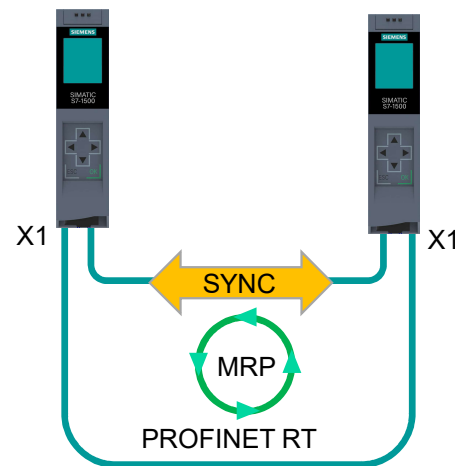
System Overview

S7-1500R System

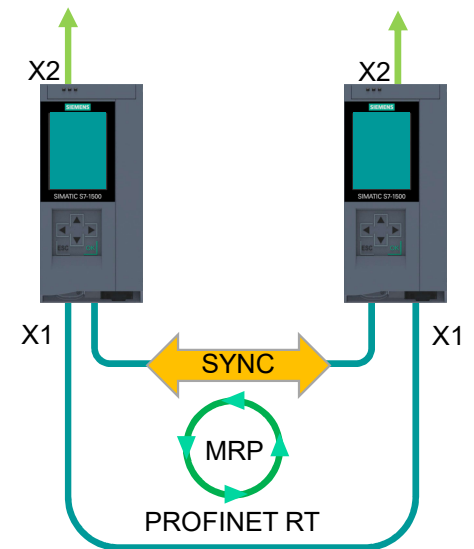
Basic Structure

- In the S7-1500R-System, **synchronization** between the redundant controller is done **via PROFINET cable**.
- A MRP ring structure is mandatory.
- CPU 1515R-2 PN offers one additional ethernet interface.

CPU 1513R-1 PN



CPU 1515R-2 PN

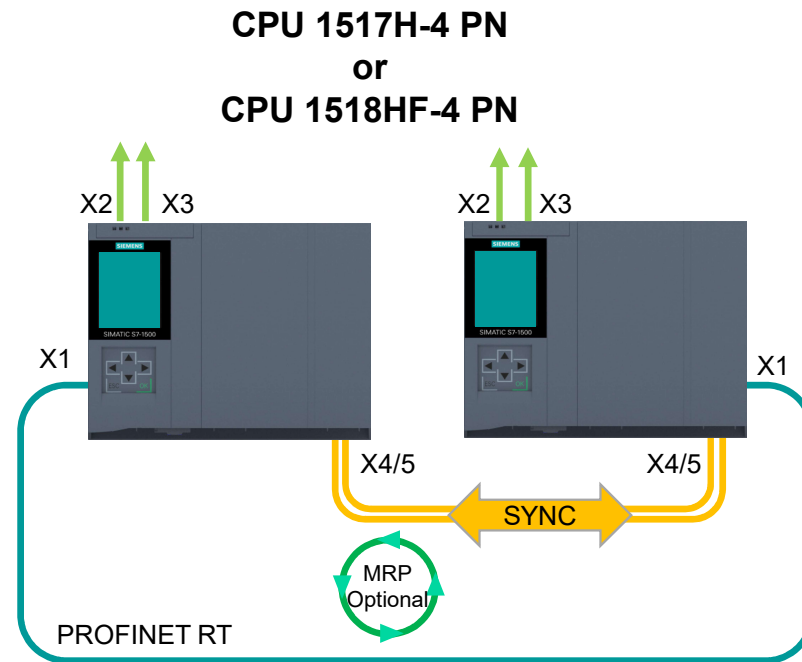


System Overview

S7-1500H System

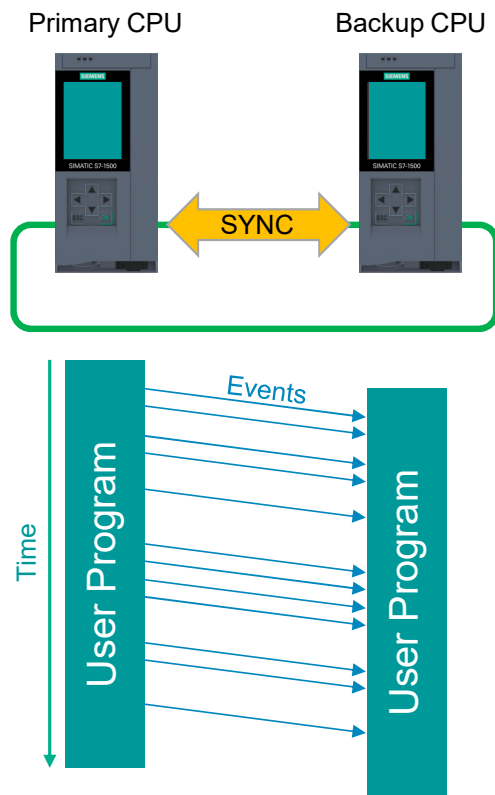
Basic Structure

- An S7-1500H system has a dedicated, redundant **fiber-optic** channel for **synchronization**.
- Any PROFINET topology can be used (Ring, Line, Star or combined).
- CPU 1518HF-4PN can be used in applications with functional safety requirements.



System Overview

How R/H synchronization works



Primary CPU and backup CPU are automatically synchronized based on events in the RUN-Redundant system state.

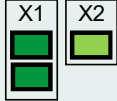
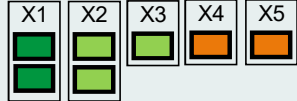
It ensures that both CPUs always have the same internal status and work synchronously.

Synchronization events:

- Cycle control point
- Direct access to IO devices
- Interruption of cyclical program processing (e.g. by interrupts or alarms)
- Updating of times, e.g., by S7 timer
- Updating the process image
- Communication functions that change data

System Overview

CPU Types

	CPU 1513R-1 PN 6ES7513-1RM03-0AB0	CPU 1515R-2 PN 6ES7515-2RN03-0AB0	CPU 1517H-4 PN 6ES7517-4HQ10-0AB0	CPU 1518HF-4 PN 6ES7518-4JT10-0AB0
Program/ Memory	600 kB code 2.5 MB of data	1 MB code 4.5 MB of data	4 MB code 50 MB data	18 MB code 150 MB data
Interfaces				
Firmware	From V3.0	From V3.0	From V4.0	From V4.0
SIPLUS Version	-	6AG1515-2RN03-7AB0	6AG1517-4HQ10-4AB0	6AG1518-4JT10-4AB0



-  PROFINET IO Controller, Supports RT, MRP, TCP/IP, Open User Communication, OPC UA, HTTPS
-  PROFINET Basic Services, TCP/IP, Open User Communication, OPC UA, HTTPS
-  SFP module slot for H-synchronization

System Overview

Interfaces of CPU 1517H and CPU 1517HF

Interface X1

- PROFINET-IO (2 Ports)
- Supports MRP

Interface X2

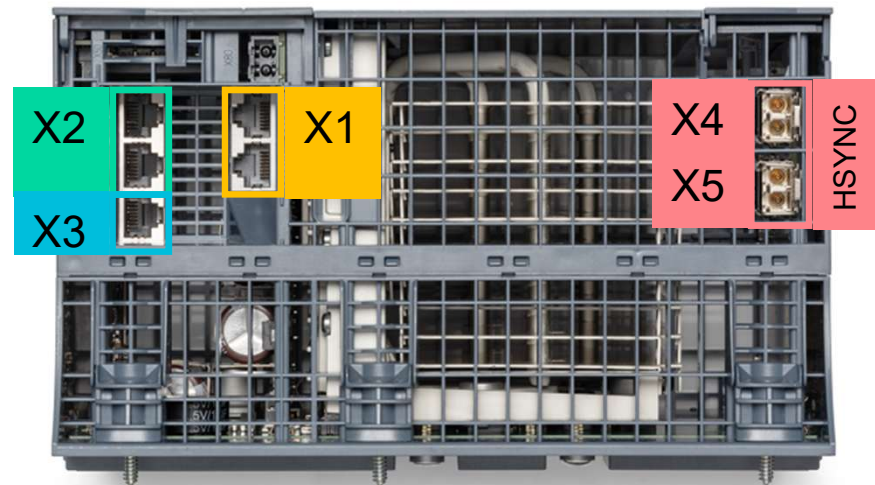
- Ethernet 100Mbit (2 Ports)
- MRP / PN-IO in later FW-versions

Interface X3

- Gigabit Ethernet (1 Port)

Interface X4/5

- Redundant Interface for H-Sync Modules



System Overview

Sync modules for S7-1500H



H-Sync Module
4 pieces per H-system



Sync Cable
2 pieces per H-system



Distance	Sync Module	Fiber Type	FO Specification	Connector
Up to 200 m	6ES7960-1CB00-0AA5	Multi Mode	Core diameter 50/125µm or 62,5/125µm	LC/PC or LC/UPC
From 2 m to 10 km	6ES7960-1FB00-0AA5	Single Mode	Core diameter 9/125µm Specification OS1 or OS2	LC/PC or LC/UPC
From 8 km to 40 km	6ES7960-1FE00-0AA5 ¹⁾	Single Mode	Core diameter 9/125µm Specification OS1 or OS2	LC/PC or LC/UPC

1) Max. ambient temperature: +55°C

System Overview

Pre-fabricated multimode sync cables for S7-1500H

Multimode cables for installation
in the control cabinet



Length	Article Number
1 m	6ES7960-1BB00-5AA5
2 m	6ES7960-1BC00-5AA5
10 m	6ES7960-1CB00-5AA5

Multimode cables for installation
between control cabinets



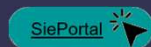
Length	Article Number
20 m	6XV1873-5AN20
30 m	6XV1873-5AN30
40 m	6XV1873-5AN40
50 m	6XV1873-5AN50
80 m	6XV1873-5AN80
100 m	6XV1873-5AT10
150 m	6XV1873-5AT15
200 m	6XV1873-5AT20

Multimode Robust Cable
Outdoor installation also possible



Length	Article Number
10 m	6XV1873-5RN10
30 m	6XV1873-5RN30
50 m	6XV1873-5RN50
100 m	6XV1873-5RT10
150 m	6XV1873-5RT15

Overview of fiber optics at Siemens

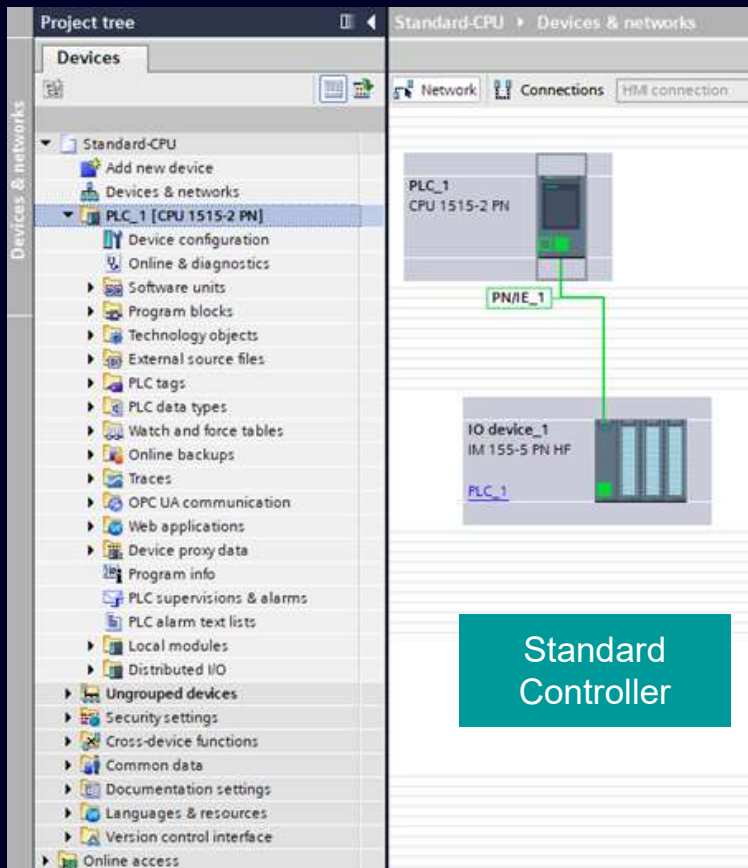


Configuration and Functional principle

System Overview

Configuration

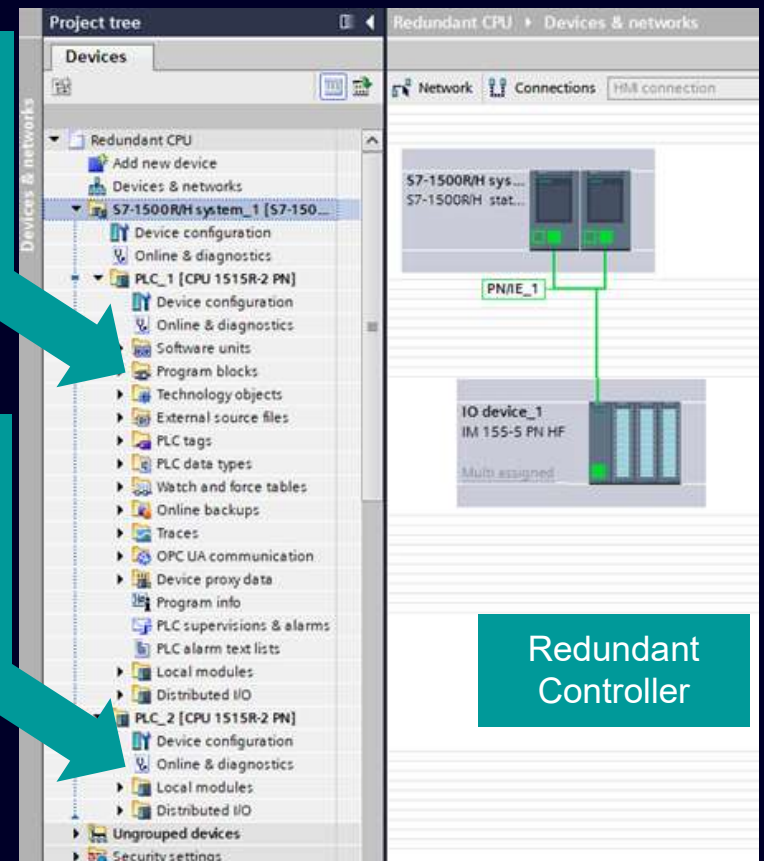
Comparison with Standard Controller



Same program structure

Only 1 CPU is configured

2nd Controller is shown for diagnosis purpose

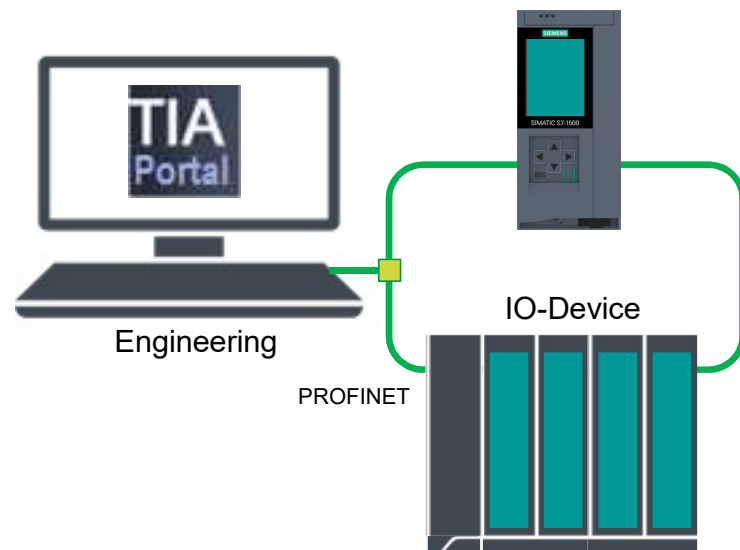


Redundant Controller

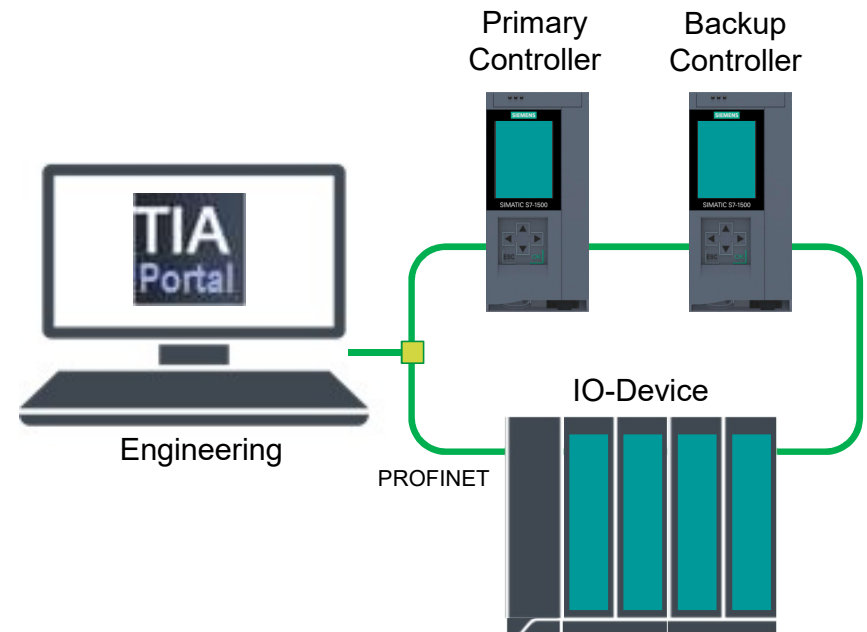
Configuration

Comparison with Standard Controller

System with Standard Controller

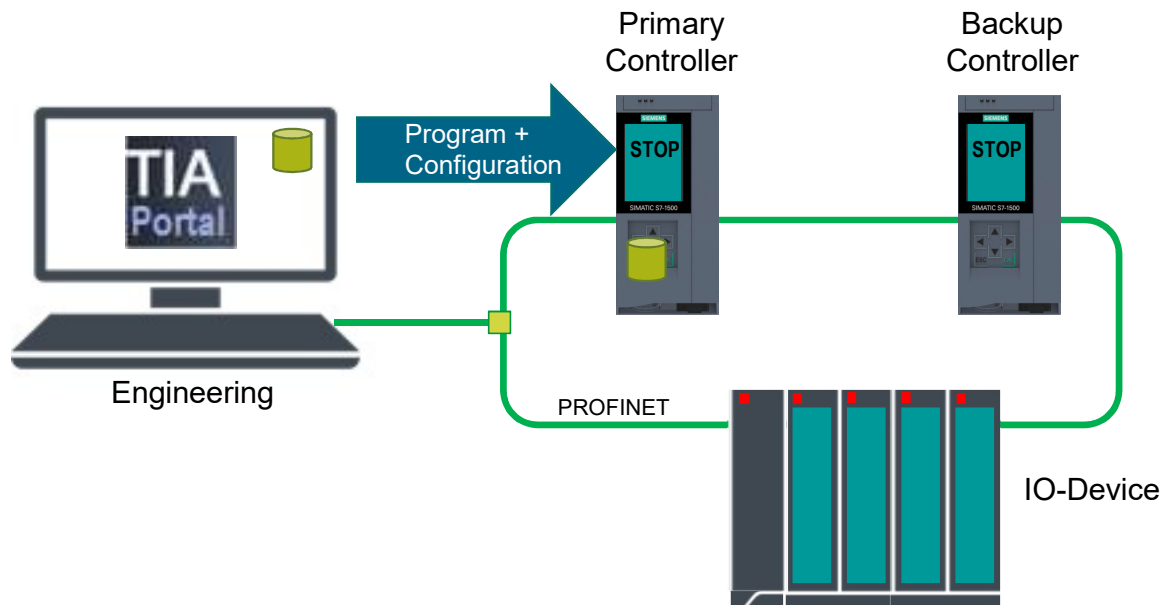


System with Redundant Controller



Functional principle

Initial commissioning

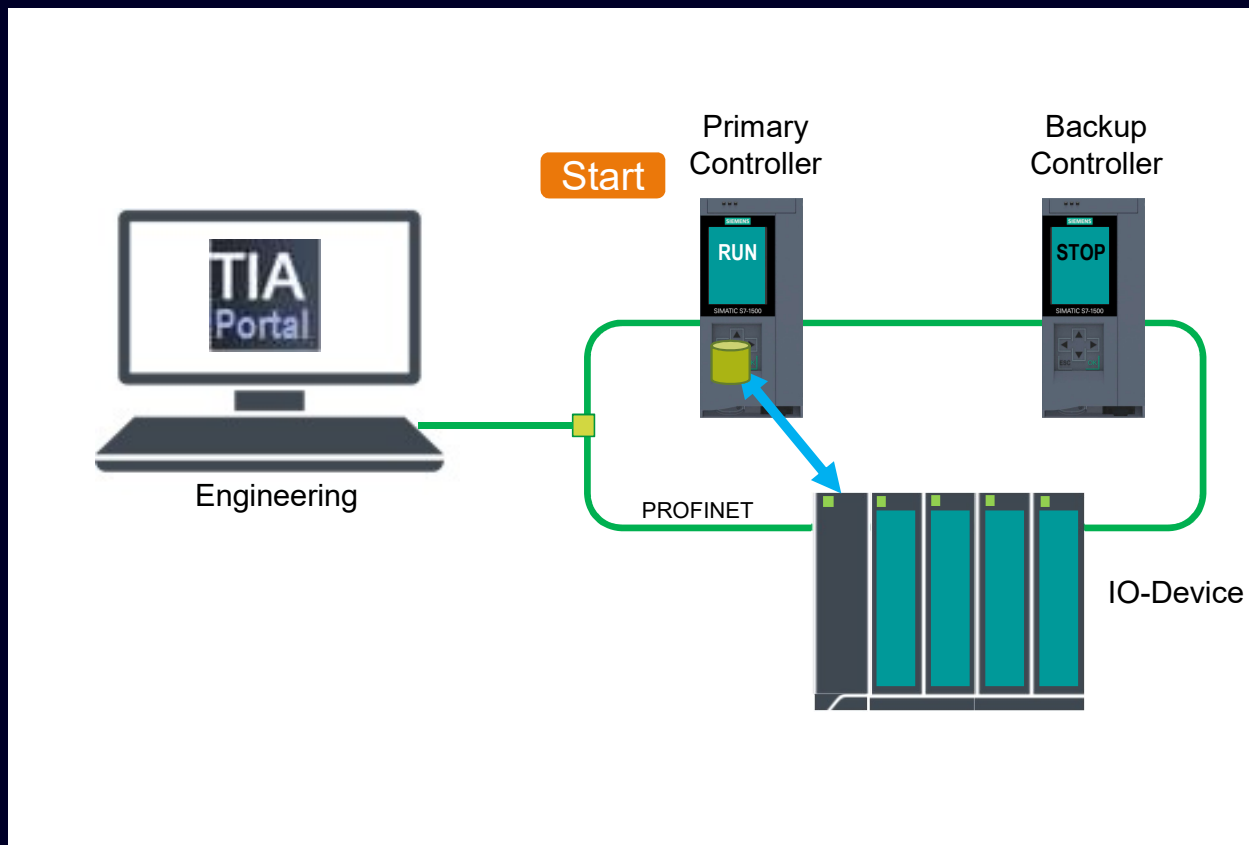


System State **STOP**

The TIA Portal project will be loaded in one of the two CPUs.
No changes are required for redundancy in the CPU program.

Functional principle

Starting a controller

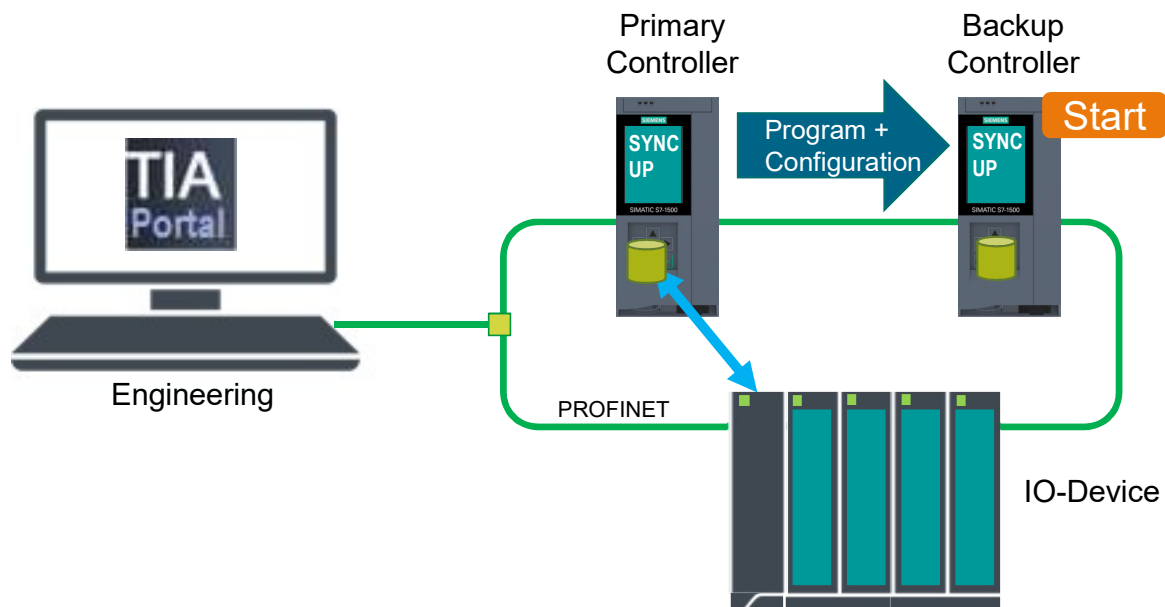


System State
RUN-Solo

The loaded controller takes over the process

Functional principle

Start the backup controller

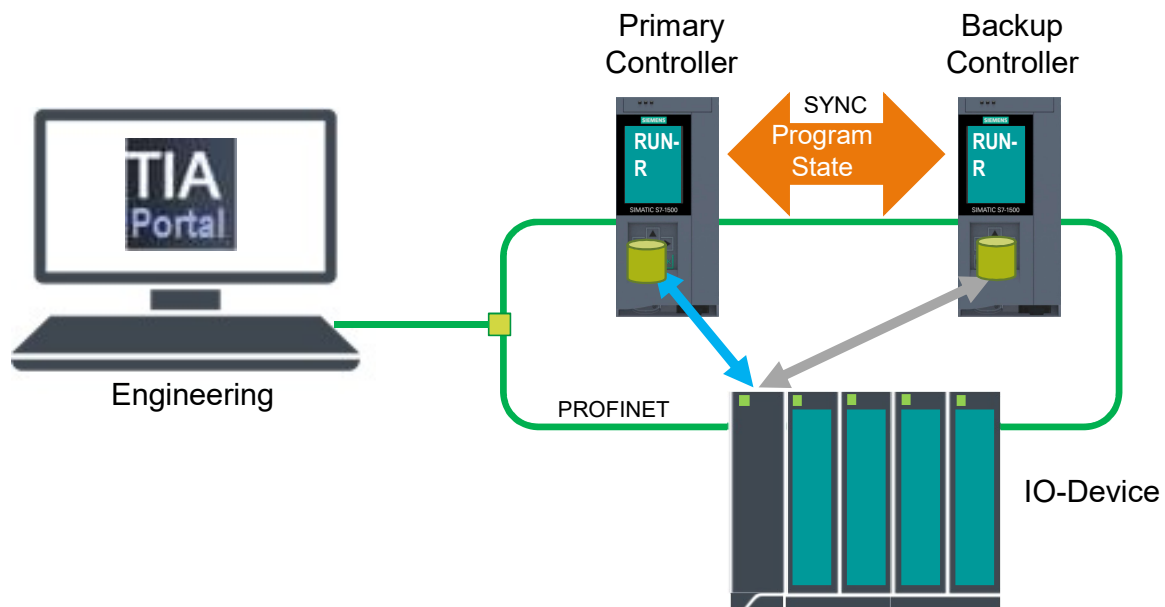


System State SYNCUP

The project is transferred
automatically to the backup CPU

Functional principle

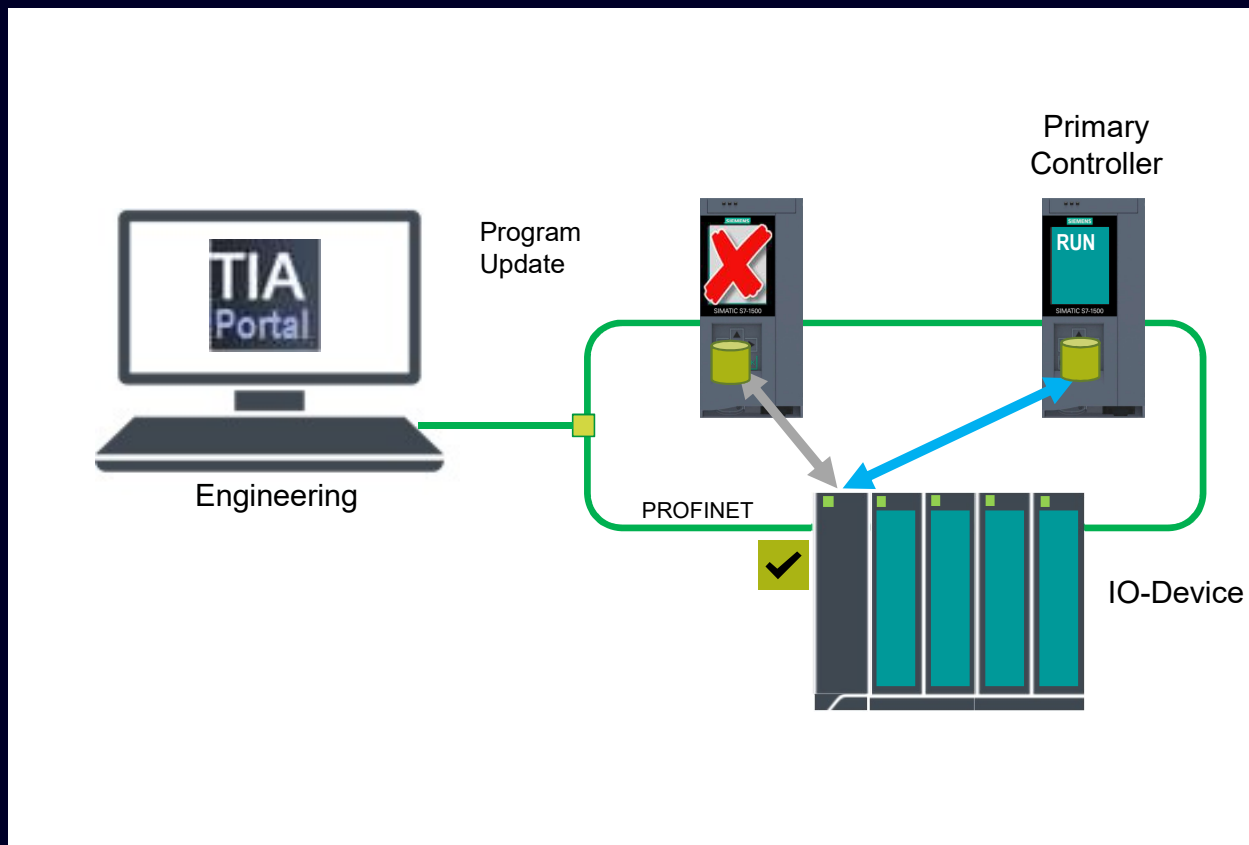
Backup controller startup completed



System State
RUN-Redundant

Both CPUs work on the same program. The state of the CPUs is synchronized.

Functional principle Fail Over



System State
RUN-Solo

If one of the controllers fail, the second takes over process control immediately

PROFINET

System Redundancy

PROFINET System Redundancy Concept

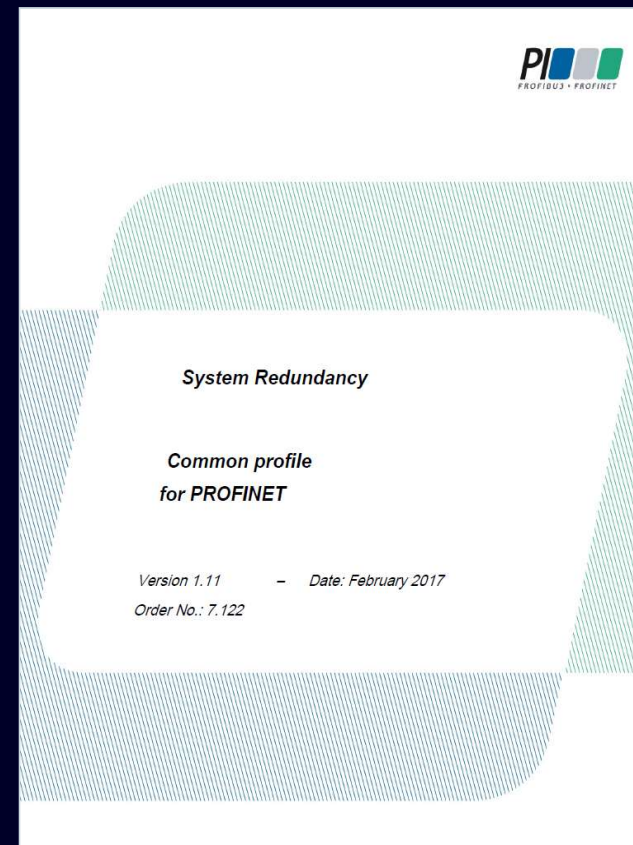
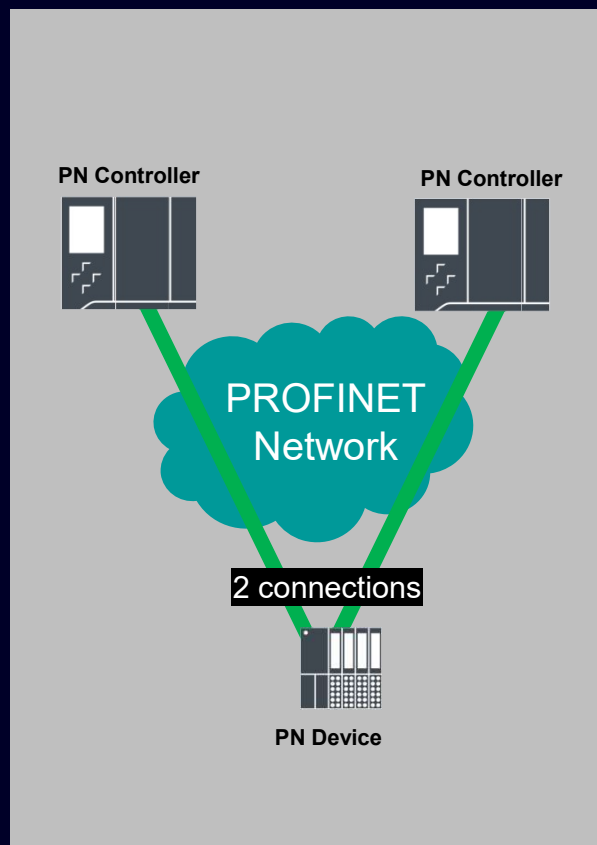
PROFINET System Redundancy

A system of redundant PN controllers and singular or redundant PN devices.

Three levels:

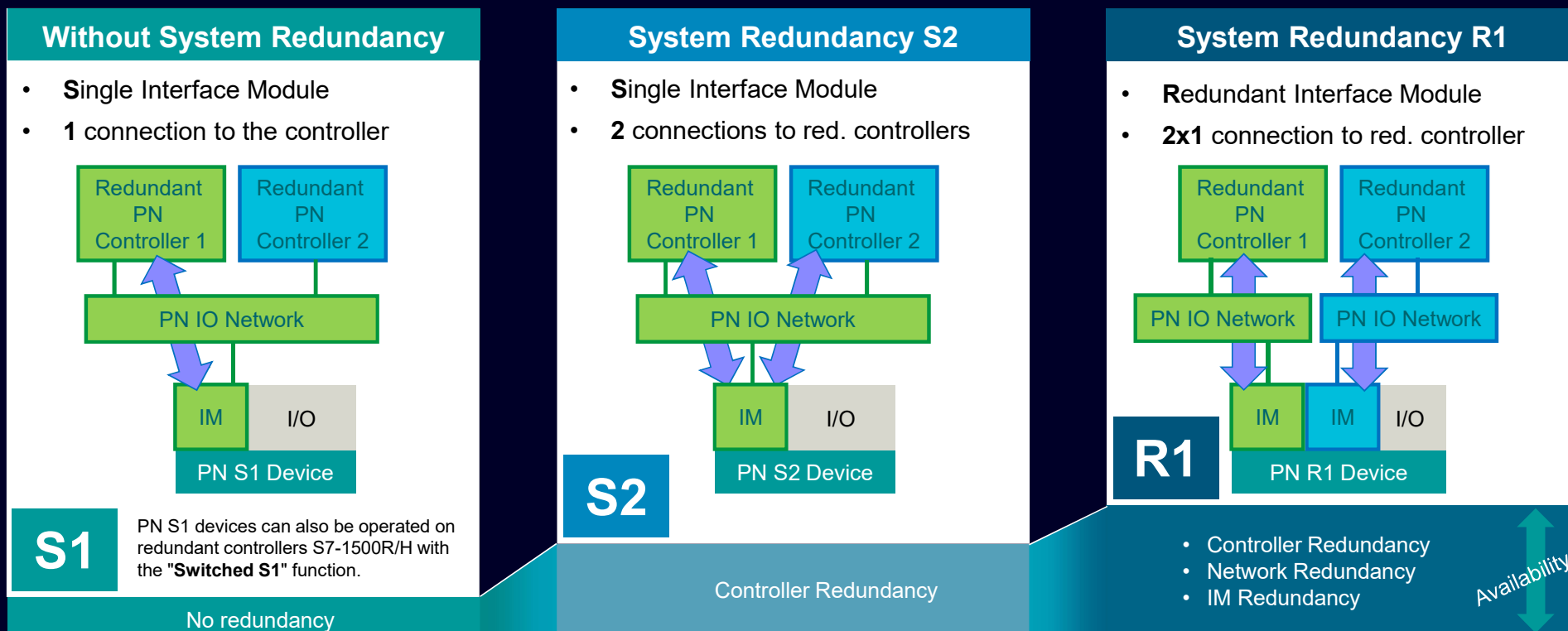
1. PN Controller
2. PROFINET network
3. PN-Device

The redundancy of one layer is independent of the other layers.



PROFINET System Redundancy

Redundancy operating modes with PROFINET



Details on PN System Redundancy operating modes: See



PROFINET System Redundancy

Redundancy operating modes with PROFINET

The system tolerates the following failures (Fault Tolerance)

Without System Redundancy (S1)

Fault tolerance:

- CPU (with bump)
- PROFINET Failure

S1

With System Redundancy S2

Fault tolerance:

- CPU (bumpless)
- PROFINET Failure

S2

With System Redundancy R1

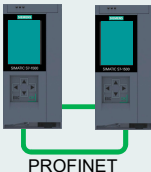
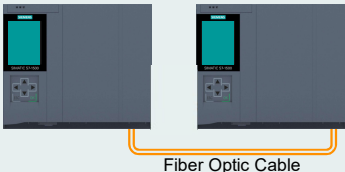
Fault tolerance:

- CPU (bumpless)
- PROFINET Failure
- Interface Module
- Subnet

R1

PROFINET System Redundancy

Support of PN System Redundancy with SIMATIC S7-1500

		S7-1500 (Non-redundant)	S7-1500R	S7-1500H
				
S1	Operation of PN devices without System Redundancy	Yes	Yes (Switched S1)	Yes (Switched S1)
S2	Operation of PN devices with S2 System Redundancy	Yes (S1 mode only)	Yes	Yes
R1	Operation of PN devices with R1 System Redundancy	No	No	Yes

Switching behavior without System Redundancy (S1)

PROFINET System Redundancy

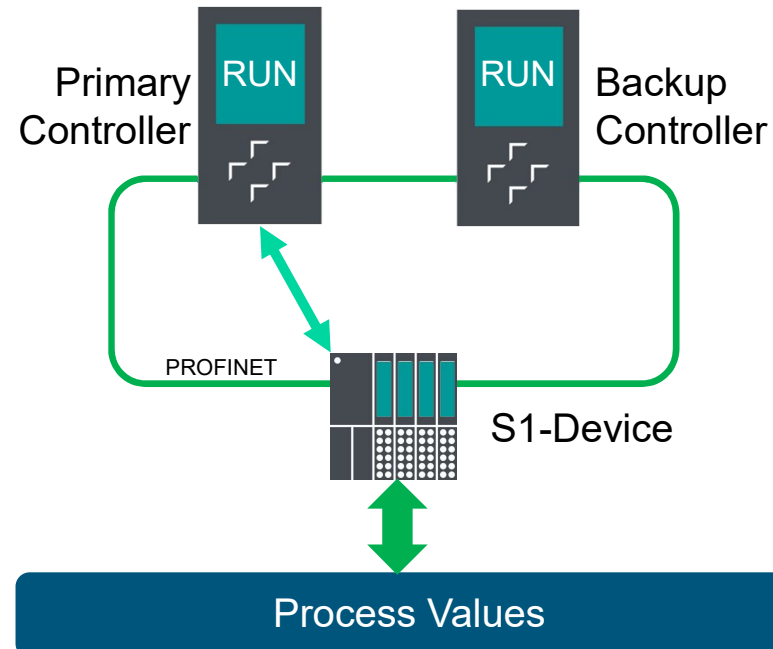
Switching behavior without PROFINET System Redundancy

Initial situation

S1

Primary CPU has a
Connection (AR) to the
S1-Device

Process values are exchanged



Initial Situation

Switching behavior

Result

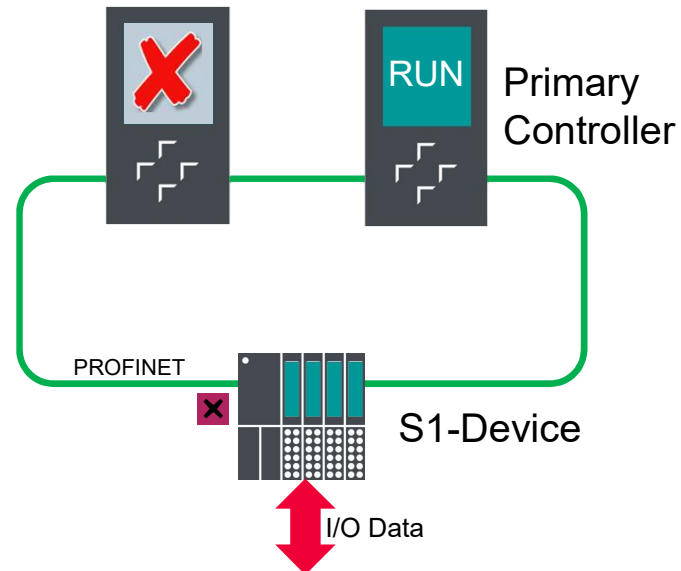
Switching behavior without PROFINET System Redundancy

Scenario: One controller fails – switching process begins

S1

Remaining controller takes over automatically.
There is no connection to the S1 device

Transfer of process values is interrupted.
Outputs go to 0 or use the configured substitute behavior.



Initial Situation

Switching behavior

Result

Switching behavior without PROFINET System Redundancy

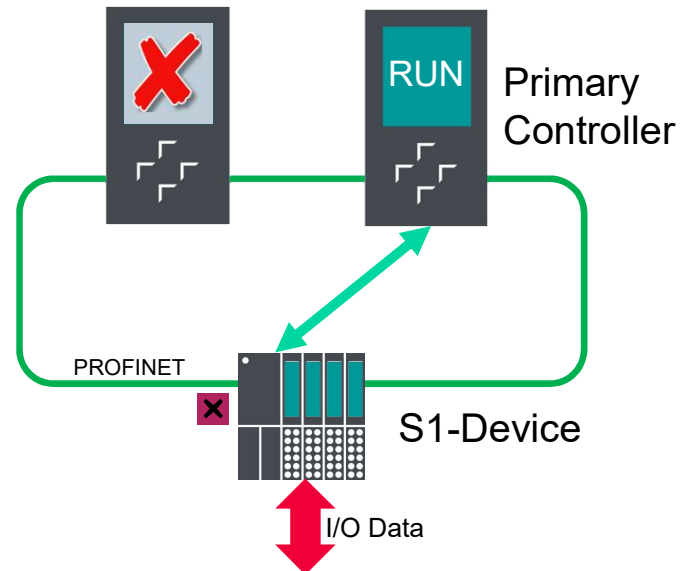
Scenario: One controller fails - connection establishment

S1

Connection to the S1 device is re-established

Transfer of process values is interrupted.
Outputs go to 0 or use the configured substitute behavior.

Duration depends on the start-up behavior of the S1 device. Minimum approx. 600ms



Initial Situation

Switching behavior

Result

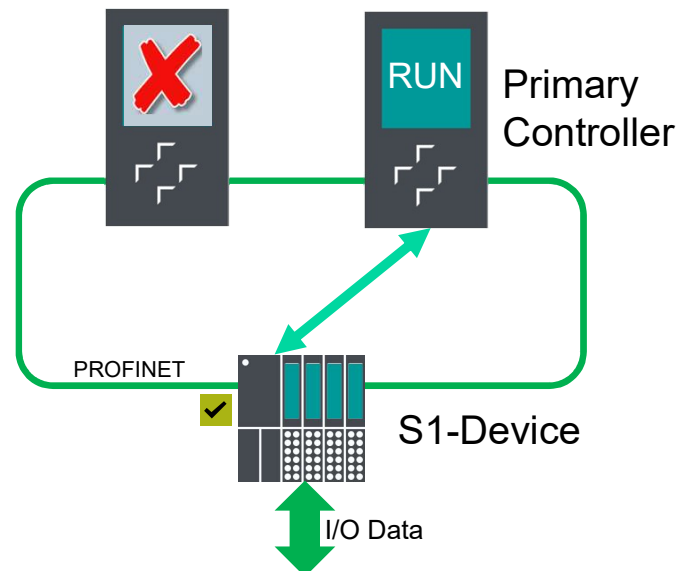
Switching behavior without PROFINET System Redundancy

Scenario: One controller fails - operation complete

S1

Remaining controller has a connection to the S1 device

Process values are exchanged



Initial Situation

Switching behavior

Result

| System Redundancy S2

PROFINET System Redundancy

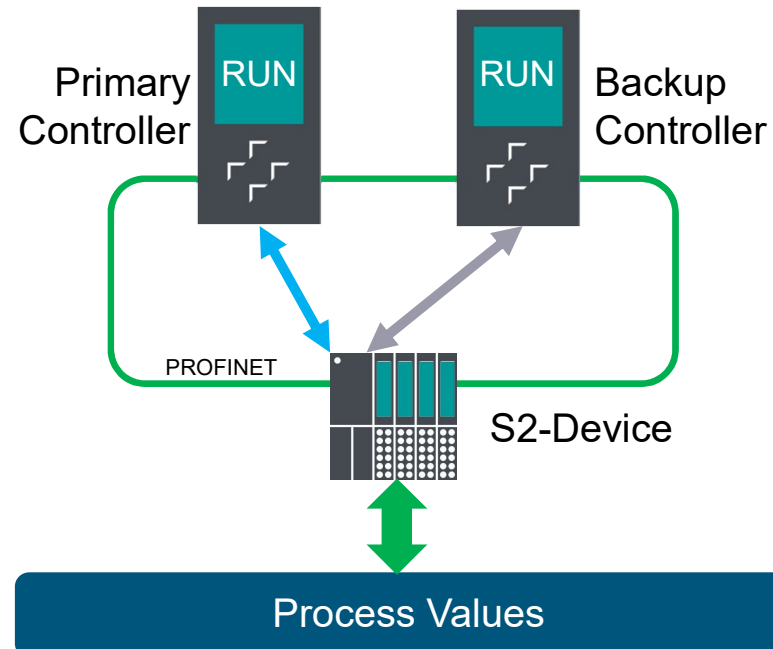
Switching behavior with PROFINET System Redundancy S2

Initial situation

S2

Both CPUs have a Connection (AR) to the S2-Device

Process values are exchanged



Initial Situation

Switching behavior

Result

Primary AR

Backup AR

Switching behavior with PROFINET System Redundancy S2

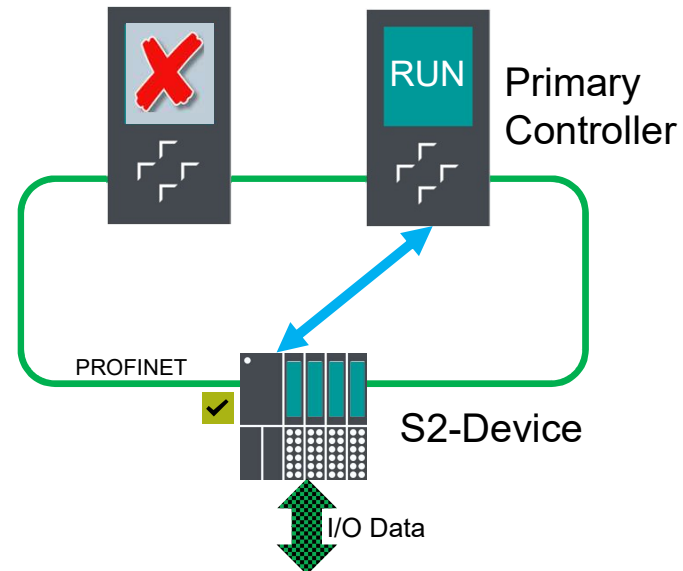
Scenario: One controller fails – switching process begins

S2

Remaining controller takes over automatically.
The existing connection is activated

Process values are frozen for a very short time¹⁾
I/O values remain at the last value

¹⁾ S7-1500R 300ms / S7-1500H 50ms



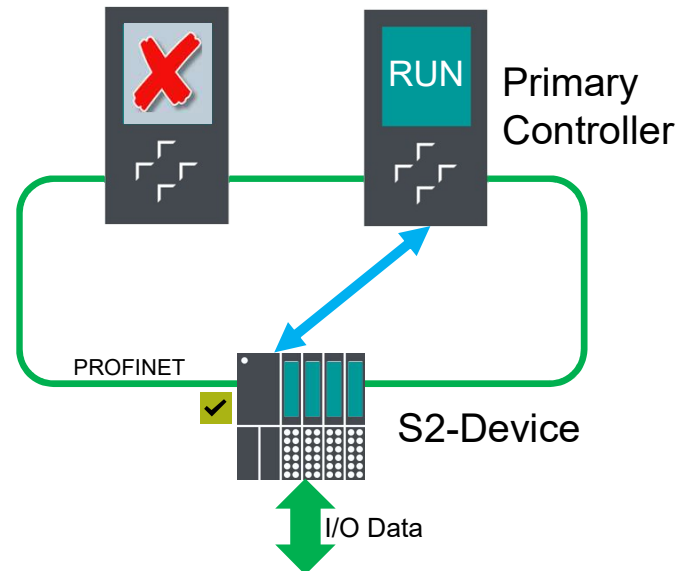
Switching behavior with PROFINET System Redundancy S2

Scenario: One controller fails - operation complete

S2

Remaining controller has a connection to the S2 device

Process values are exchanged



Initial Situation

Switching behavior

Result

Primary AR
Backup AR

Siemens Products with System Redundancy S2

PROFINET System Redundancy

Siemens Products with System Redundancy S2

Siemens I/O systems (I)

S2

Product		Article Number
ET 200SP IM155-6 PN HF IM 155-6 MF HF		6ES7155-6AU01-0CN0 6ES7155-6AU30-0CN0 6ES7155-6MU01-0CN0
ET 200MP IM 155-5 MF HF		6ES7155-5MU00-0CN0 Active backplane: 6ES7590-0BL00-0AA0 The active backplane bus allows the pulling and plugging of ET 200MP modules while the CPU is in operation.
PN/PN coupler		6ES7158-3AD10-0XA0

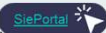
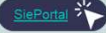
PN functions in detail: See



Siemens Products with System Redundancy S2

Siemens I/O systems (II)

S2

Product		Article Number
ET 200SP HA  IM155-6 PN HA (with Single IM) New: Support of F-Modules in V21 		6DL1155-6AU00-0PM0
ET 200iSP  IM 152-1PN (with Single IM) For use up to Ex zone 1		6ES7152-1BA00-0AB0
SIMATIC CFU  Compact Field Unit CFU PA: Connection of PROFIBUS PA field devices CFU DIQ: Digital I/O CFU CIO HART: Configurable I/O	 	6ES7655-5PX11-1XX0 (CFU PA) 6ES7655-5PX31-1XX0 (CFU DIQ) 6ES7655-5PX41-1XX0 (CFU CIO HART) With Alu housing: 6ES7655-5PX11-1AX0 (CFU PA) 6ES7655-5PX31-1AX0 (CFU DIQ) 6ES7655-5PX41-1AX0 (CFU CIO HART)

PN functions in detail: See



Siemens Products with System Redundancy S2

Siemens I/O systems IP67

S2

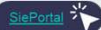
Product		Article Number
ET 200eco PN M12-L  (from FW 1.1)		6ES7 14*-6**00-0BB0
ET 200AL  IM 157-1 MF Hint: ET 200AL IO modules can also be connected in S2-Mode to an ET 200SP device via BA Send adapter		6ES7157-1MA00-0AB0
ET 200clean 		6ES7 141-7BH00-0BB0 (DI16x24VDC, 8xM12) 6ES7 143-7BH00-0BB0 (DIQ 16x24VDC, 8xM12) 6ES7 148-7JH00-0BB0 (CM 8xIO-Link, 8xM12)

PN functions in detail: See



Siemens Products with System Redundancy S2 SINAMICS Drives

S2

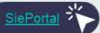
Product		Article Number
SINAMICS S120, S150, G130, G150  with CU310-2PN (FW >=5.1 HF1) (as GSDML)		6SL3040-1LA01-0AA0
SINAMICS S120, S150, G130, G150  with CU320-2PN (FW >= 5.1 HF1) (as GSDML)		6SL3040-1MA01-0AA0
SINAMICS G220 		6SL4112-0C...

Application example for configuring G220 with S2-Redundancy: See 

Application example for configuring SINAMICS drives: See 

Siemens Products with System Redundancy S2 Motor Management System and Motor Starters

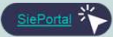
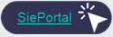
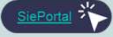
S2

Product		Article Number
SIMOCODE pro V PN  SIMOCODE pro V PN GP FW V3.0		3UF7 011.. New hardware is required for S2 function (No firmware update possible on existing hardware)
Soft Starter SIRIUS 3RW55 / 3RW55 Failsafe Communication Module “High Feature” FW V3.0 		3RW5950-0CH00

Siemens Products with System Redundancy S2

SCALANCE Switches (I)

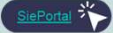
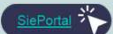
S2

Product	
SCALANCE XC-200 series 	
SCALANCE XP-200 Series 	
SCALANCE XC-300 series 	
SCALANCE XR-300 series 	

Siemens Products with System Redundancy S2

SCALANCE Switches (II)

S2

Product	
SCALANCE XF204-2BA SCALANCE XF204-2BA DNA (Y-Switch) 	
SCALANCE X400 from FW 6.3 	
SCALANCE X500 from FW 6.3 	

| System Redundancy R1

PROFINET System Redundancy

Switching behavior with PROFINET System Redundancy R1

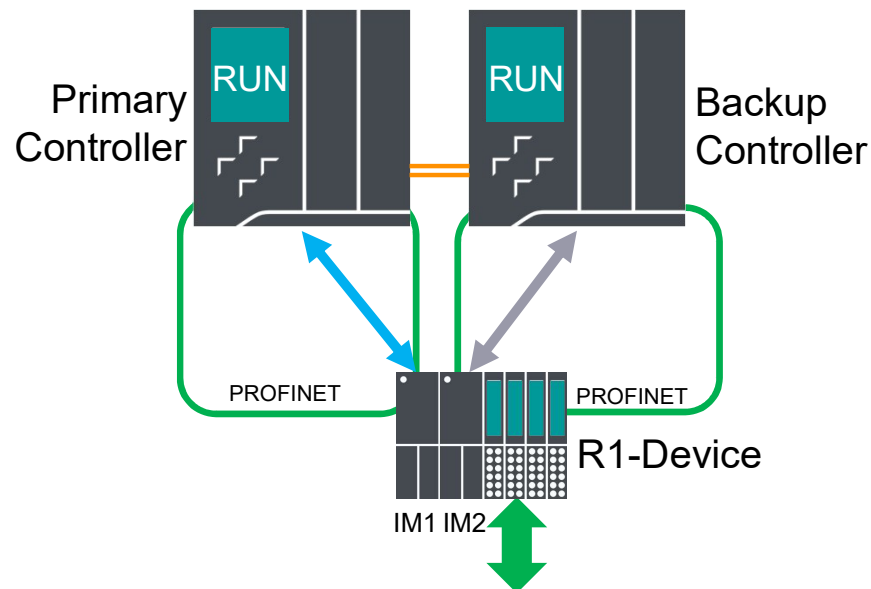
Initial situation

R1

Primary Controller is connected to
Interface Module 1 (IM1)

Backup Controller is connected to
Interface Module 2 (IM2)

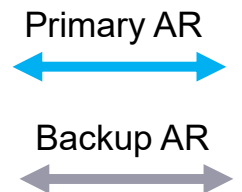
Process values are exchanged



Initial Situation

Switching behavior

Result



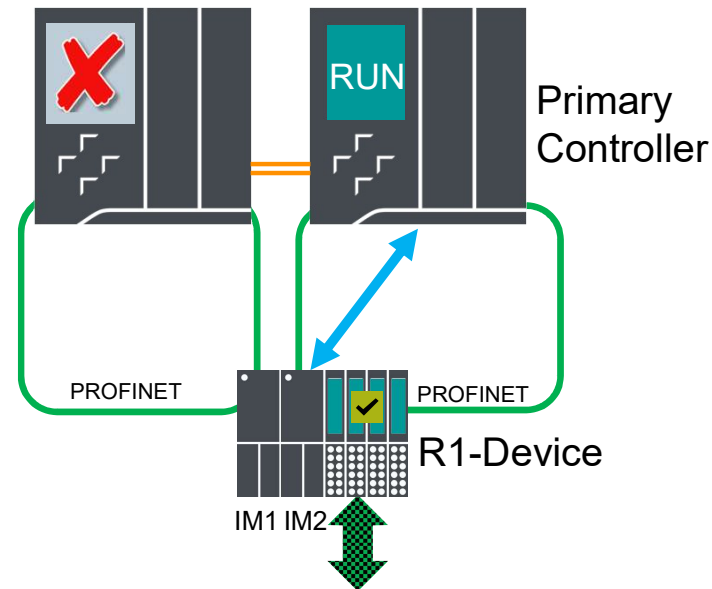
Switching behavior with PROFINET System Redundancy R1

Scenario: One controller fails – switching process begins

R1

Remaining controller takes over
automatically.
The existing connection is activated

Process values are used for a very short
time frozen:
I/O values remain at the last value



Initial Situation

Switching behavior

Result

Primary AR
Backup AR

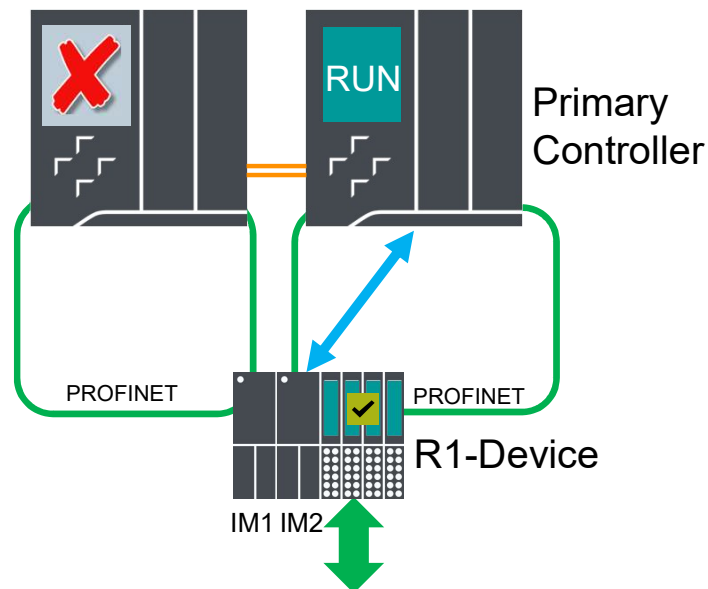
Switching behavior with PROFINET System Redundancy R1

Scenario: One controller fails - operation complete

R1

Remaining controller has a connection to interface module 2

Process values are exchanged



Initial Situation

Switching behavior

Result

Switching behavior with System Redundancy R1

Scenario: Failure of a PROFINET interface module

Switching behavior with PROFINET System Redundancy R1

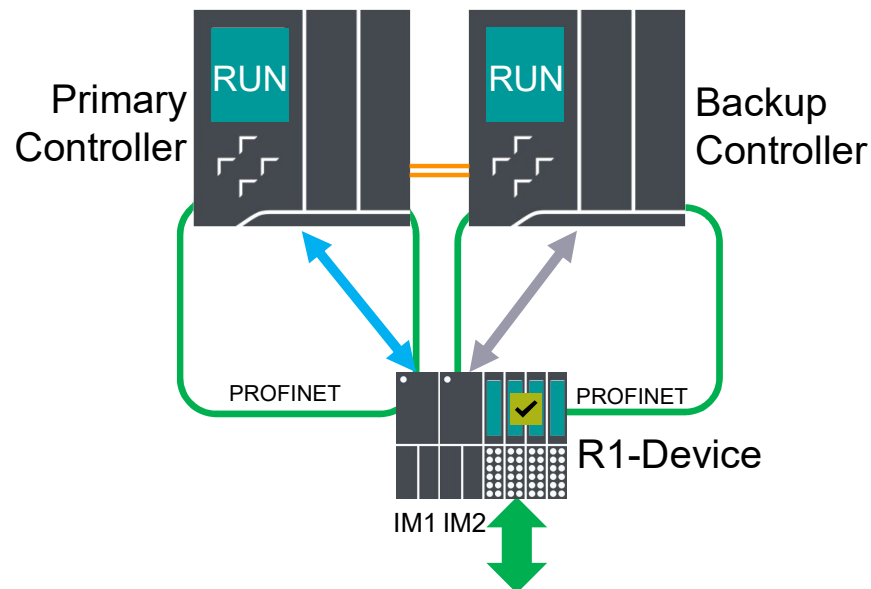
Initial situation

R1

Primary Controller is connected to
Interface Module 1 (IM1)

Backup Controller is connected to
Interface Module 2 (IM2)

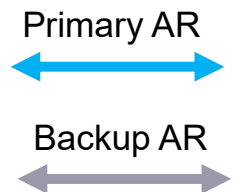
Process values are exchanged



Initial Situation

Switching behavior

Result



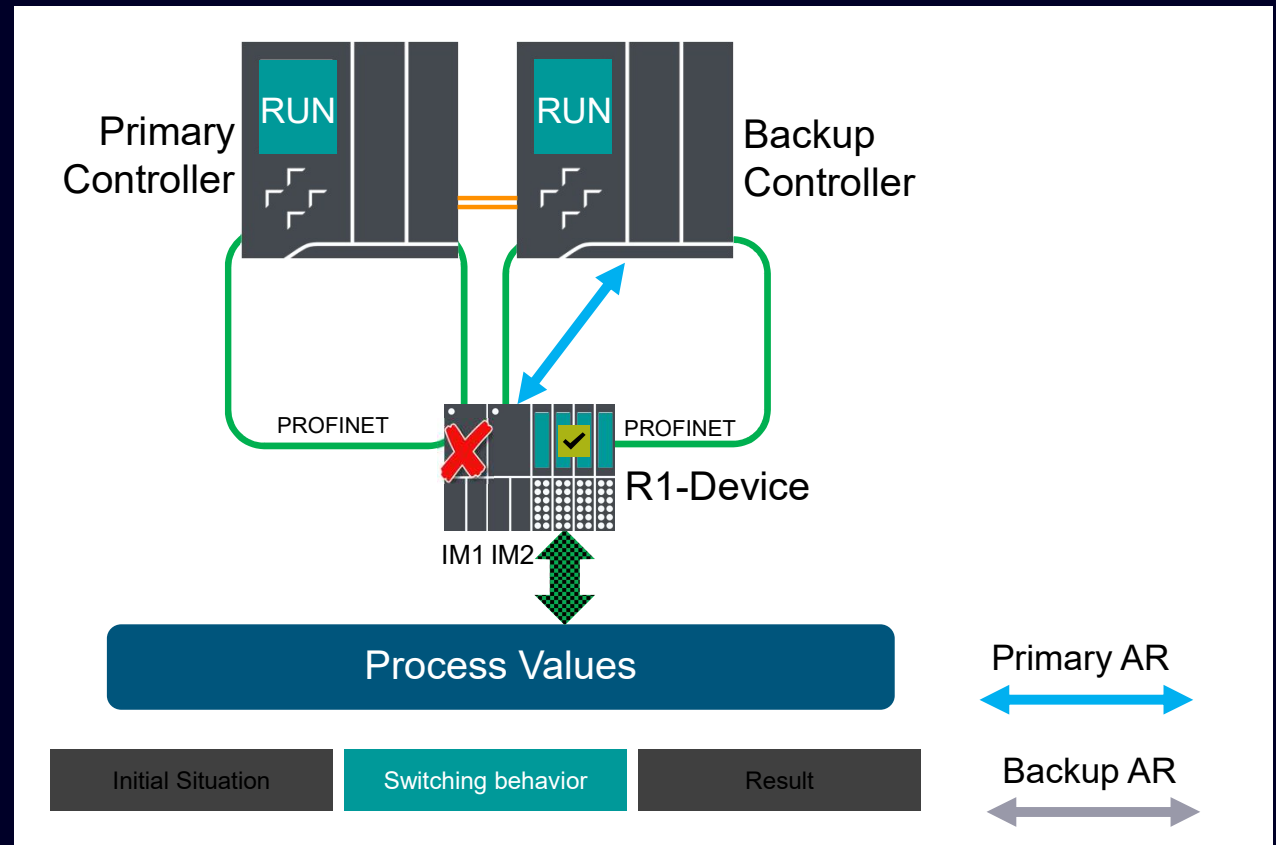
Switching behavior with PROFINET System Redundancy R1

Scenario: An interface module fails – switching process begins

R1

Backup Controller enables connection to
Interface-Module 2 (IM2)

Process values are frozen for a very short
time:
I/O values remain at the last value



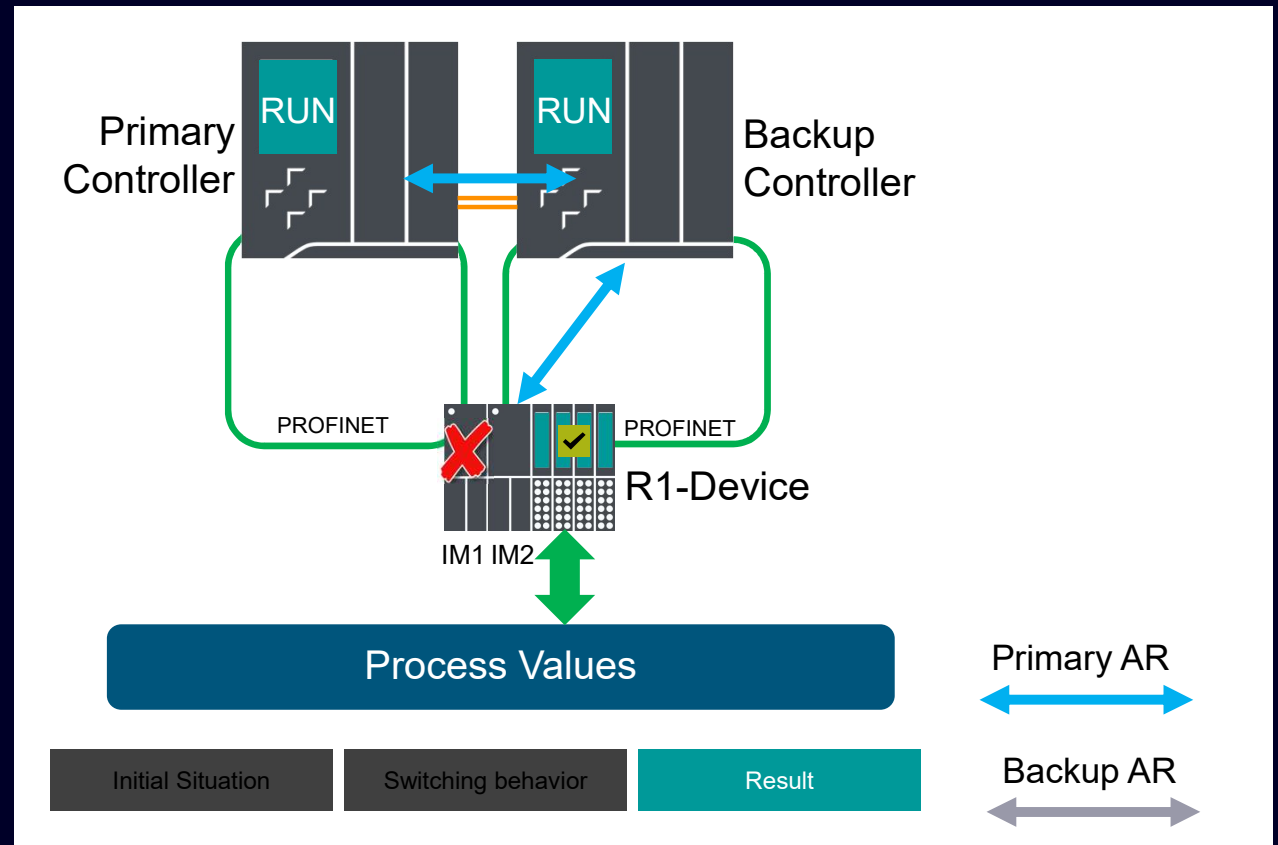
Switching behavior with PROFINET System Redundancy R1

Scenario: An interface module fails - operation complete

R1

I/O data is transferred via sync line via the backup controller to the Primary controller.
There is no controller switching.

Process values are exchanged



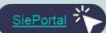
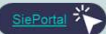
Siemens Products with System Redundancy R1

PROFINET System Redundancy

Siemens Products with System Redundancy R1

Siemens I/O systems

R1

Product		Article Number
ET 200SP  IM155-6 PN R1(with redundant IM)		6ES7155-6AU00-0HM0
ET 200SP HA  IM155-6 PN HA (with redundant IM) V21  New: Support of F-Modules in V21		6DL1155-6AU00-0PM0
ET 200iSP IM 152-1PN (with redundant IM) 		6ES7152-1BA00-0AB0

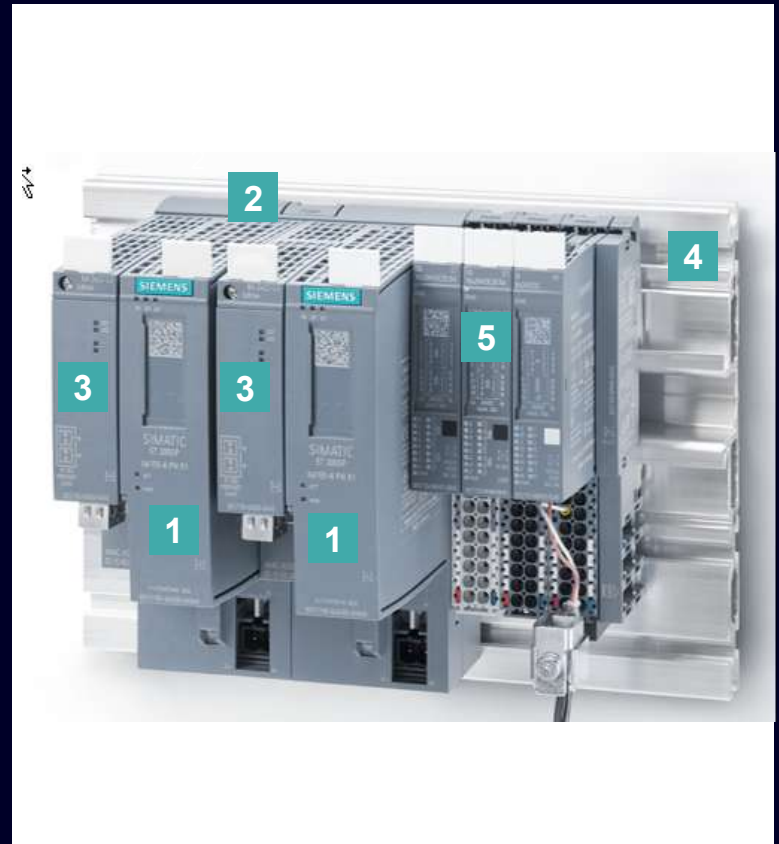
Siemens Products with System Redundancy R1

ET 200SP – IM155-6 PN R1

R1

An ET 200SP R1 system consists of:

- 1** 2x IM 155-6 PN R1
- 2** 1x BU-Type M0
- 3** 2x same bus adapters from the ET 200SP spectrum
- 4** 1x SIMATIC system rail
- 5** Any IO modules



Siemens Products with System Redundancy R1

ET 200SP – IM155-6 PN R1 - Ordering data

R1



IM 155-6 PN R1
6ES7155-6AU00-0HM0



BU Type M0
6ES7193-6BR00-0



SIMATIC System Rail
6ES7193-6MR00-0AA0 (19")
6ES7193-6MR00-0BA0 (530mm)
6ES7193-6MR00-0CA0 (850mm)
6ES7193-6MR00-0DA0 (2 m)

The **SIMATIC System Rail** is required when using the IM 155-6 PN R1

- Tight dimensional tolerances for operating conditions under high mechanical stress (significantly better than dimensional tolerance standard EN 60715)
- long-term stable surface treatment for optimum interference dissipation
- stable design for self-supporting construction
- integrated Bosch profile grooves (6mm profile grooves, B-type)

Configuration of System Redundancy R1

PROFINET System Redundancy

Configuration of System Redundancy R1

Option 1

R1

Subnet separation

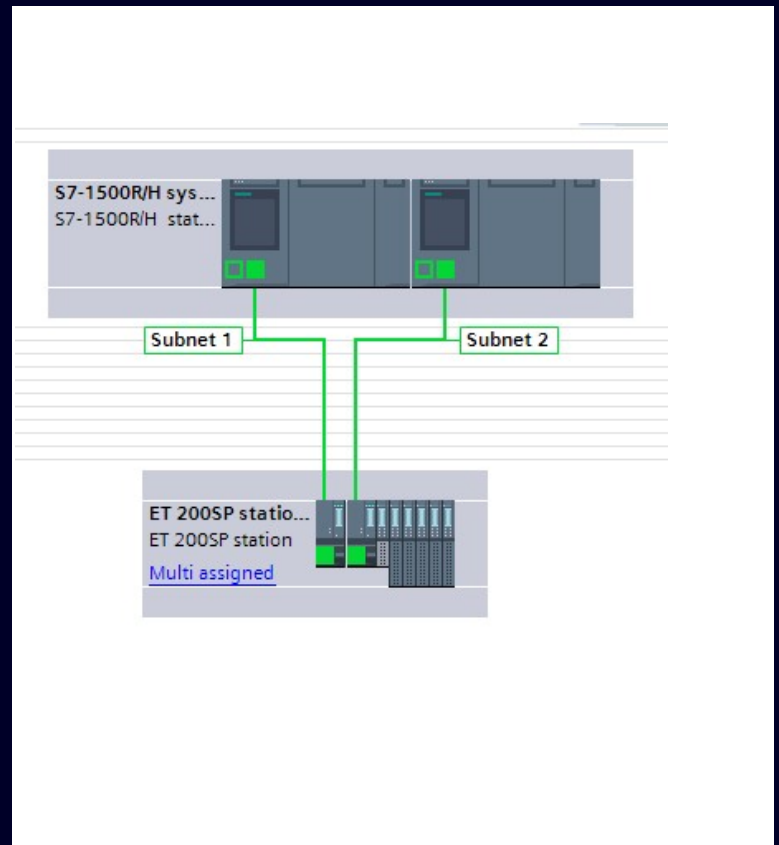
2 separate subnets are used between controller and IO device.

Advantage:

Same (symmetric) IP addresses in both networks can be used.

Disadvantage:

No double-sided connection of S1 or S2 devices possible (via Y-Switch).



Configuration of System Redundancy R1

Option 2

R1

Shared Subnet

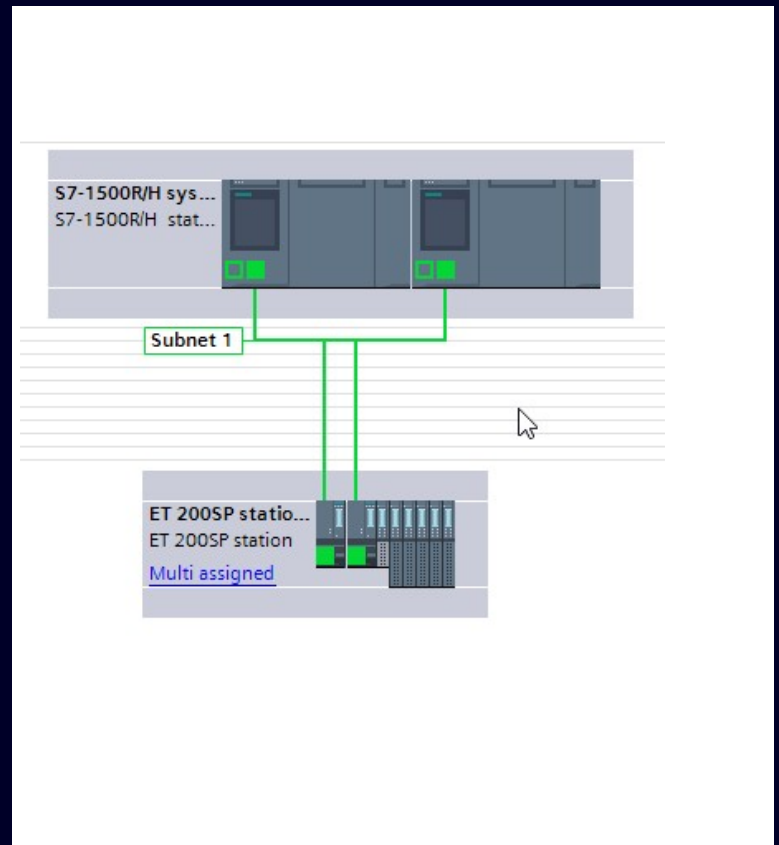
Controller and IO-devices are connected to a common subnet

Advantage:

Allows two-sided connection of S1 or S2 devices via Y-Switch.

Disadvantage:

No symmetric IP addresses possible.



Configuration of System Redundancy R1

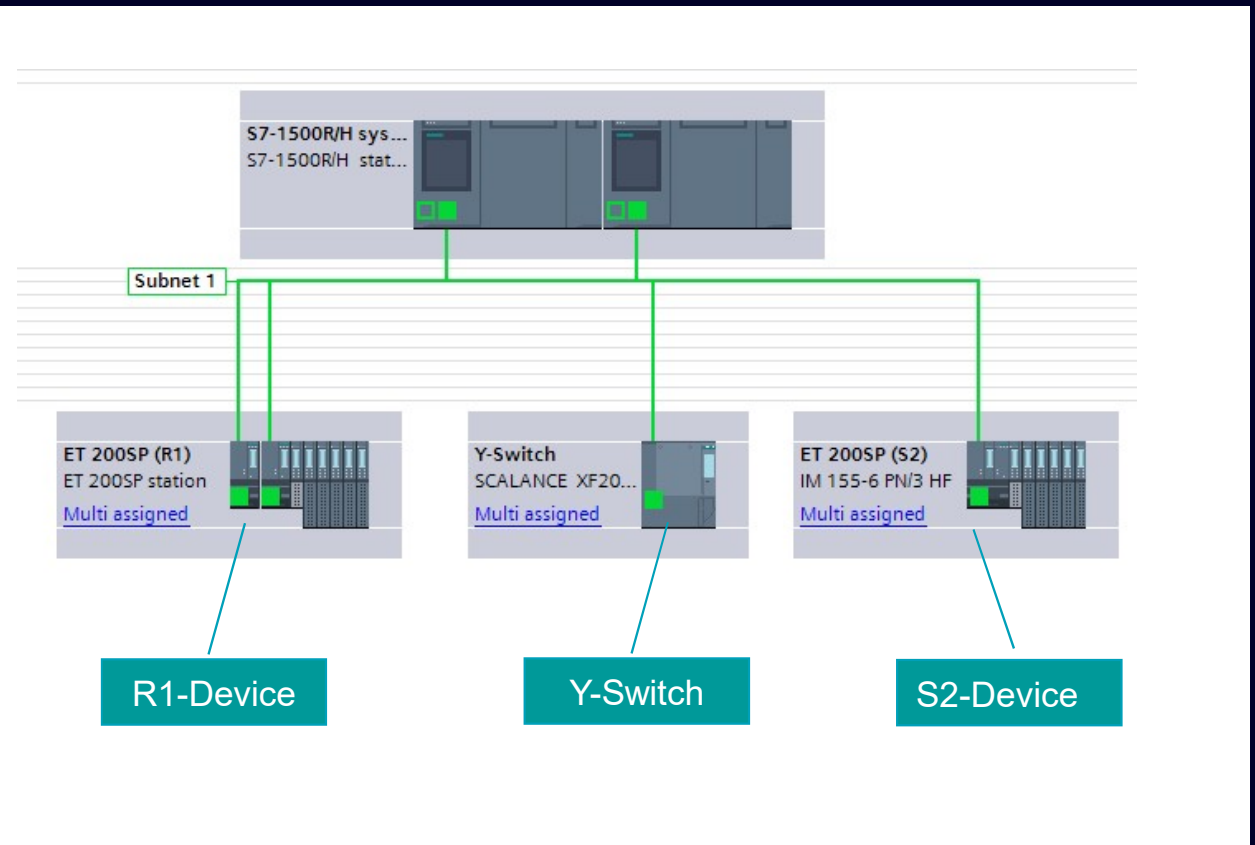
Option 2: Shared subnet – configuration with S1 and S2 IO-devices

R1

Combination of R1- with S2- and/or S1-Devices

S2 and S1 devices can be connected to a R1 system by using the Y-Switch.

Details: See [“Configuration Examples S7-1500H with R1 Redundancy”](#)



Configuration of System Redundancy

Display of the redundancy mode in the TIA Portal

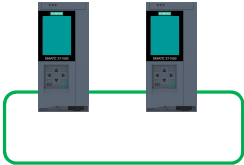
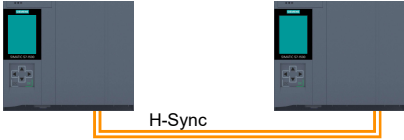
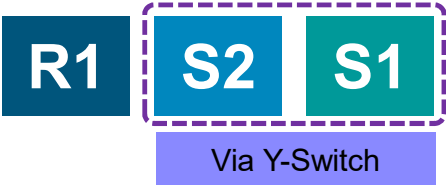
The screenshot displays the Siemens TIA Portal interface for a project named "S7-1500H_Ring_R1-Devices_Y-Switch". The left pane shows a network topology diagram with various components including S7-1500H stations, ET 200SP stations, switches, and I/O modules. A callout points to the ET 200SP stations, stating: "In the network view, S1, S2 and R1 devices are displayed as 'Multi-Assigned'".

The right pane shows the "I/O communication" table, which lists the connections between the PLC and various I/O devices. A callout points to the "Mode" column, stating: "The distinction S1/S2/R1 is displayed in the 'Mode' column in the 'I/O communication' table. Here: ET 200MP: S2 Redundancy".

Partner 1	Partner 2	Interface partner 2	Mode	Optional IO-Device
1	PLC_1			
2	PROFINET interface_1			
3	X1	IO device_1	IO device(R1)	
4	X1	IO device_3	IO device(R1)	
5	X1	IO device_5	IO device(R1)	
6	X1	Switch_Ring1	IO device(S1)	
7	X1	Switch_Ring2	IO device(S1)	
8	X1	Y-Switch_DNA	IO device(S2)	
9	X1	Device_S1	IO device(S1)	
10	X1	Device_S2	IO device(S2)	
11	PLC_2			
12	PROFINET interface_1			
13	X1	IO device_2	IO device(R1)	
14	X1	IO device_4	IO device(R1)	
15	X1	IO device_6	IO device(R1)	
16	X1	Switch_Ring1	IO device(S1)	
17	X1	Switch_Ring2	IO device(S1)	
18	X1	Y-Switch_DNA	IO device(S2)	
19	X1	Device_S1	IO device(S1)	
20	X1	Device_S2	IO device(S2)	
21				

Network Configurations

Network configuration with S7-1500R/H Overview

	S7-1500R	S7-1500H/HF	
Controller			
PROFINET Topology	MRP Ring 	MRP Ring 	Line  Redundant MRP Ring  Redundant line 
Additional Topologies	Ring coupling	Ring coupling, combined topology	
IO-Devices System Redundancy			

Network Configuration S7-1500R

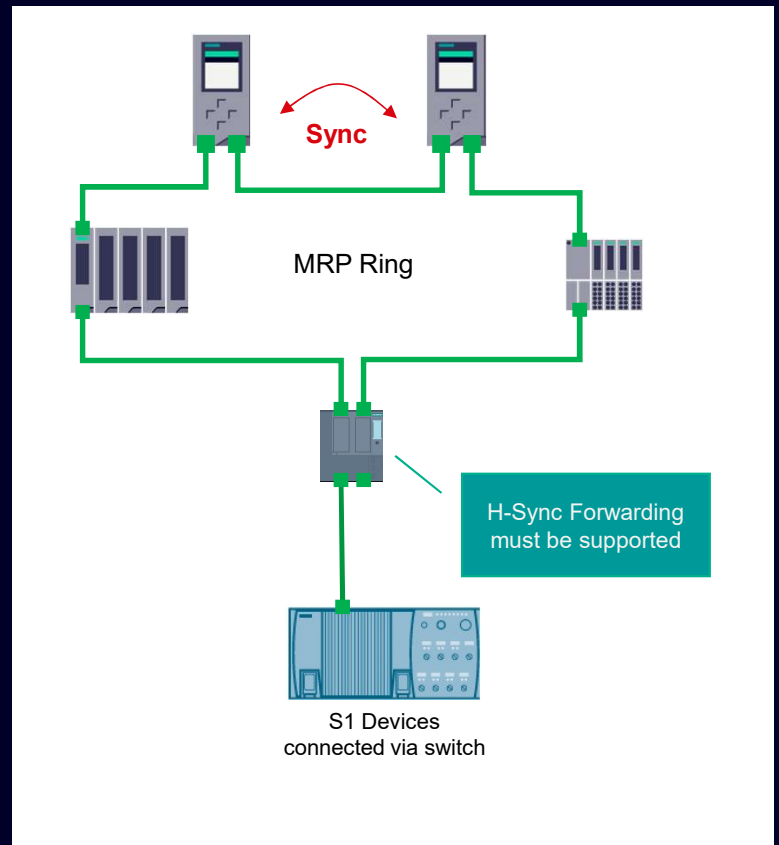
Network Configuration Examples S7-1500R

Ring Topology with MRP

Ring Topology

- A MRP ring is mandatory – the redundant CPUs (R-CPU) must be part of the ring. R-CPU must be configured as “MRP Auto-Manager”.
- If the ring is interrupted, an MRP reconfiguration is automatically triggered (up to 200ms).
- No devices are allowed in the direct connection between the two R-CPU.
- Connecting S1 Devices via a Switch ¹⁾
- Devices with more than two ports (e.g., switches) must support H-Sync forwarding.
- Recommendation: Maximum 16 devices in the ring for optimal performance.

1) Reason: S1 devices do not route the H-Sync telegrams during an MRP reconfiguration phase. This would lead to a high CPU cycle time if the segment was interrupted. See chapter "H-Sync Forwarding" in the system manual S7-1500R/H

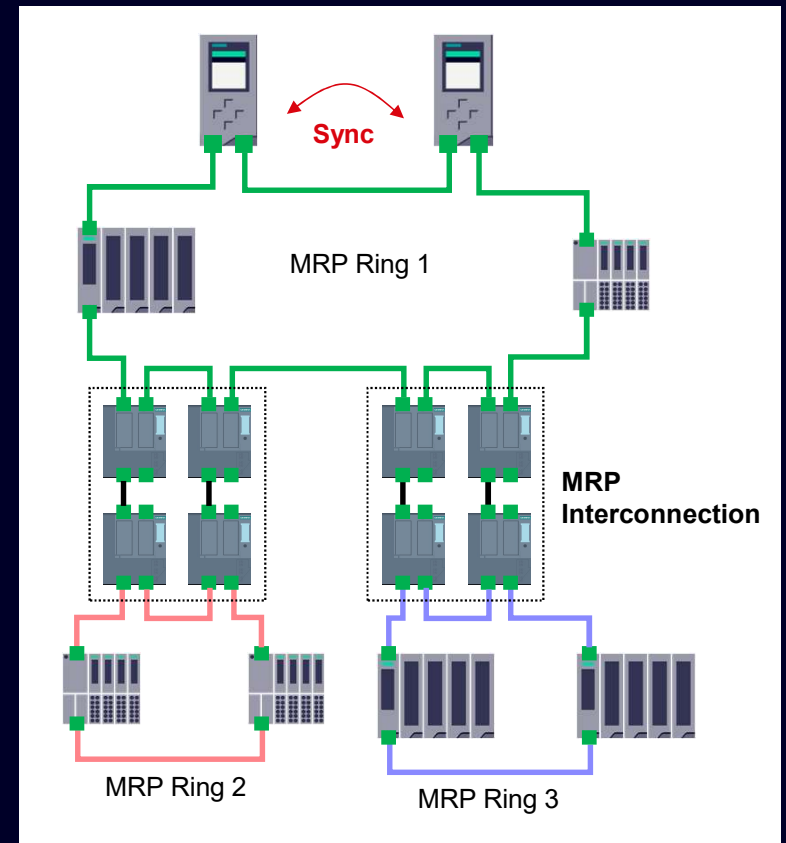


Network Configuration Examples S7-1500R

Ring Topology – Supplemented with MRP Interconnection

Ring coupling via MRP interconnection

- **MRP Interconnection** extends the Media Redundancy Protocol (MRP) by redundantly coupling multiple rings in PROFINET networks through pairs of switches.
- The MRP rings are interconnected in a redundant manner, ensuring continuous operation even if one switch fails.
- Up to 5 MRP Interconnection pairs can be connected to the main ring. Each pair supports up to 2 MRP Interconnection instances, allowing the integration of **up to 10 additional MRP rings with the main ring**.
- All PROFINET devices can be connected via MRP rings (up to 50 devices per ring), which significantly increases the overall availability of the solution.
- Only the switches located in the main MRP ring (with CPU1500R) must support the H-Sync Forwarding function. Reason: There are no H-Sync telegrams in the additional MRP rings.
- Configuration of the MRP-Interconnection switches via **Web-Server**.

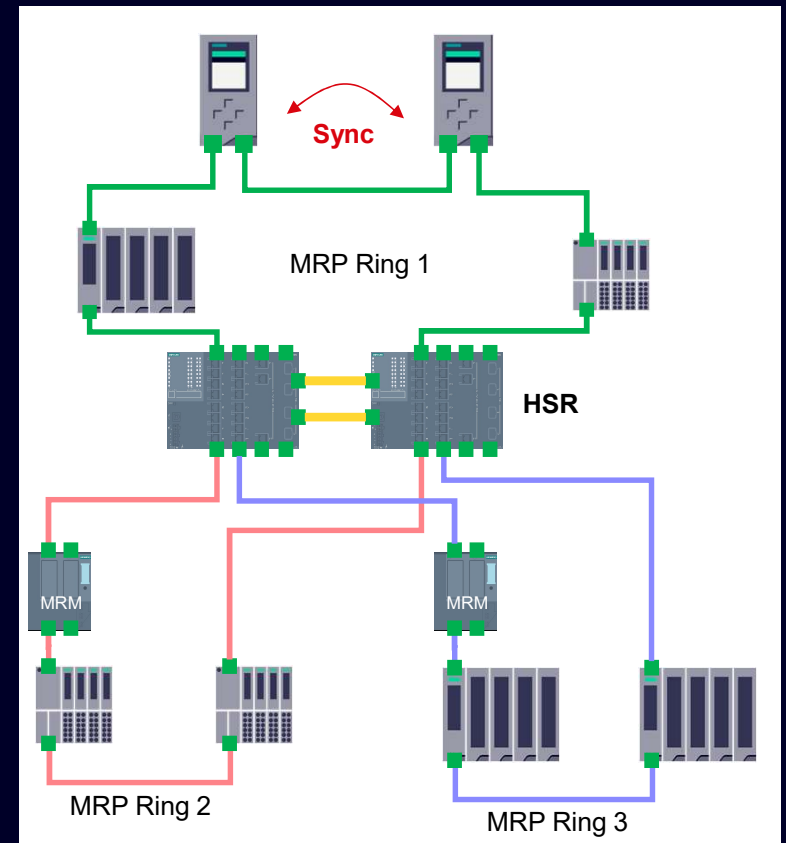


Network Configuration Examples S7-1500R

Ring Topology – Supplemented with HSR + Protocol Bridge

Ring coupling with HSR + Protocol Bridge

- It connects **High-availability Seamless Redundancy (HSR)** rings with Media Redundancy Protocol (MRP) networks. All MRP rings are bridged/closed through the HSR switches. Up to 16 rings can be configured for each pair of HSR switches.
- The MRP rings are redundantly coupled ensuring continued operation even if one switch fails.
- All PROFINET devices can be connected via MRP rings (up to 50 devices per ring). This increases the overall availability of the solution.
- HSR switches are XC-300, XR-300, XC-400, XR-500WG, and XR-500 ("H-Sync Forwarding" is supported). Since each MRP ring requires an MRM (MRP Manager), the solution becomes economically attractive starting from 3/4 MRP rings.
- Configuration of the HSR + Protocol Bridge switches in **TIA Portal**.



Network Configuration S7-1500H with S2 Redundancy

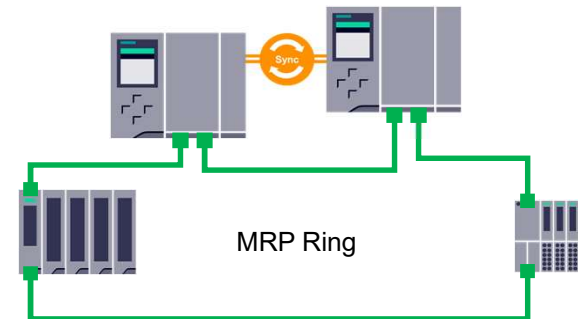
Network Configuration Examples S7-1500H

Example with ring topology

S2

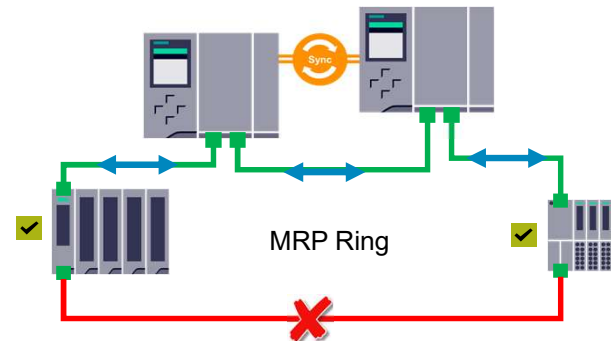
Ring topology with S2 Redundancy

- H-CPU's must be set as "MRP Auto-Manager".



Behavior in case of ring interruption

- If the ring is interrupted, MRP reconfiguration is triggered (up to 200ms).
- All devices continue to work without communication failure.



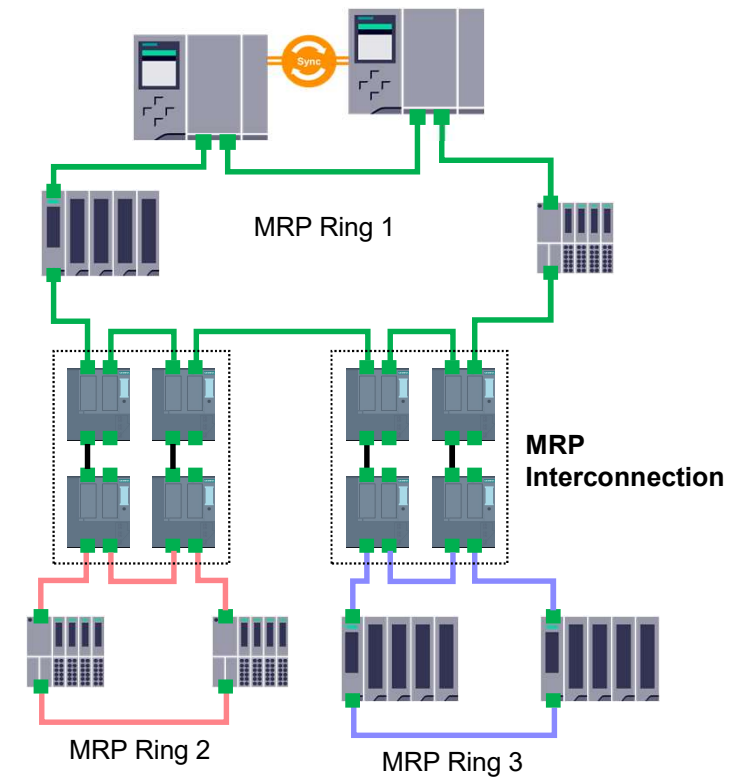
Network Configuration Examples S7-1500H

Ring Topology – Supplemented with MRP Interconnection

S2

Ring coupling via MRP interconnection

- **MRP Interconnection** extends the Media Redundancy Protocol (MRP) by redundantly coupling multiple rings in PROFINET networks through pairs of switches.
- The MRP rings are interconnected in a redundant manner, ensuring continuous operation even if one switch fails.
- Up to 5 MRP Interconnection pairs can be connected to the main ring. Each pair supports up to 2 MRP Interconnection instances, allowing the integration of **up to 10 additional MRP rings with the main ring**.
- All PROFINET devices can be connected via MRP rings (up to 50 devices per ring), which significantly increases the overall availability of the solution.
- Configuration of the MRP-Interconnection switches via **Web-Server**.



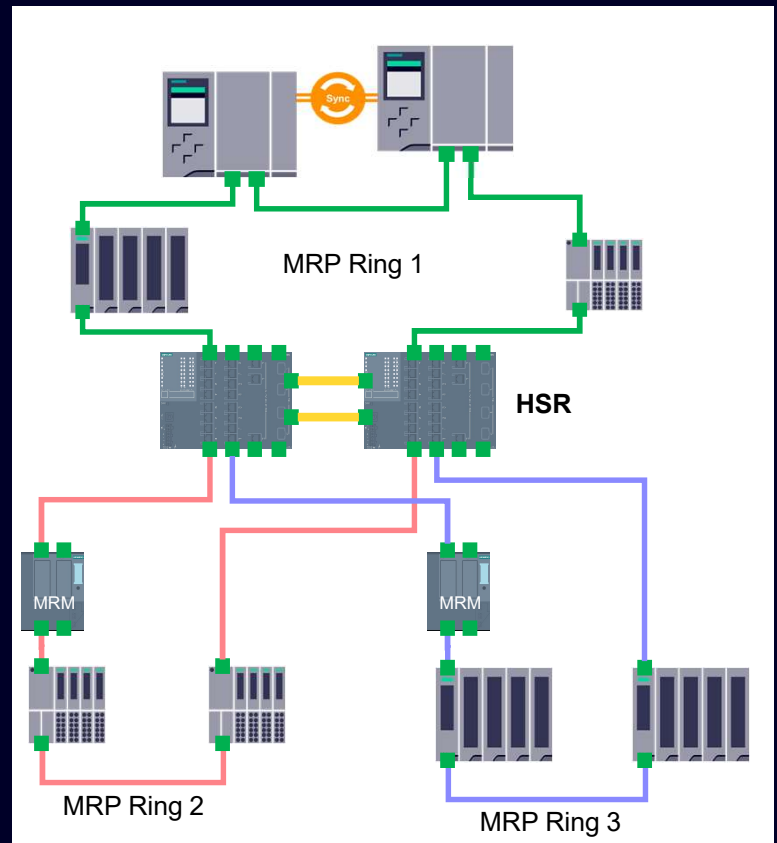
Network Configuration Examples S7-1500H

Ring Topology – Supplemented with HSR + Protocol Bridge

S2

Ring coupling with HSR + Protocol Bridge

- It connects **High-availability Seamless Redundancy (HSR)** rings with Media Redundancy Protocol (MRP) networks. All MRP rings are bridged/closed through the HSR switches. Up to 16 rings can be configured for each pair of HSR switches.
- The MRP rings are redundantly coupled ensuring continued operation even if one switch fails.
- All PROFINET devices can be connected via MRP rings (up to 50 devices per ring). This increases the overall availability of the solution.
- HSR switches are XC-300, XR-300, XC-400, XR-500WG, and XR-500. Since each MRP ring requires an MRM (MRP Manager), the solution becomes economically attractive starting from 3/4 MRP rings.
- Configuration of the HSR + Protocol Bridge switches in **TIA Portal**.



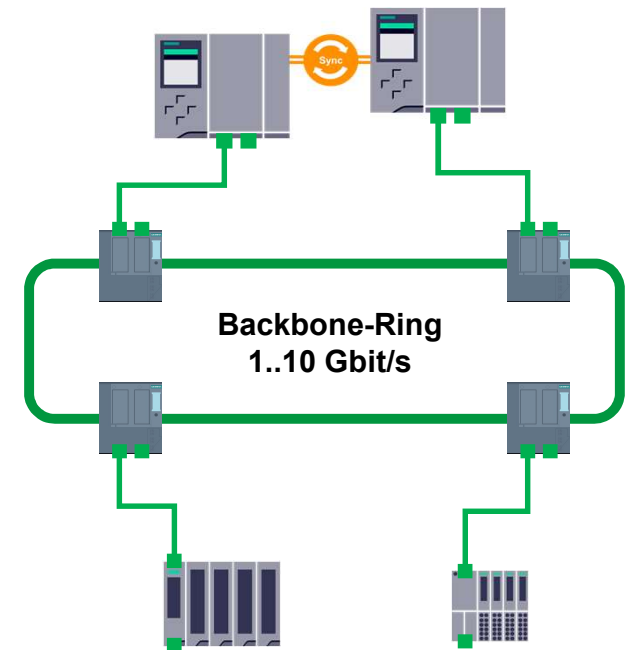
Network Configuration Examples S7-1500H

Example for combined topology – For example: Backbone-Ring

S2

Connection to backbone ring

- In that case the ring is not managed by the CPU 1500H.
- This allows connection to existing infrastructure.
- In the ring there is no restriction to 100MBit/s transmission speed.
- Only one requirement, a PROFINET communication must be possible between the S7-1500H and its PN IO devices. Which ring technology is used, is here not relevant because not managed by the H-system.



Network Configuration Examples S7-1500H

Example with line topology

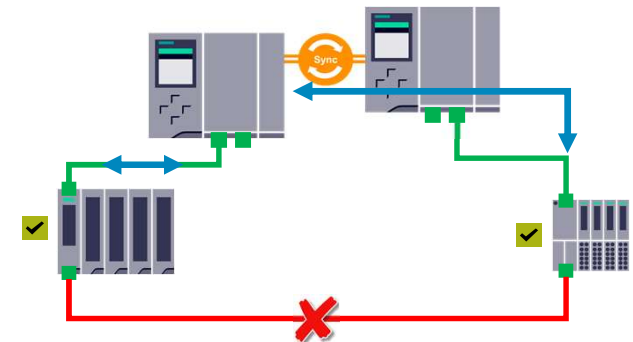
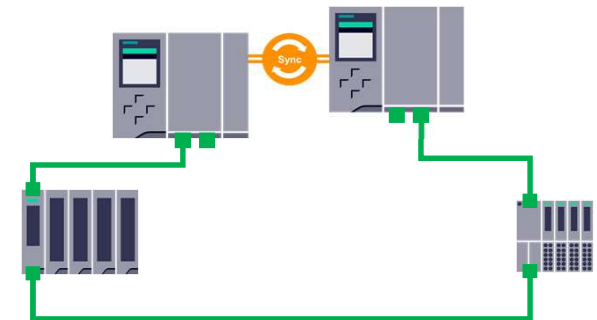
Line topology with S2 Redundancy

- MRP protocol is not used.
- If the line is somewhere interrupted, then the PROFINET IO communication will take place via the H-Sync lines.

Behavior in case of line interruption

- PN IO-Data is transferred via H-Sync line.
- All devices continue to work without failure.

S2



Network Configuration Examples S7-1500H

Example with line/star topology

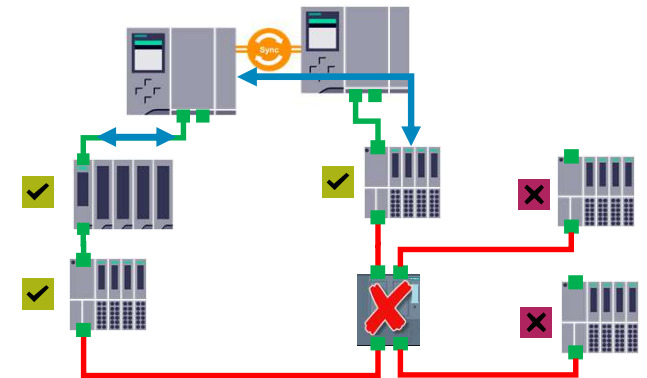
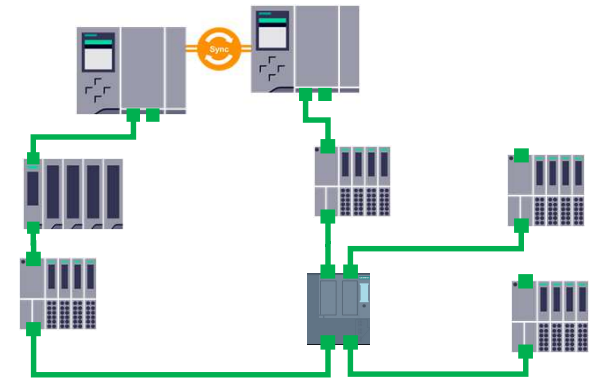
Line/star topology with S2 Redundancy

- MRP protocol is not used.
- If the line is somewhere interrupted, then the PROFINET IO communication will take place via the H-Sync lines.

Behavior in case of a fault in the switch

- Devices connected via the switch, have no further connection to any controllers → lost of the communication

S2



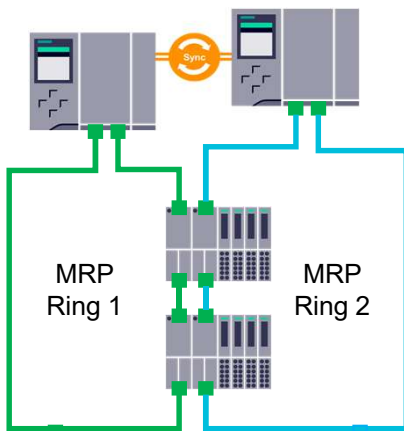


Network Configuration S7-1500H with R1 Redundancy

Network Configuration Examples S7-1500H

Ring topology for S7-1500H

R1



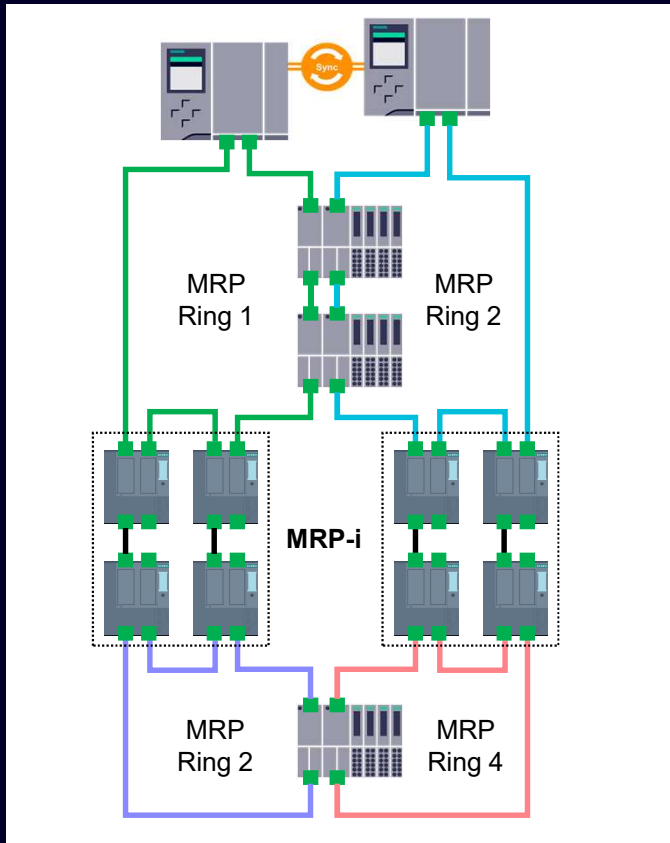
Ring topology with R1 Redundancy

- A separate MRP ring is configured for each H-CPU.
- H-CPU's must be set as "MRP Auto-manager".
- If the ring is interrupted, an MRP reconfiguration is automatically triggered (up to 200ms).
- Separate networks – same (mirror-image) IP addresses can be used.

Network Configuration Examples S7-1500H

Ring topology for S7-1500H – extended with MRP Interconnection

R1



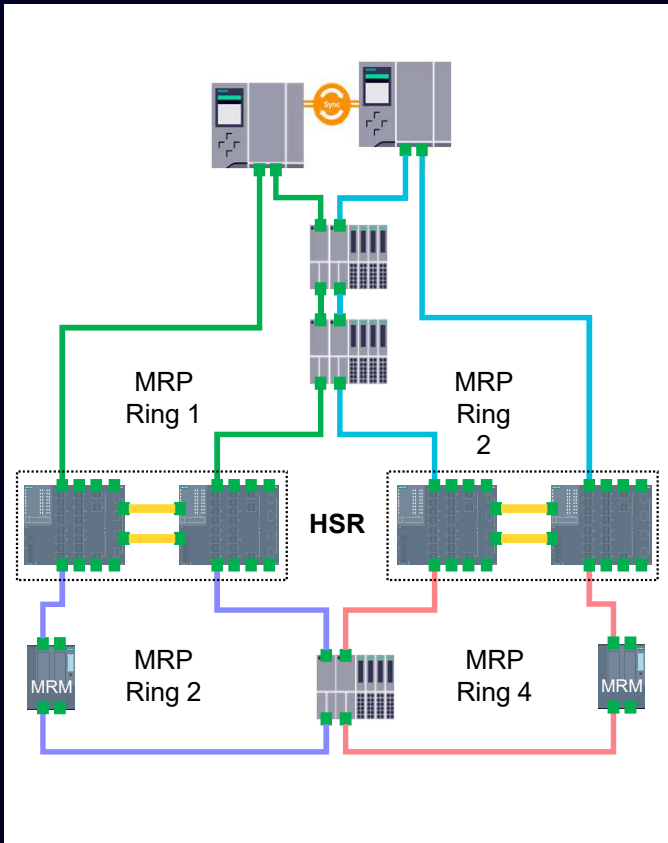
Redundant ring coupling via MRP interconnection

- **MRP Interconnection** extends the Media Redundancy Protocol (MRP) by redundantly coupling multiple rings in PROFINET networks through pairs of switches.
- Also supported with R1 configuration.
- The MRP rings are interconnected in a redundant manner, ensuring continuous operation even if one switch fails.
- Up to 5 MRP Interconnection pairs can be connected to the main R1 rings. Each pair supports up to 2 MRP Interconnection instances, allowing the integration of **up to 10 additional MRP rings with the main R1 rings**.
- All PROFINET devices can be connected via MRP rings (up to 50 devices per ring), which significantly increases the overall availability of the solution.
- Configuration of the MRP-Interconnection switches via **Web-Server**.

Network Configuration Examples S7-1500H

Ring Topology – Supplemented with HSR + Protocol Bridge

R1



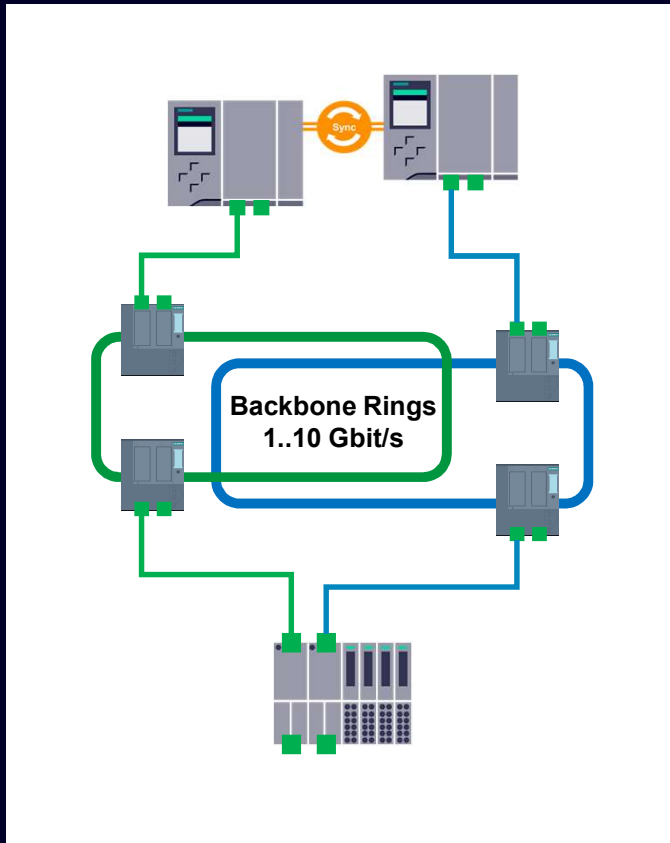
Ring coupling with HSR + Protocol Bridge

- It connects **High-availability Seamless Redundancy (HSR)** rings with Media Redundancy Protocol (MRP) networks. All MRP rings are bridged/closed through the HSR switches. Up to 16 rings can be configured for each pair of HSR switches.
- The MRP rings are redundantly coupled ensuring continued operation even if one switch fails.
- All PROFINET devices can be connected via MRP rings (up to 50 devices per ring). This increases the overall availability of the solution.
- HSR switches are XC-300, XR-300, XC-400, XR-500WG, and XR-500. Since each MRP ring requires an MRM (MRP Manager), the solution becomes economically attractive starting from 3/4 MRP rings.
- Configuration of the HSR + Protocol Bridge switches in **TIA Portal**.

Network Configuration Examples S7-1500H

Example for Combined Topology – For example: Backbone-Ring

R1



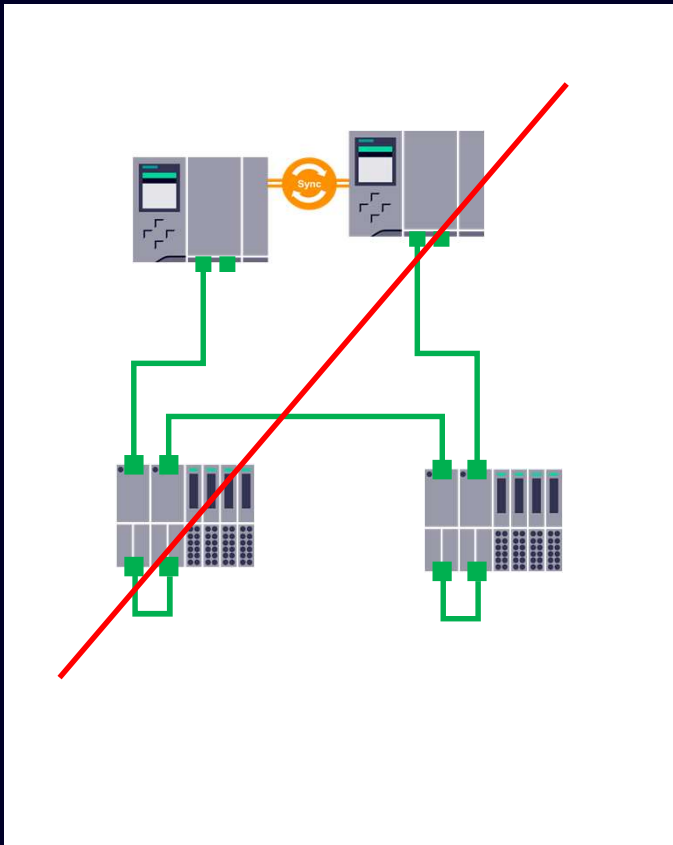
Connection to separate ring

- R1 rings do not have to be managed by the H-CPU.
- This enables connection to existing redundant network infrastructure.
- In the rings, there is no restriction to 100MBit/s transmission speed.
- Only one requirement, a PROFINET communication must be possible between the S7-1500H and its PN IO devices. Which ring technology is used, is here not relevant because not managed by the H-system.

Network Configuration Examples S7-1500H

Unsupported Topology - Single R1 Ring

R1



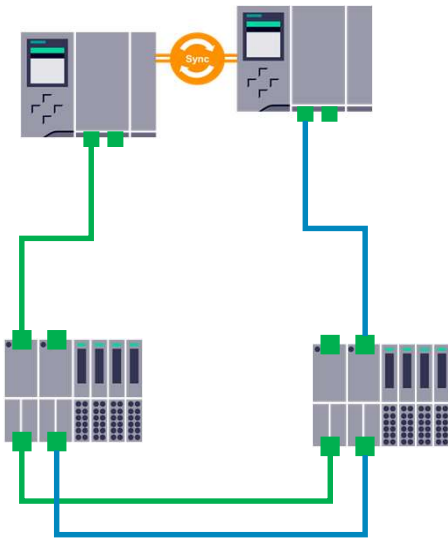
This topology is not supported because its availability does not meet requirements :

The probability of failure within the ring is twice that of an S2 setup because the number of IMs is doubled.

Network Configuration Examples S7-1500H

Example with line topology

R1



Line topology with R1 Redundancy

- Line topology is also supported for R1 setup.
- Up to 512 devices can be connected within this topology.
- The feeding of the PN line from different sides increases the availability in case of double line interruption.

Single-sided feeding

If both lines are interrupted, devices behind them fail



Feeding from two sides

If both lines are interrupted, all devices can be reached



Network Configuration Examples S7-1500H

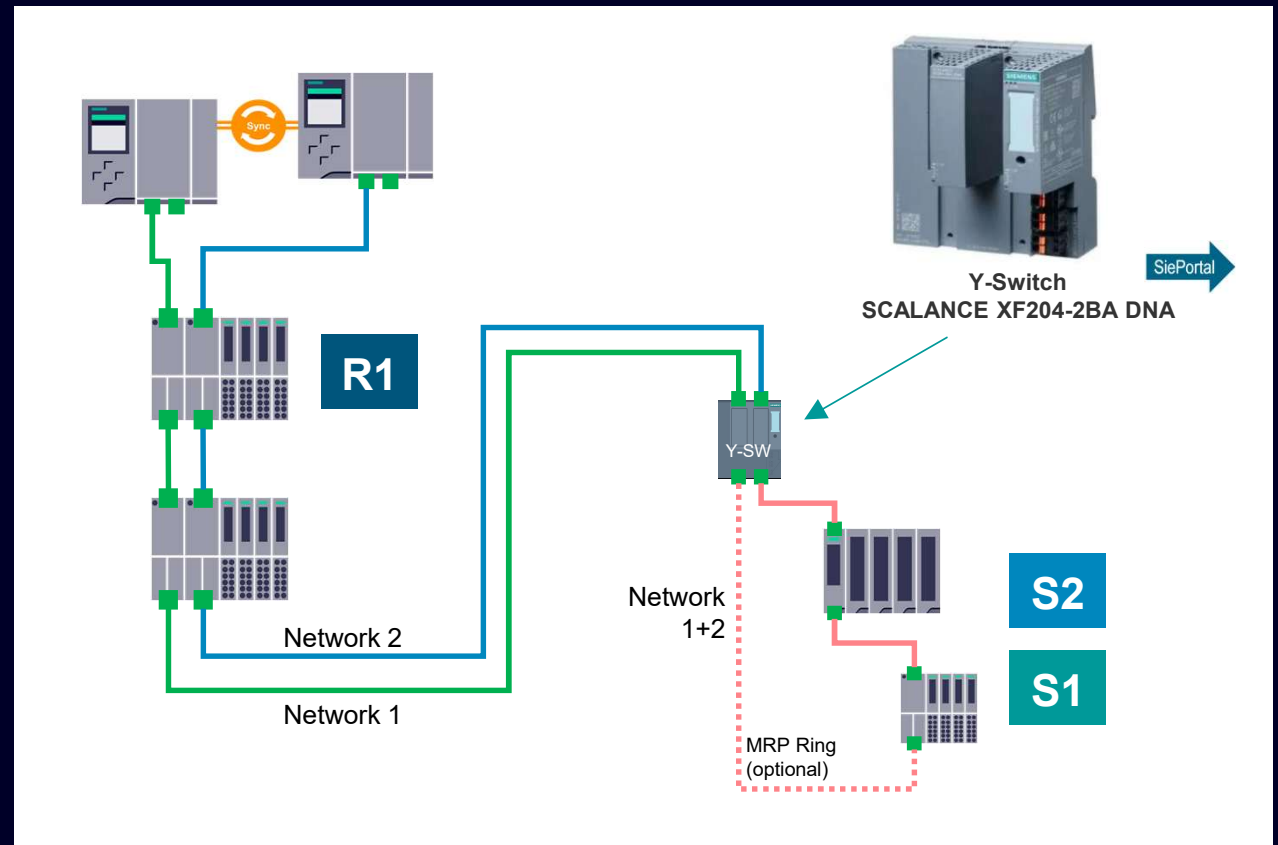
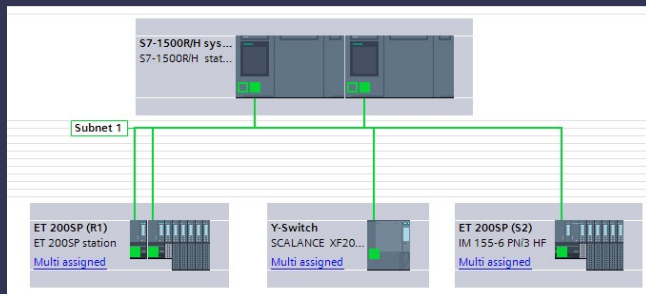
Combination of R1 and S1/S2 devices

R1

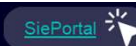
R1 configurations can be extended with S2 or S1 devices using a Y-Switch.

The prerequisite for this is the configuration in the TIA Portal on 1 subnet.

Result: Unique IP addresses in Network 1+2 are required.



Application examples with Y-Switch: See



Network Configuration Examples S7-1500H

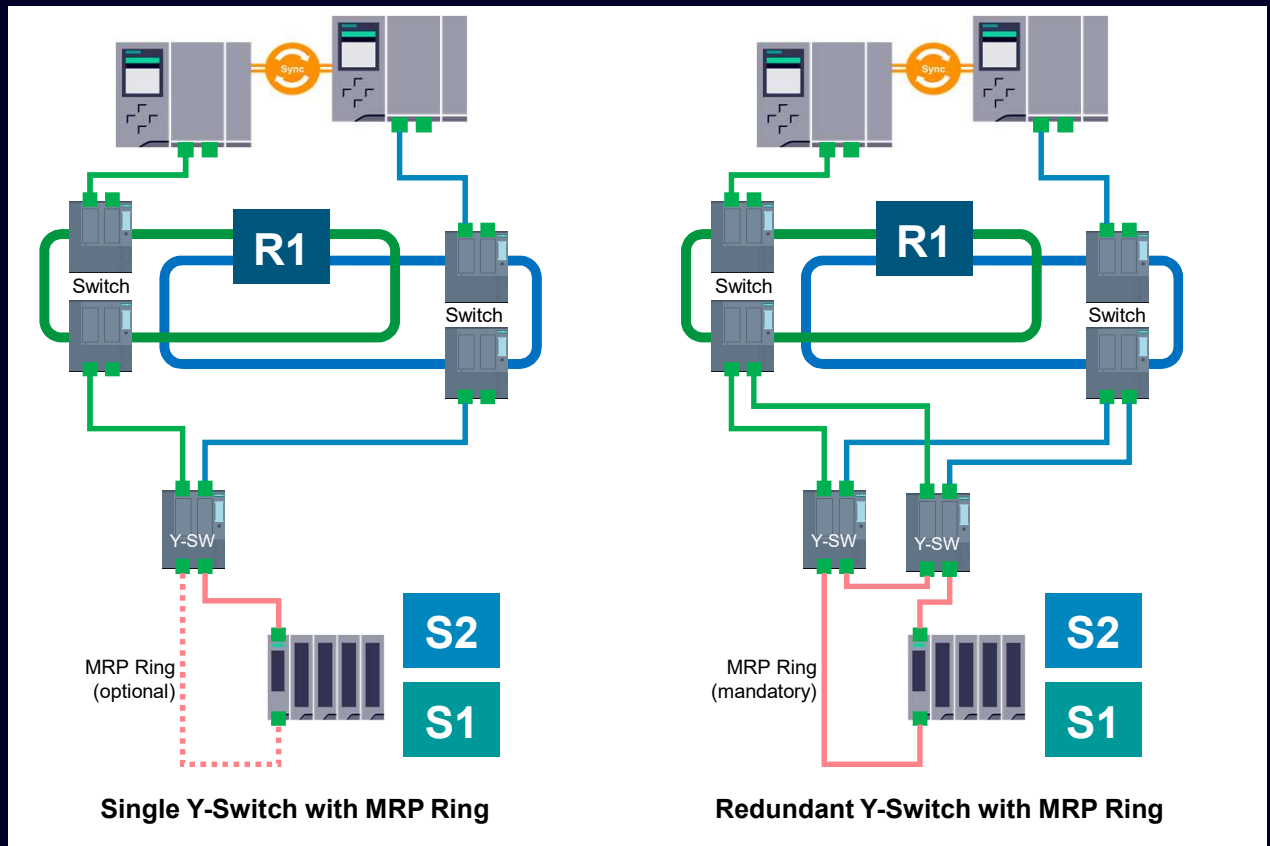
Combination of R1 and S1/S2 devices on ring structures

To connect the Y-Switch to MRP rings, additional switches are required.

To increase availability, the Y-Switch can also be designed redundantly "Redundant DNA".

In this case, a MRP ring must be used on the S1/S2 side.

R1

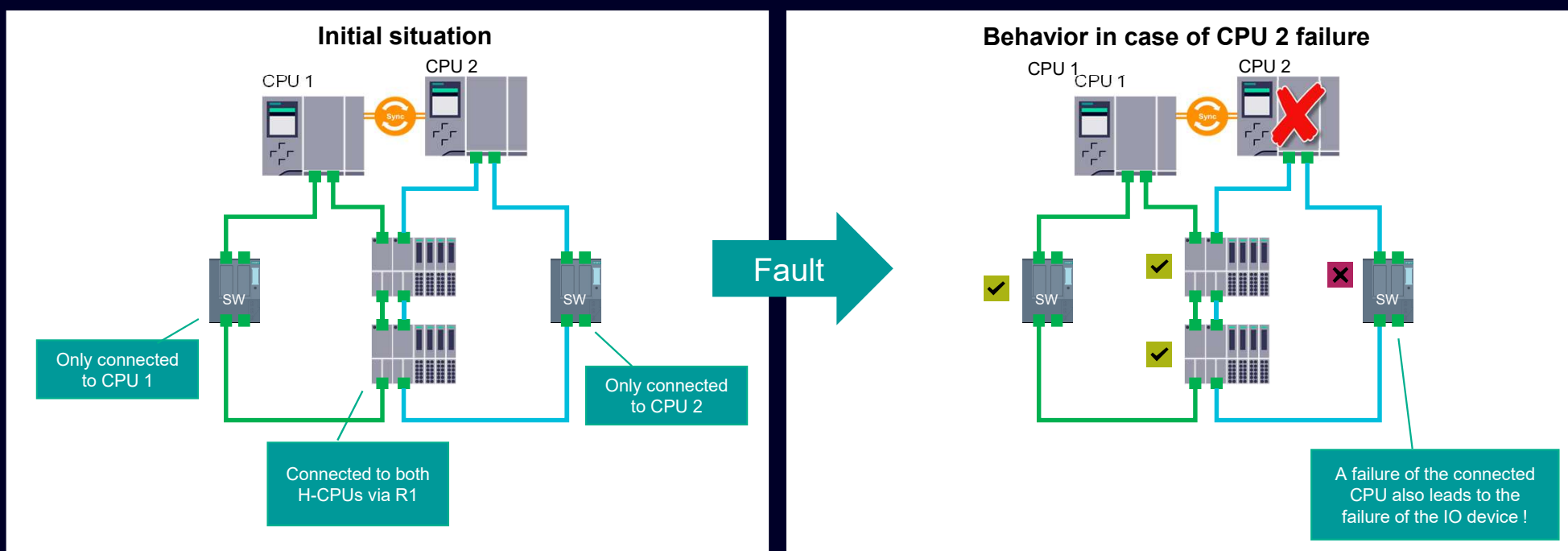


Network Configuration Examples S7-1500H

IO-devices that are connected to only one CPU

R1

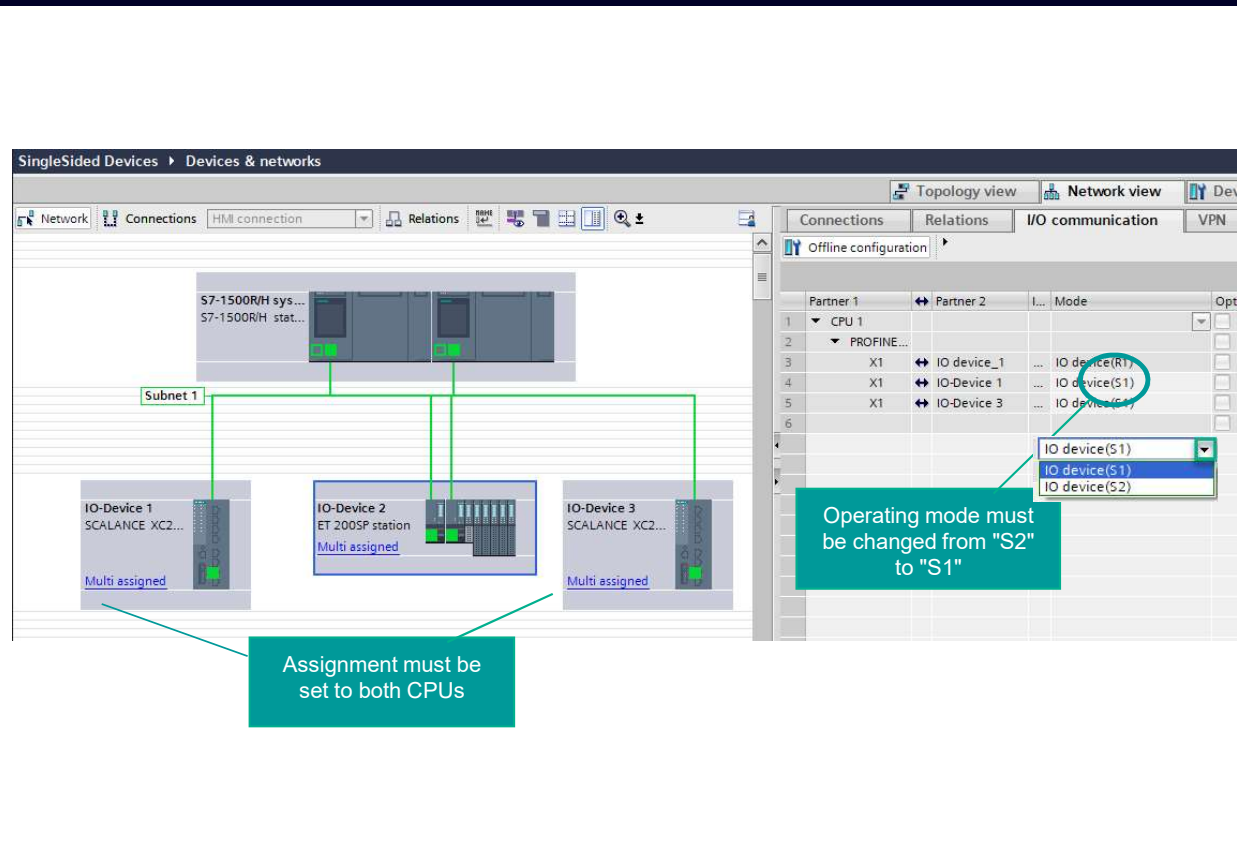
In some cases, it is sufficient if S1/S2 devices are connected to only one of the two CPUs. Examples of this, are switches that send diagnostic data to the controller via PROFINET.



Network Configuration Examples S7-1500H

IO-devices that are connected to only one CPU - Configuration

R1



Configuration rules

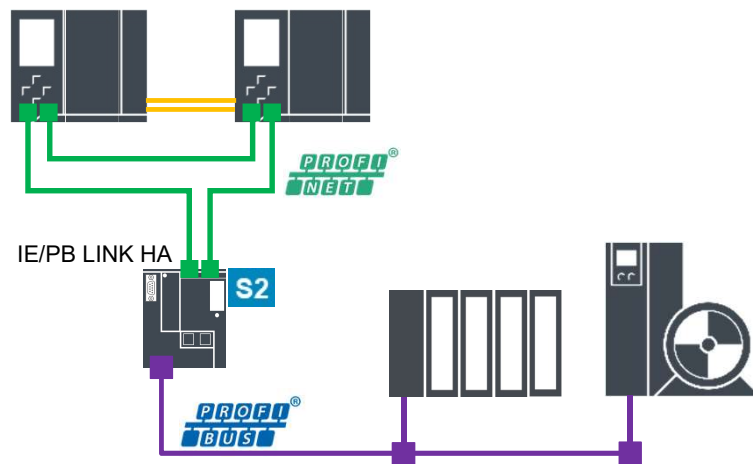
- Even if devices are connected to only one CPU, they must be assigned to both CPUs.
- Operation Mode must be set to IO device (S1).
- If the operating mode remains unchanged (S2 instead of S1), the “disconnected” CPU will report a communication error.

Network Configuration for PROFIBUS Devices

Network Configuration for PROFIBUS Devices

Connection of PROFIBUS DP devices via IE/PB LINK HA

S2



Support of IE/PB LINK HA

- A connection of PROFIBUS devices without bump is possible via the IE/PB LINK HA device.
- GSD based and integrated DP devices can be connected.
- With TIA Portal V20 also Safety DP devices are supported.
- **To be noted:** a redundant IE/PB LINK HA is not possible/supported.

Network Configuration for PROFIBUS Devices

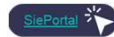
Connection of PROFIBUS DP devices via IE/PB LINK HA



IE/PB LINK HA
MLFB: 6GK1411-5BB00



C-PLUG
MLFB: 6GK1900-0AB10



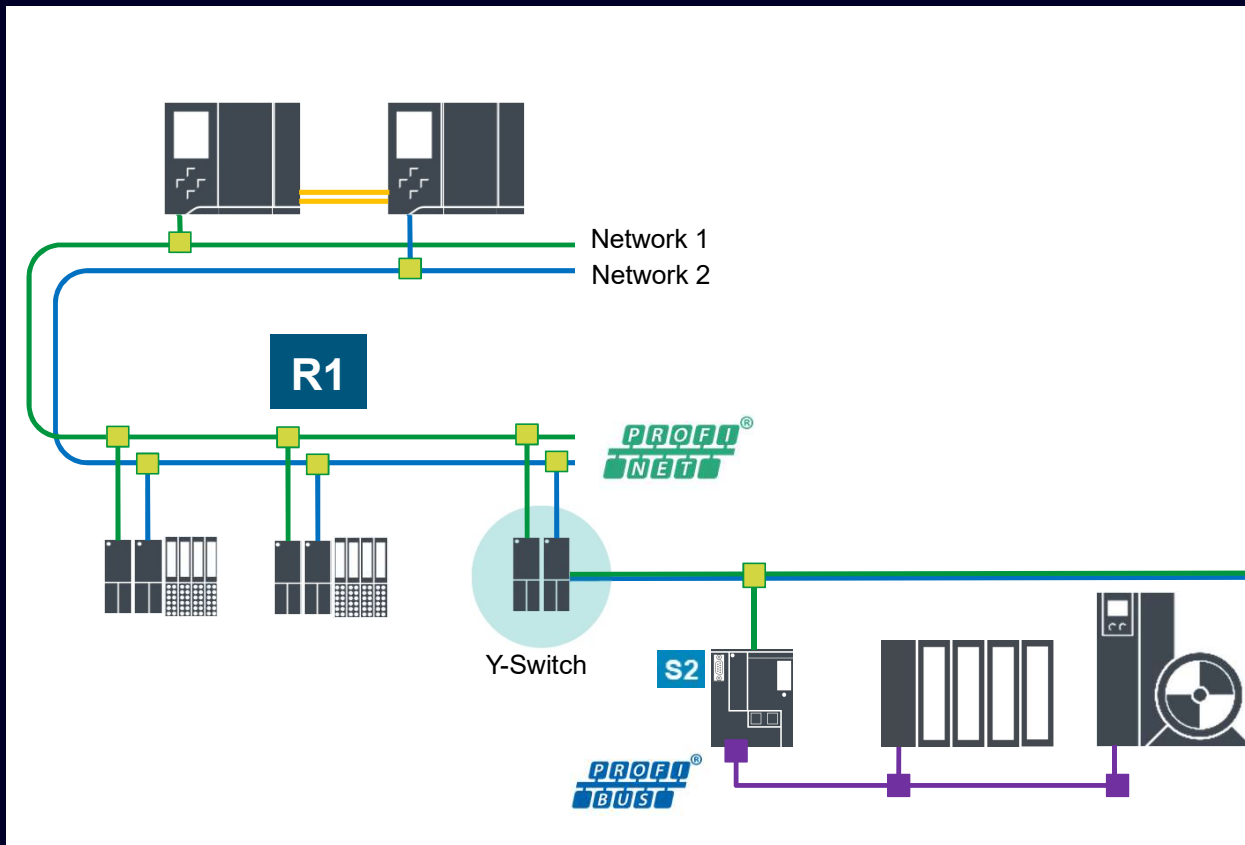
Preconditions

- R/H-CPU Firmware from V3.1
- IE/PB LINK HA Firmware V4.1
- Without **C-PLUG** the IE/PB LINK HA cannot be used in combination with the S7-1500R/H System! It is not optional.

Network Configuration for PROFIBUS Devices

Connection of PROFIBUS DP devices on a R1 System

R1

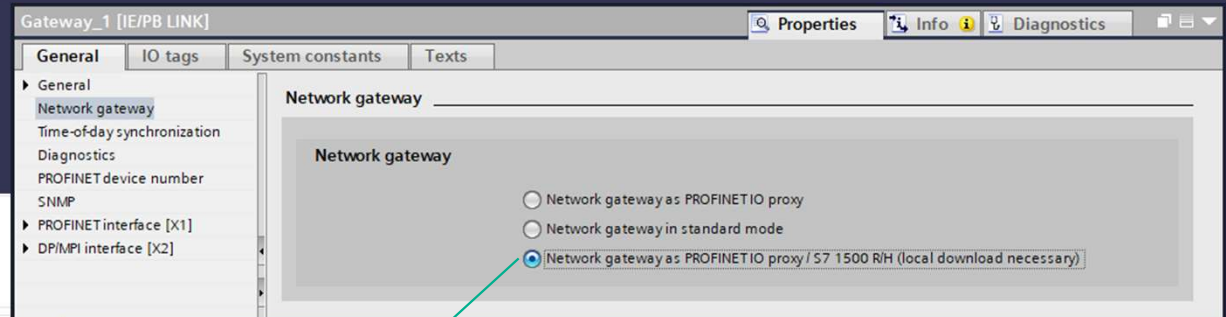
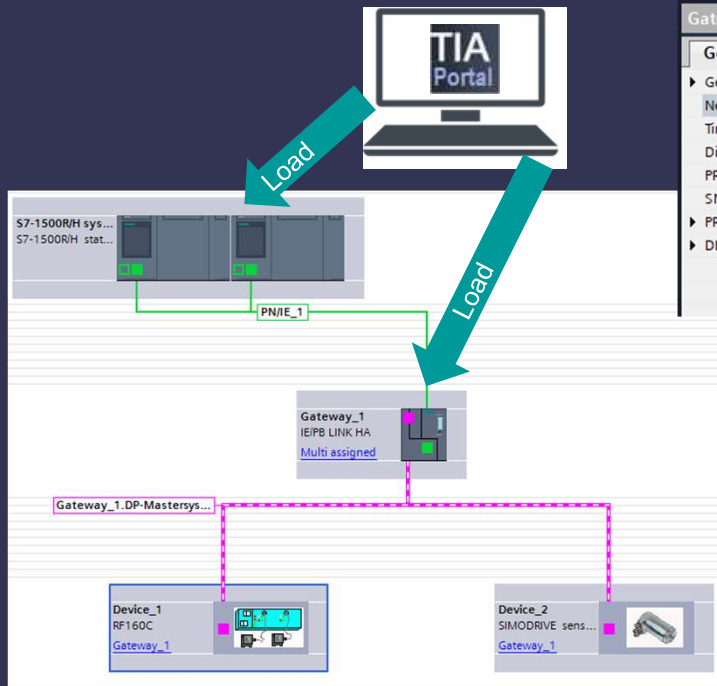


R1 with PROFIBUS

A combination of Y-Switch and IE/PB LINK HA enable the connection of PROFIBUS devices to a R1 system.

Network Configuration for PROFIBUS Devices

Connection of PROFIBUS DP devices - Configuration



This setting is required when using IE/PB LINK HA with S7-1500R/H controller

Please note:

- Each IE/PB LINK HA must be loaded separately from TIA Portal.
- Additional diagnosis for IE/PB LINK HA is available via context menu (NCM diagnosis)

Configuration in Network View

Communication between Controller

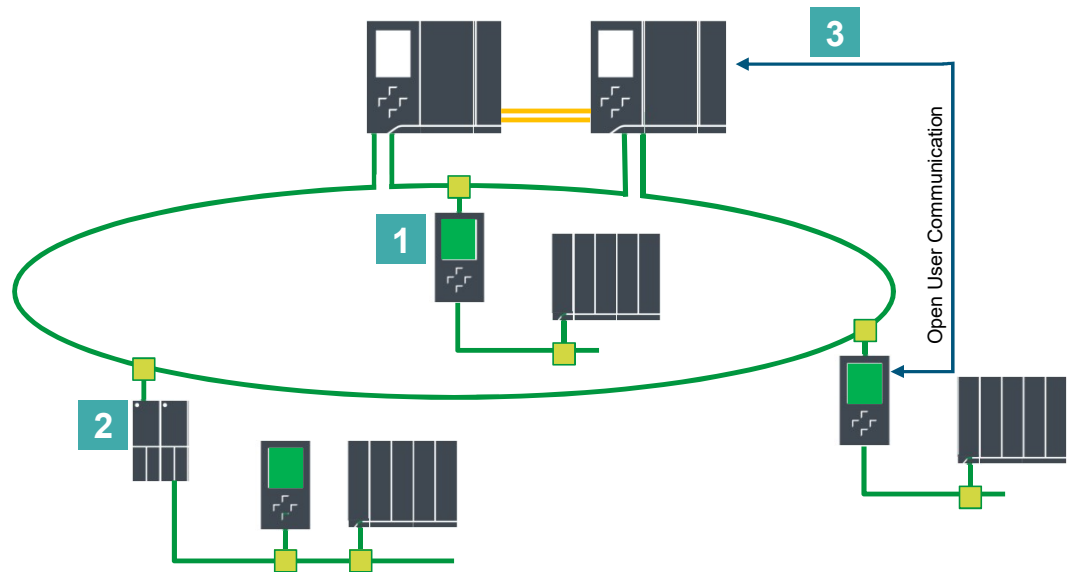
Communication between Controller

Connection of subordinate controllers (I-Devices)

S2

- 1** Connection as iDevice (S1)
 - Not bumpless
 - currently only possible via GSD
- 2** Via PN/PN Coupler (S2)
 - Bumpless switching
- 3** Open user communication between H-System and subordinate controller
 - With short interrupt

Bumpless means:
When switching the H-system, the outputs remain at the last value. The switching time is very short.



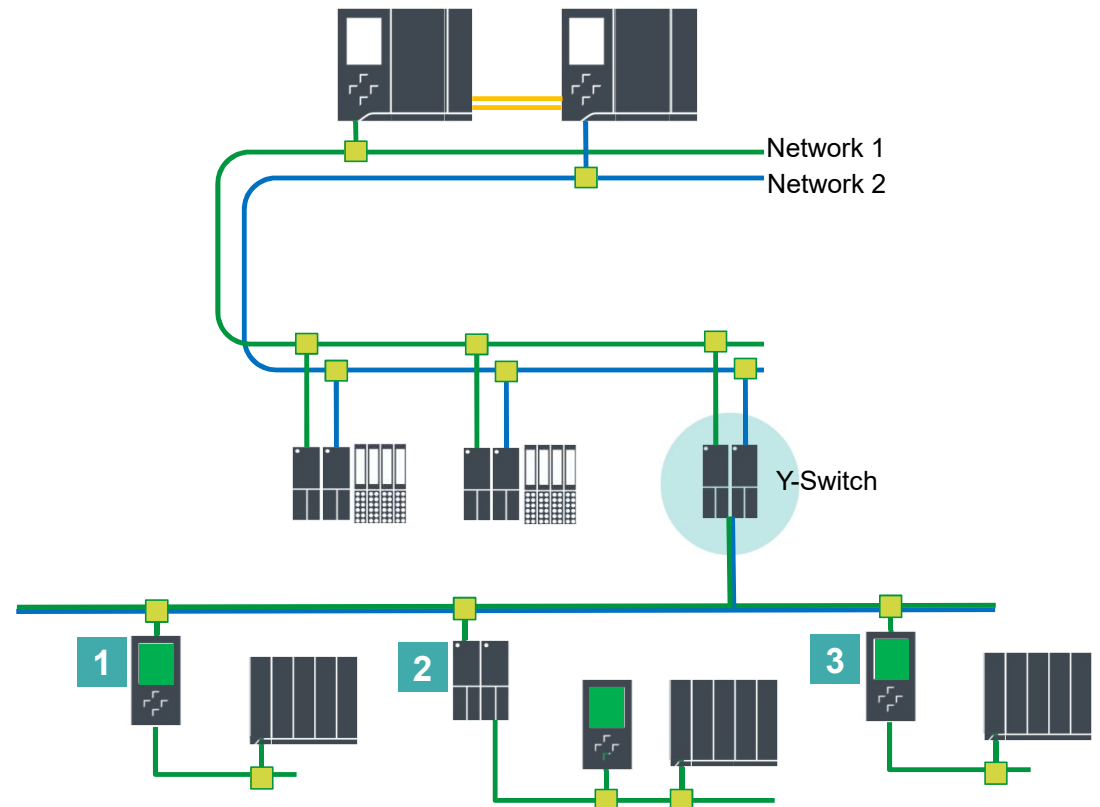
Communication between Controller

Connection of subordinate controllers (I-Devices)

R1

- 1** Connection as iDevice (S1)
 - Not bumpless
 - currently only possible via GSD
- 2** Via PN/PN Coupler (S2)
 - Bumpless switching
- 3** Open user communication between H-System and subordinate controller
 - With short interrupt

Bumpless means:
When switching the H-system, the outputs remain at the last value. The switching time is very short.

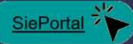


Communication between Controller

Redundant Open User Communication

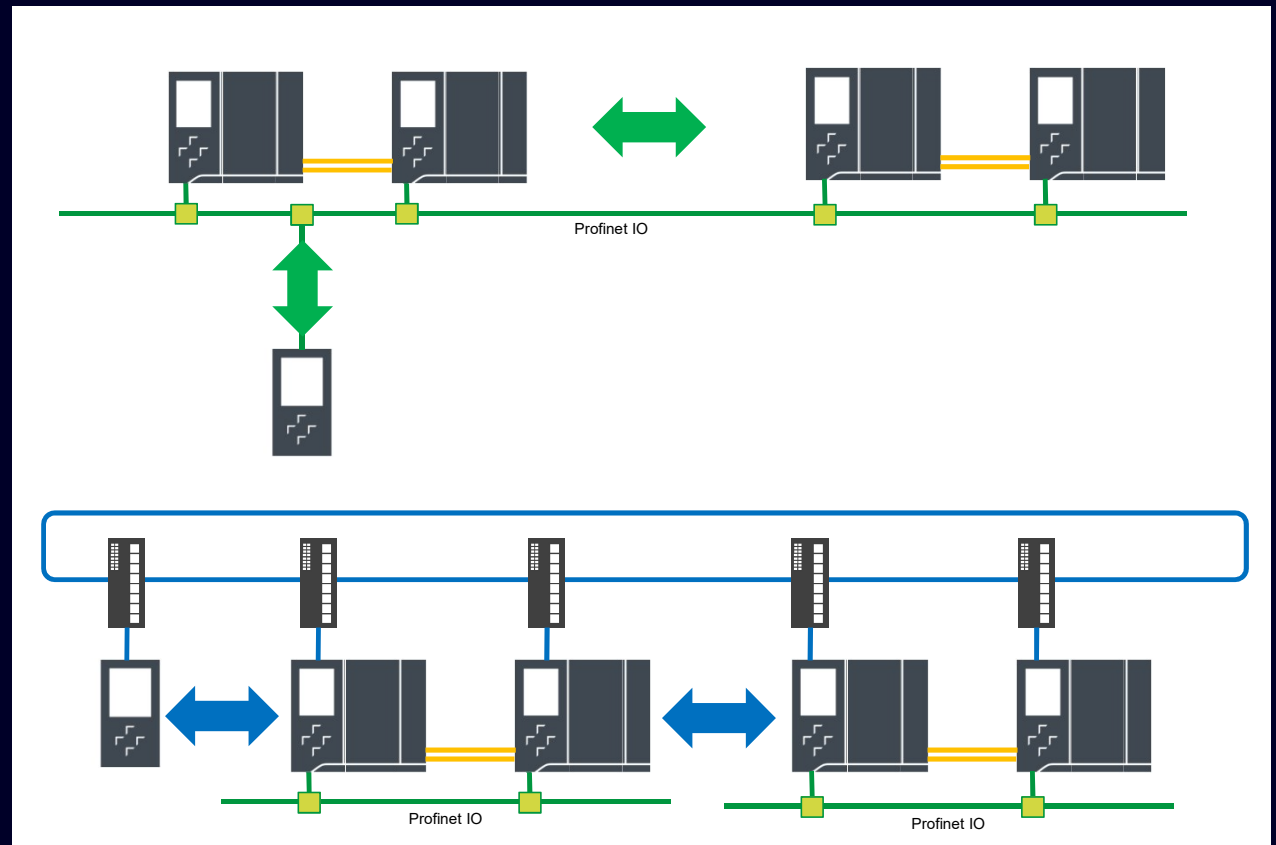
Redundant Open User Communication

Monitoring and automatic connection switchover for up to 4 lower-level connections based on Open User Communication can be found in the following application example:



It covers communication between a S7-1500R/H system and

- Another S7-1500 R/H-System,
- S7-1200 / 1500 CPU,
- S7-300 / 400 CPU,
- S7-400H System,



Support of Ethernet CPs & System Power Supply

Support of Ethernet CP 1543-1

Increased number of ethernet interfaces / Active Backplane for H-CPU's

Support of CP1543-1 module (Ethernet only)

- Support of **CP1543-1** for northbound communication
- Extended Security Features (VPN, Firewall, ...) of CP can be used.
- The setup depends on the system (R/H).
- One pair of CP1543-1 can be configured with a virtual address IP (System IP).

	R-System	H-System
Max. Number of CPs	2	6
Backplane support	U-Connector	Active Backplane
Hot swap of CPs	No	Yes ¹⁾
Redundant System Power Supply	No	Yes

¹⁾ When CP module is pulled plugged in the H-System, OB83 will be called



R-CPU with CP 1543-1

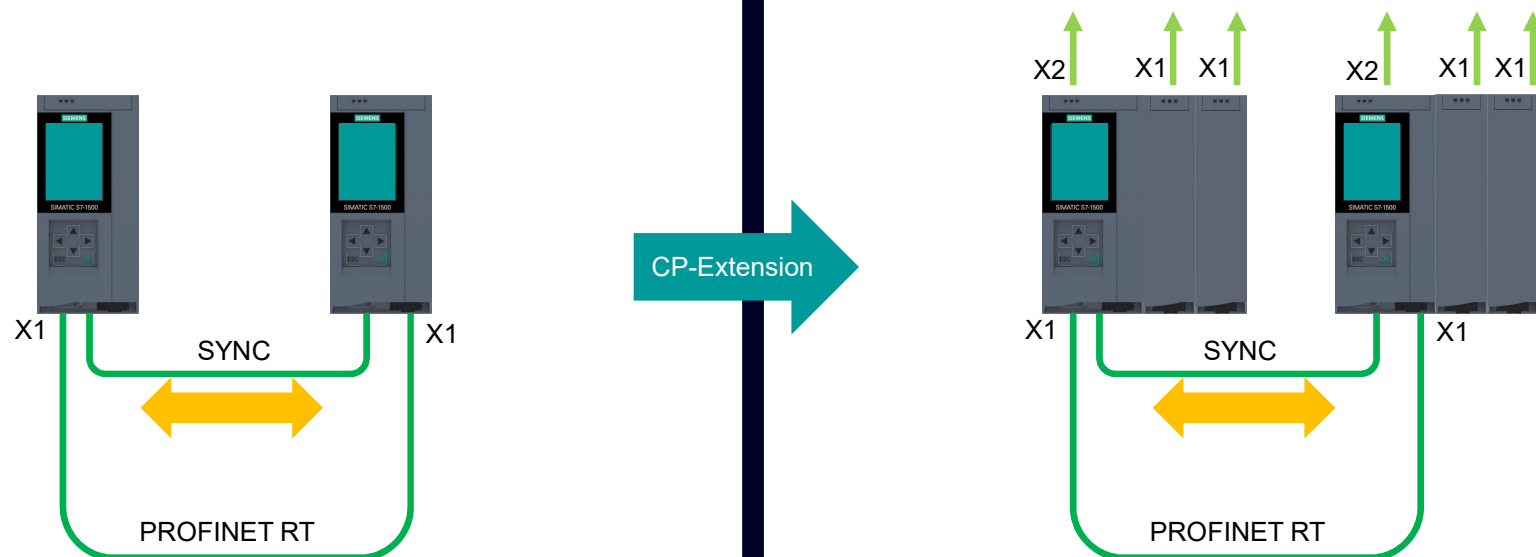


H-CPU with CP 1543-1

Support of Ethernet CP 1543-1 Extension for S7-1500R System

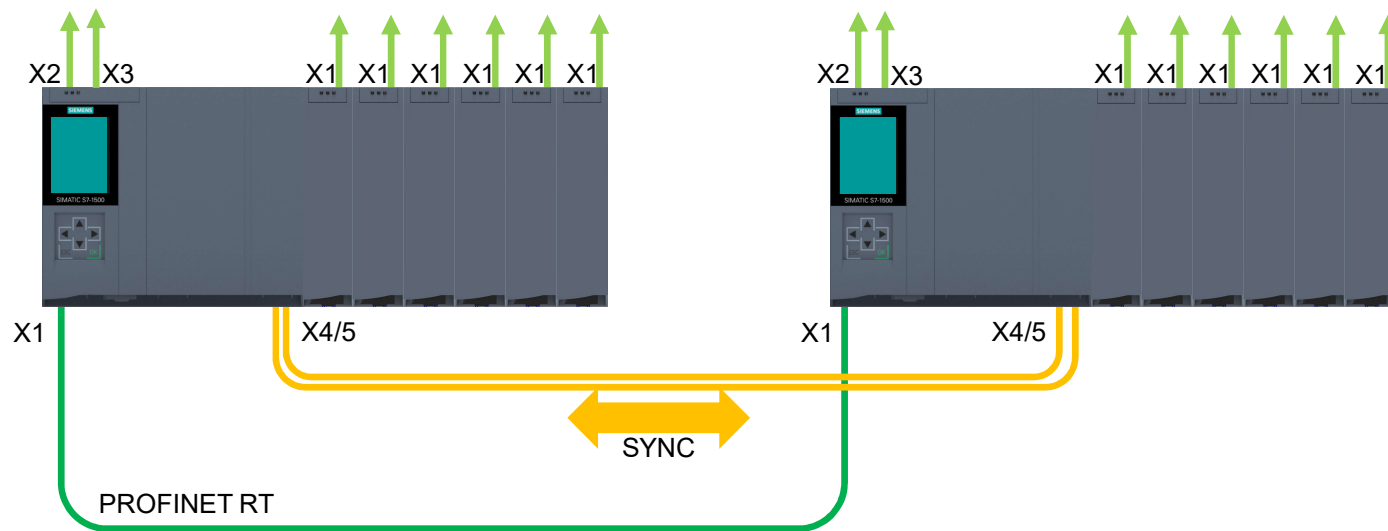
A S7-1500R system can be extended with up to 2 CP 1543-1 modules (per CPU).

Standard U-Connector is used between R-CPU and CP Modules.



Support of Ethernet CP 1543-1 Extension for S7-1500H System

A S7-1500H/HF System can be extended with up to 6 CP 1543-1 modules (per CPU).
Mandatory Active Backplane allows **hot swap of CP Modules**.

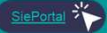


Active Backplane for S7-1500H

Updated Hardware necessary

Supported versions of Active Backplane

- S7-1500H uses the same Active Backplane Bus Module as ET 200MP

Version	Article Number 	Required FS for H-System support
4 Slots	6ES7 590-0BD00-0AA0	FS04 (Firmware V1.1)
8 Slots	6ES7 590-0BH00-0AA0	FS04 (Firmware V1.1)
12 Slots	6ES7 590-0BL00-0AA0	Not supported



8 Slots for CP / PS Modules

Slot 1 for H-CPU



FS04 is required

Support of SIMATIC System Power Supplies

System Power Supplies for S7-1500H

For connection of 3 or more CP modules, the integrated Power Supply of H/HF CPUs is not sufficient

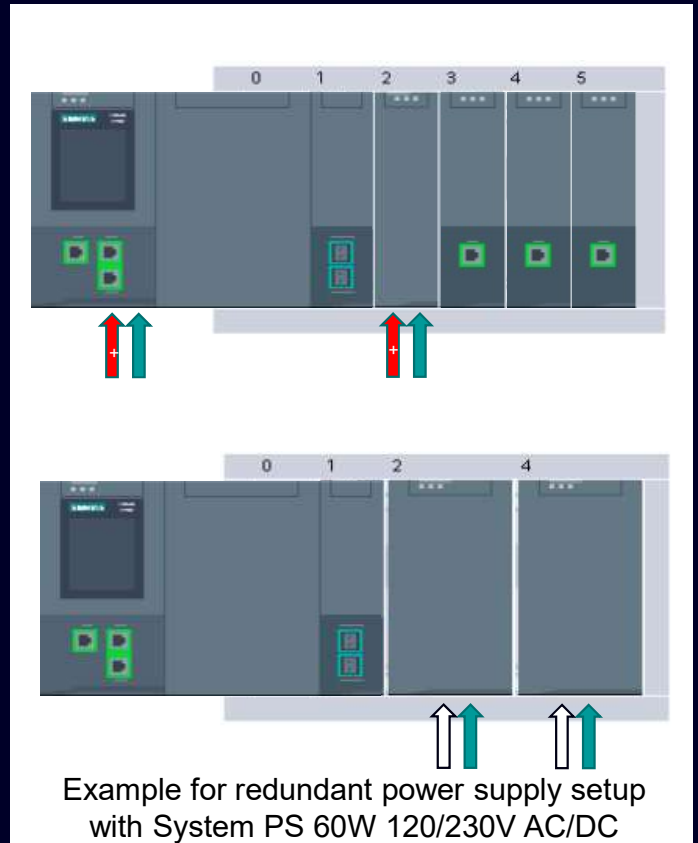
→ Use of System Power Supplies is required

→ TIA Portal shows power consumption of the module in the Power Segment Overview

Power segment overview		
Module	Slot	Power consumption
Active backplane bus ST 1+4 Slot_1	0	-2.00W
PLC_1	1	12.00W
PS 25W 24VDC_1	2	25.00W
CP 1543-1_1	3	-5.25W
CP 1543-1_2	4	-5.25W
CP 1543-1_3	5	-5.25W
Summary		19.25W

Since multiple System Power Supplies can be used in one rack, they can also be used in a redundant way for increased availability.

The dimensioning of the PS must be done in such a way that if one PS fails, the supply of the rack is sufficient.



Support of System Power Supplies

Extended Data Persistence with High Feature System Power Supply

If the integrated retentive memory is not sufficient, the size can be increased by using the High-Feature System Power Supply (HF PS):

CPU Type	Data Memory	Retentive Data	Extended Retentive Data
CPU 1513R-1 PN	2,5 MB	256 kB	2,5 MB
CPU 1515R-2 PN	4,5 MB	512 kB	4,5 MB
CPU 1517H-4 PN	50 MB	2,25 MB	50 MB
CPU 1518HF-4 PN	150 MB	4,5 MB	100 MB

Note: a redundant power supply setup is not supported with the High-Feature System Power Supply (HF PS).



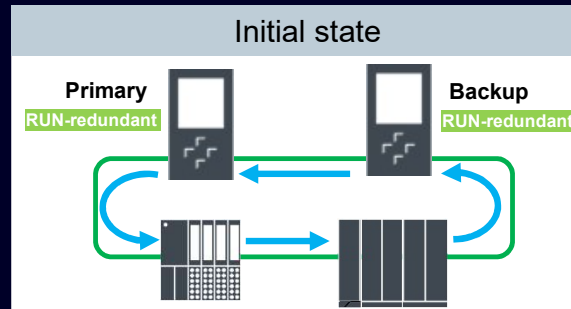
High Feature
System Power Supply

Redundant Controller
S7-1500R/H

Failure Scenarios

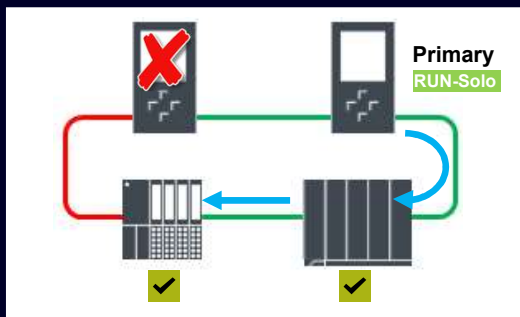
Failure scenarios for S7-1500R/H IO connection via Ring topology

S2



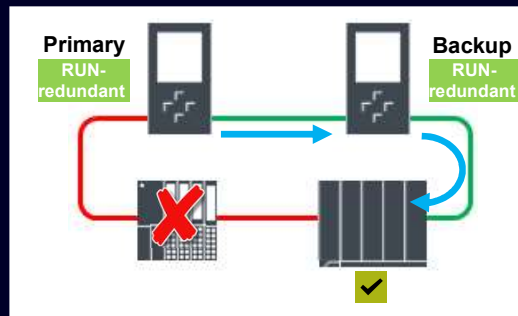
Primary CPU or backup CPU failure

IO communication is switched to the remaining CPU



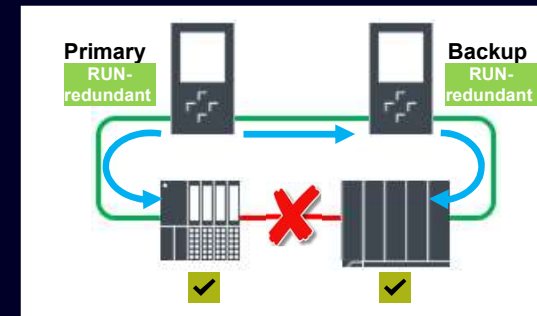
Failure of an IO device

IO communication takes place via MRP ring from Primary CPU to remaining IO-Devices



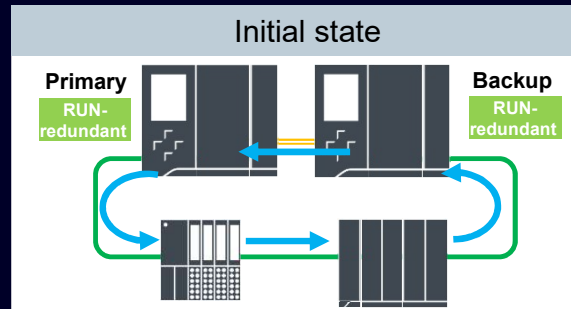
Failure of a PROFINET cable in the ring

IO communication continues via MRP Ring.



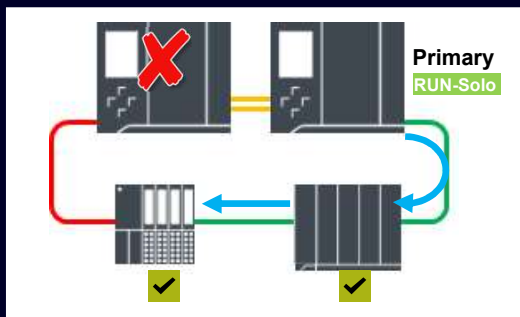
Failure scenarios for S7-1500H IO connection via Line topology

S2



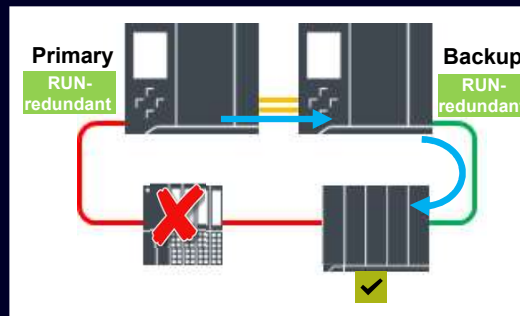
Primary CPU or backup CPU failure

IO communication is switched to the remaining CPU



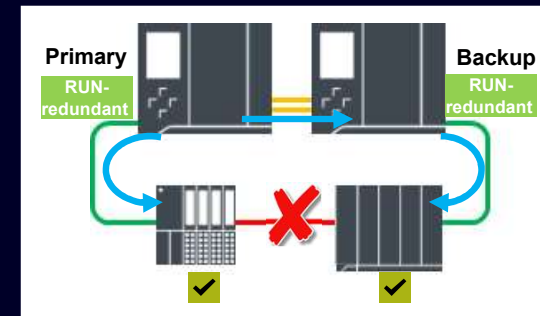
Failure of an IO device

IO communication takes place via Fiber H-Sync Connection from Primary CPU to remaining IO-Devices



Failure of a PROFINET line

IO communication takes place directly or via Fiber H-Sync Connection from Primary CPU

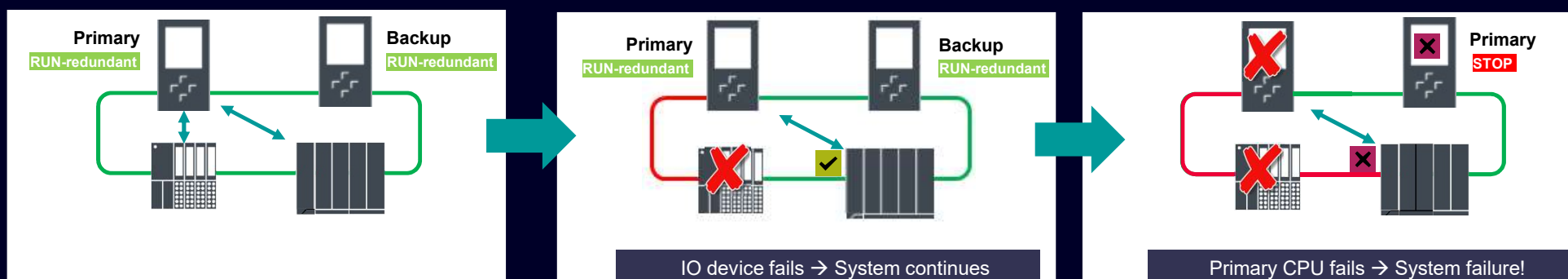


Failure scenarios for S7-1500R/H

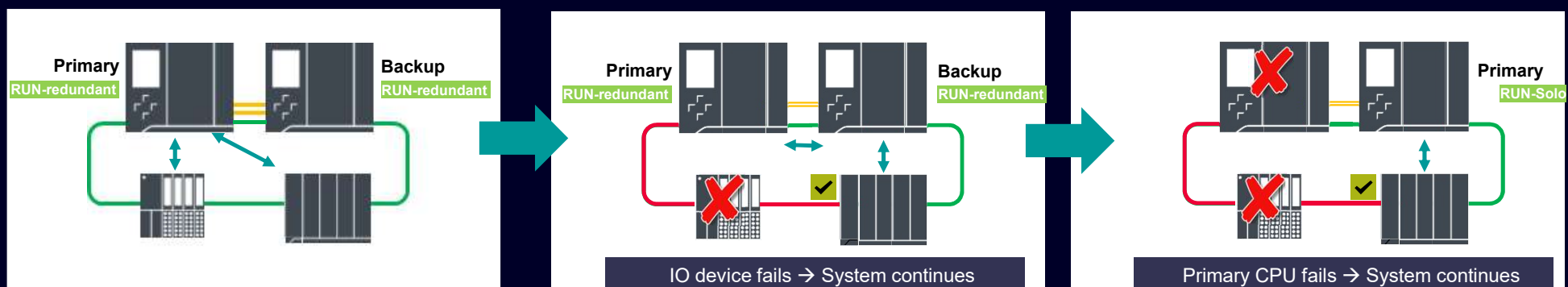
Double fault: Failure of an IO device AND der Primary-CPU

S2

Behavior with S7-1500R



Behavior with S7-1500H

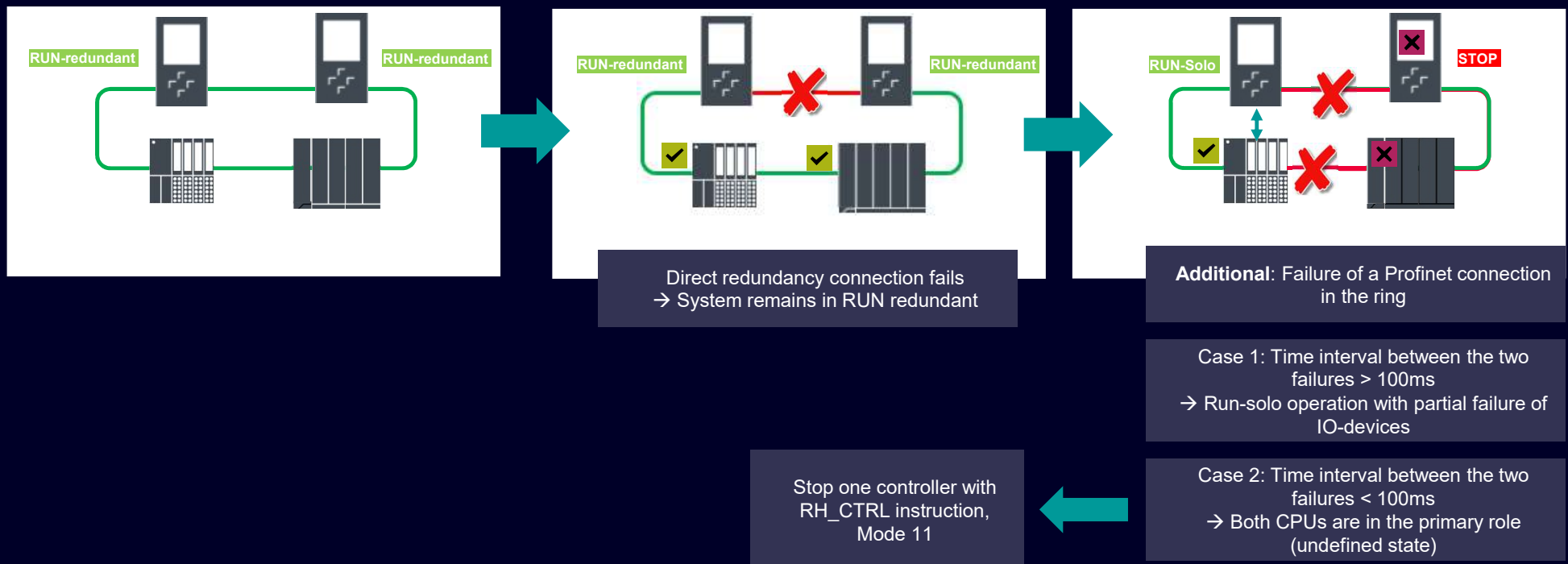


Failure scenarios for S7-1500R

Failure of redundancy connections to CPU synchronization

S2

Behavior with S7-1500R

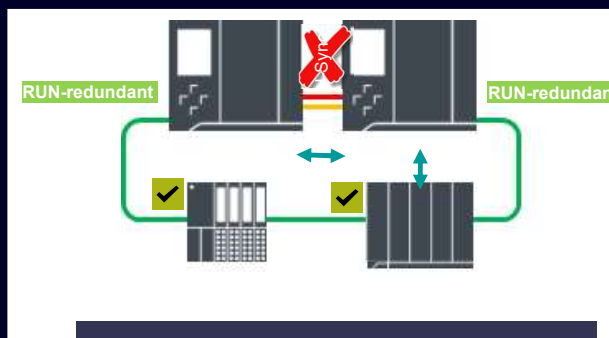


Failure scenarios for S7-1500H

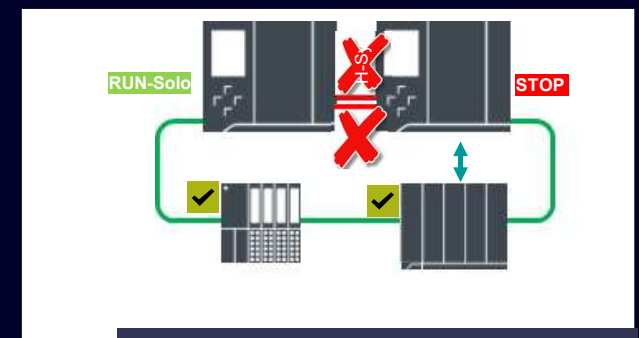
Failure of redundancy connections to CPU synchronization

S2

Behavior with S7-1500H



1 fiber optic redundancy connection fails
→ System remains in RUN redundant



Additional: Failure of the second fiber optic redundancy connection

Case 1: Time interval between the two failures > 55ms
→ Run-Solo operation

Case 2: Time interval between the two failures < 55ms
→ Both CPUs are in the primary role (undefined state)

Recommendation:
Routing sync lines
separately on the H-
System

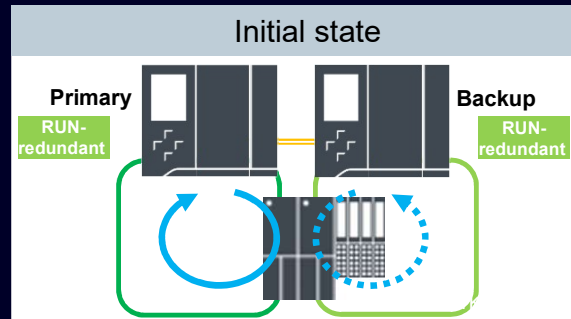
Stop one controller with
RH_CTRL instruction,
Mode 11

New

Failure scenarios for S7-1500H

Design with system redundancy R1 and MRP rings

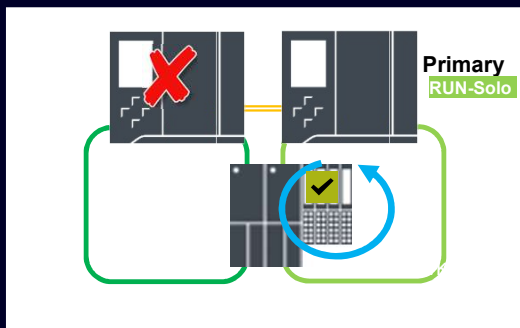
R1



I/O data

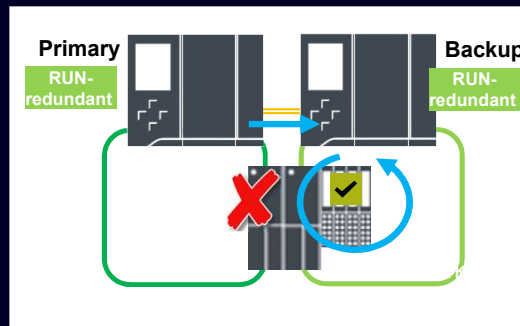
Primary CPU or backup CPU failure

IO communication is switched to the remaining CPU and the second interface module.



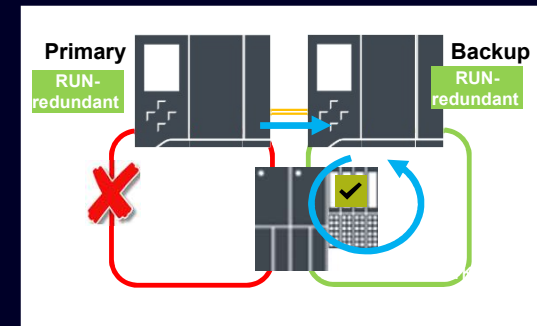
Failure of an interface module

IO communication takes place via Fiber H-Sync Connection from Primary CPU to remaining IO-Devices



Failure of a PROFINET network

IO communication takes place directly or via Fiber H-Sync Connection von Primary CPU

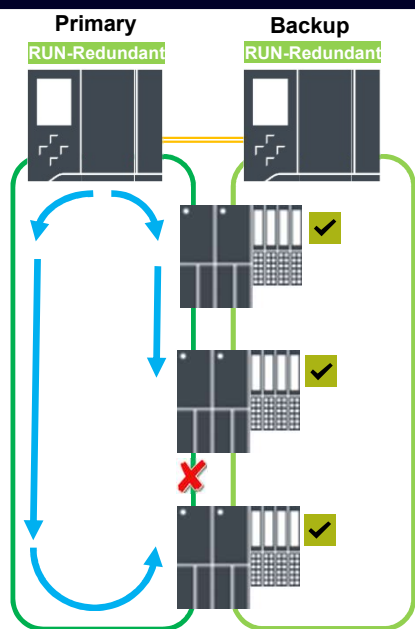


Specific failure scenarios for S7-1500H

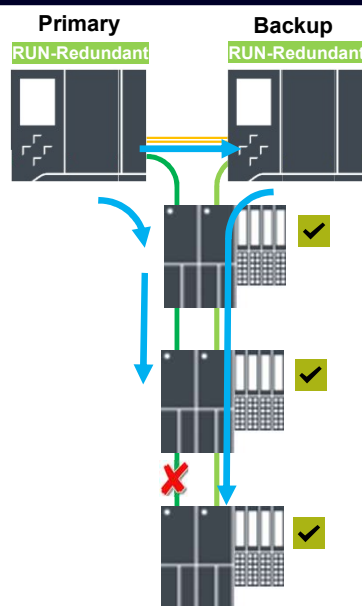
Setup with system redundancy R1 – failure of one line

R1

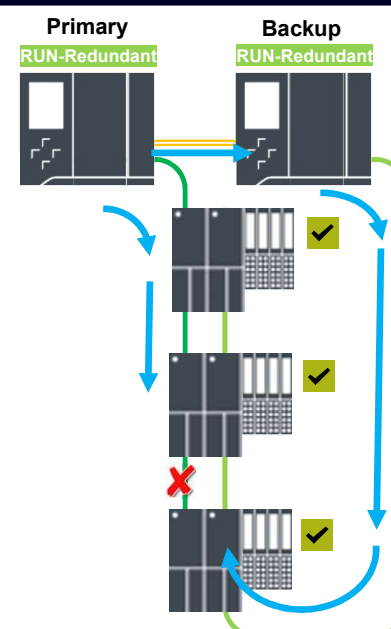
2 MRP Rings



Line topology - with feeding from the same direction



Line topology – with feeding from different directions

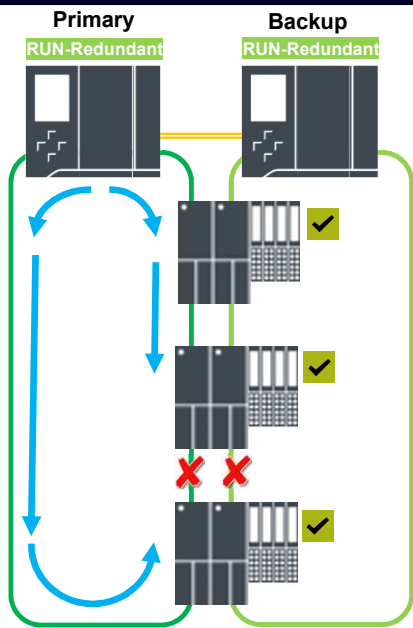


Specific failure scenarios for S7-1500H

Setup with system redundancy R1 – failure of two lines

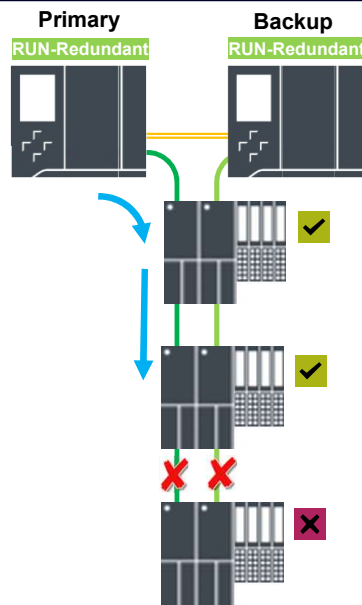
R1

2 MRP Rings



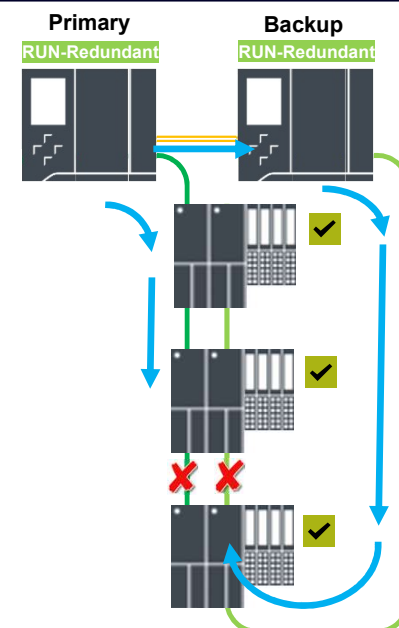
After MRP reconfiguration, all IO-devices continue to work

Line topology - with feeding from the same direction



IO-devices fail after interruption

Line topology – with feeding from different directions



IO data is forwarded via the backup CPU.
All IO-devices continue to work

Communication via System IP and Device IP

Communication via System IP

IP addresses for the R/H system

In addition to the Device IP addresses, for each interface pair of the R/H system, another **System IP address** can be activated.

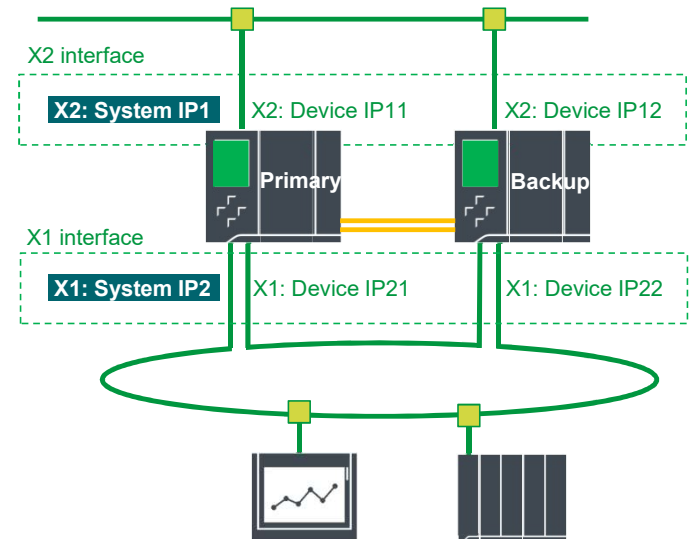
System IP address for switched communication

☒ Enable the system IP address for switched communication

IP address: 192 . 168 . 0 . 3

Subnet mask: 255 . 255 . 255 . 0

Virtual MAC address: 00- 00- 5E- 00- 01- 1

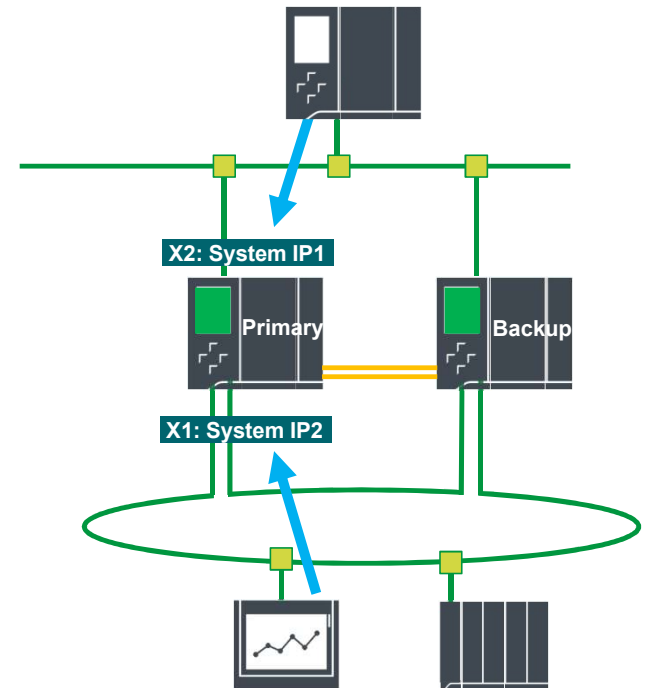


Communication via System IP Behavior

The System IP address is automatically assigned to the primary CPU.

For a communication partner (e.g. standard CPU or HMI), the R/H system behaves like a "normal" (non-redundant) connection partner.

In some cases, it can be necessary to move the System IP address to the backup controller. This can be reached by calling the RH_CTRL instruction.

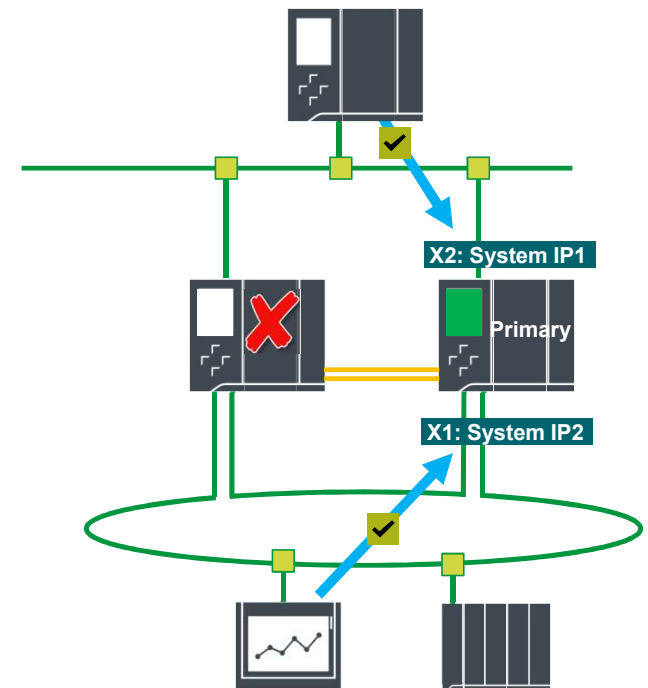


Communication via System IP

Behavior with Primary Backup Failover

If the primary controller fails, the System IP addresses are automatically moved to the Backup CPU

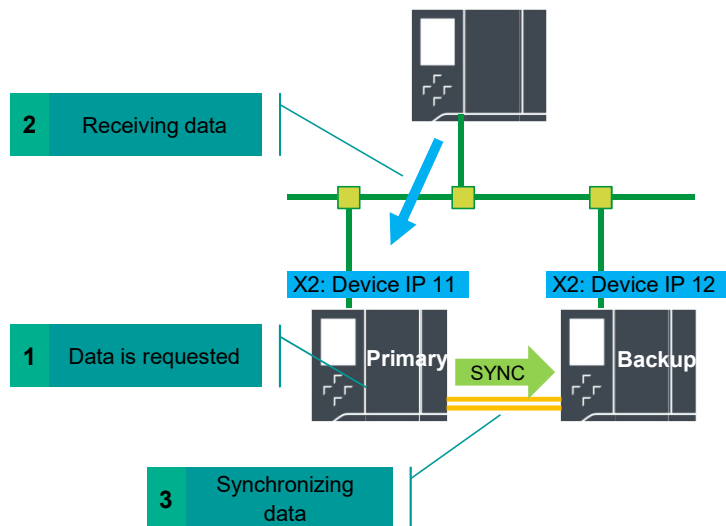
→ A standard controller / HMI can continue to communicate via the same IP address



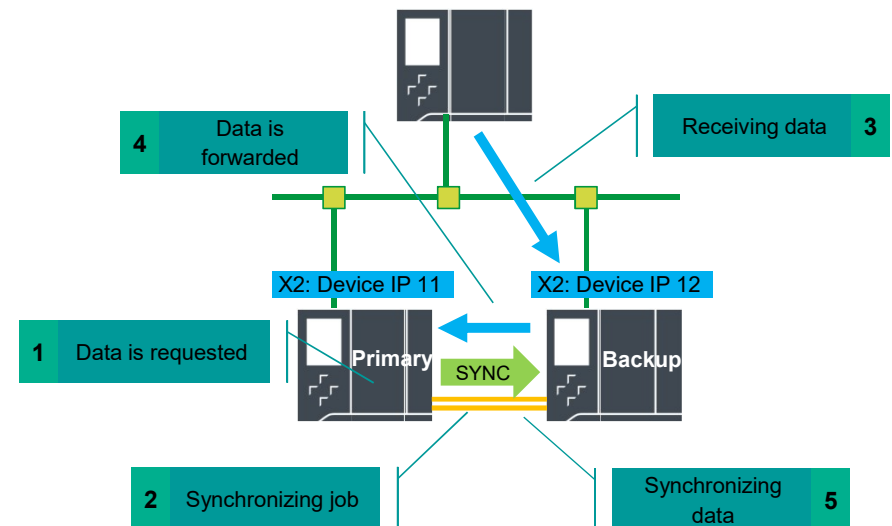
Communication via Device IP addresses Behavior

Communication via Device IP addresses works both via the Primary as well as via the backup CPU.
Please note: Using the connection via the Backup-CPU leads to higher sync load in the system.

Communication via Primary CPU



Communication via the Backup CPU



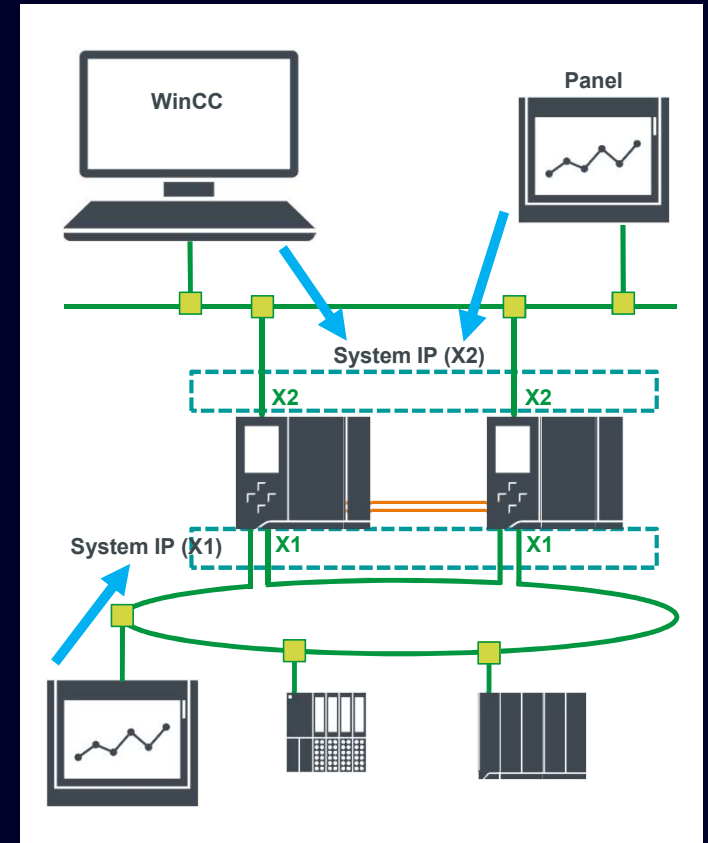
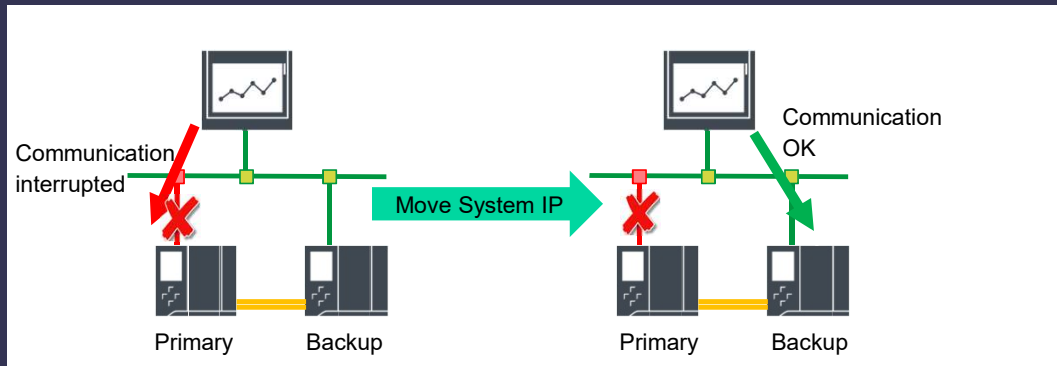
HMI Connection via System IP and Device IP

HMI Connection via a non-redundant network using System IP address

An HMI connection via the System IP addresses for a non-redundant network is possible in all cases.

Note when connected via X2/X3:

If the connection to the Primary CPU is interrupted, communication with the R/H system is no longer possible because the System IP address remains with the Primary CPU. In this case, the System IP address can be moved programmatically to the Backup CPU. See slide [RH_CTRL Instruction](#)



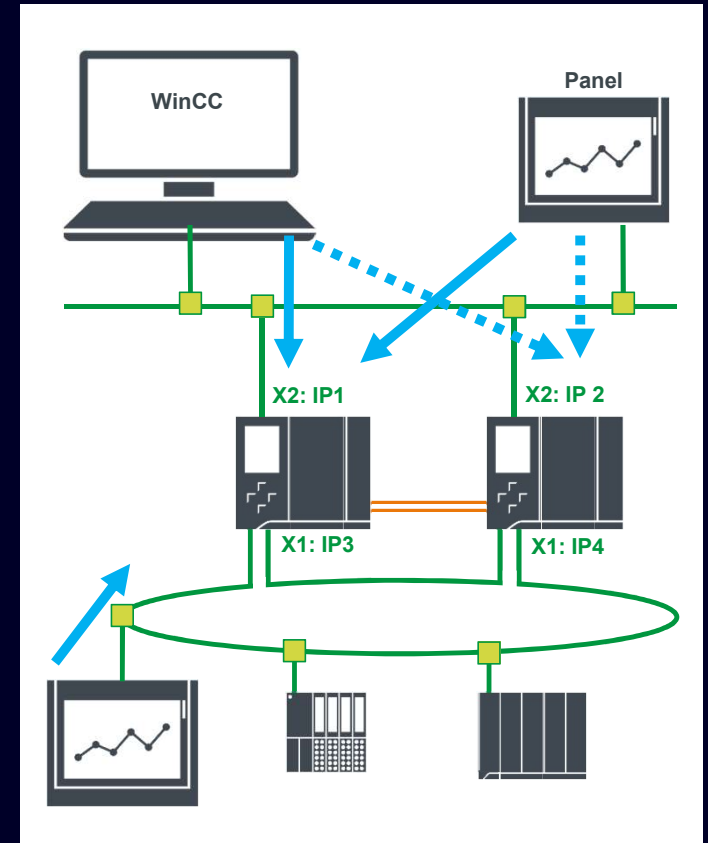
HMI Connection

via a non-redundant network using Device IP addresses

An HMI connection via the Device IP addresses requires a switching option on the HMI side. This is possible with

- WinCC OA from V3.18
- WinCC V7.5 SP1 via Scripting
- SIMATIC Panels via Scripting

The following application example is available for connecting SIMATIC panels to an R/H system:



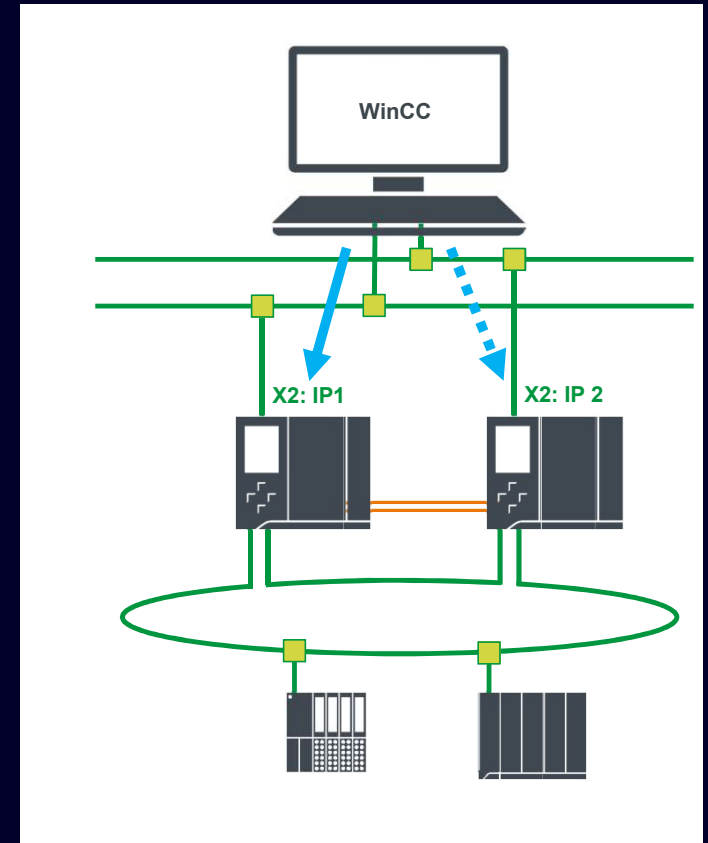
HMI Connection via redundant network using Device IP addresses

The HMI connection via a redundant network using Device IP addresses is possible with

- WinCC OA ab V3.18
- WinCC V7.5 SP1 via Scripting

The communication connection is switched by WinCC in the event of a fault.

See 



HMI Connection

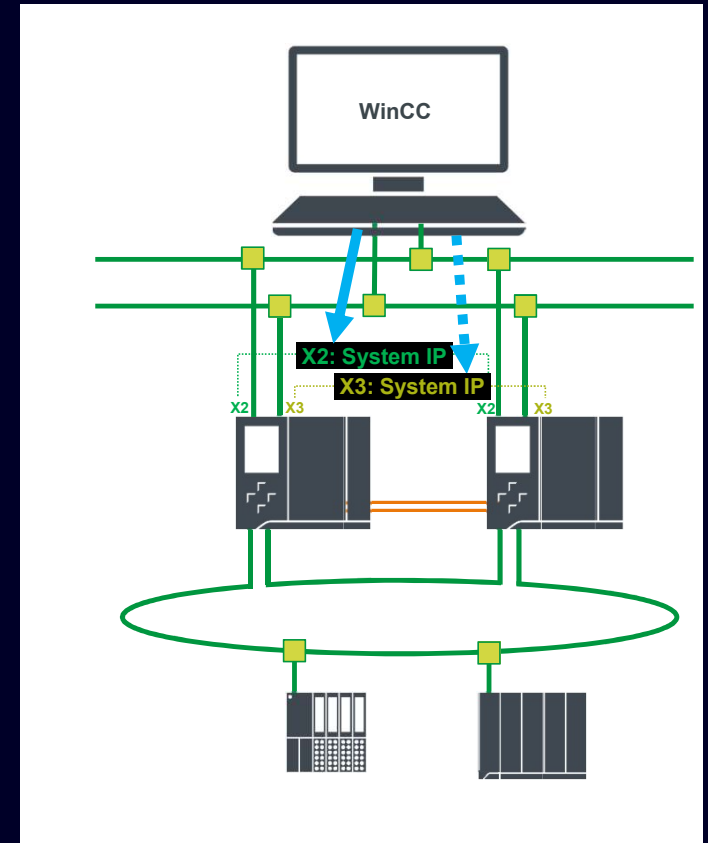
via redundant network using System IP addresses

The HMI connection via a redundant network using System IP addresses is possible with

- WinCC OA from V3.18
- WinCC V7/V8 from V7.5 SP1 via Scripting

Behavior in case of error:

- If the primary CPU fails, the RH system switches over by moving the System IP addresses
- If a network fails, WinCC switches over



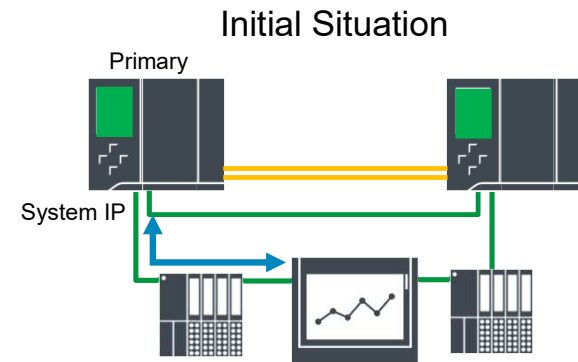
HMI Connection in S2-Rings

Use of System IP address with MRP Ring

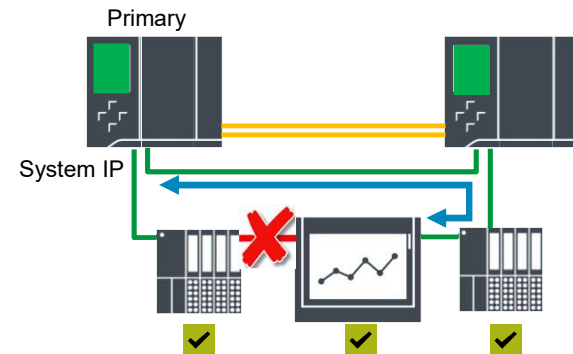
In case of ring interruption...

- Reconfiguration of the MRP Ring
- IO data continues to be transferred
- Communication between HMI panel and primary CPU is done via the remaining route

→ HMI communication continues



Behavior in case of ring interruption



HMI Connection in S2-Rings

Use of System IP address with line topology

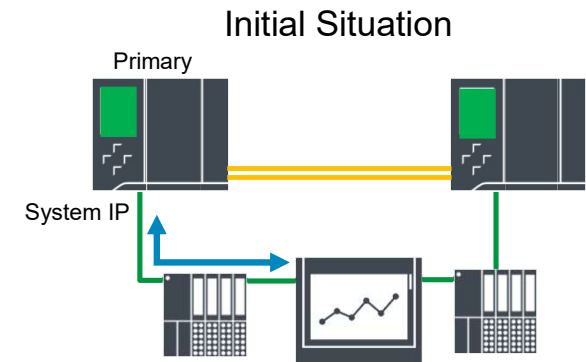
In case of line interruption...

- IO data continues to be transferred
- No more connection between HMI panel and primary CPU
- Only IO data is forwarded via the H-Sync line but no IP communication

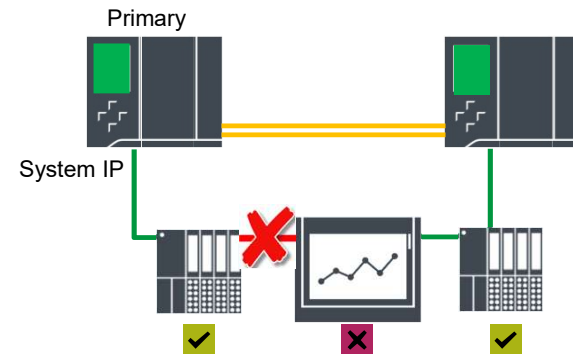
→ Communication to HMI panel fails

Recommendation:

Use the device IP address of the CPUs for HMI communication



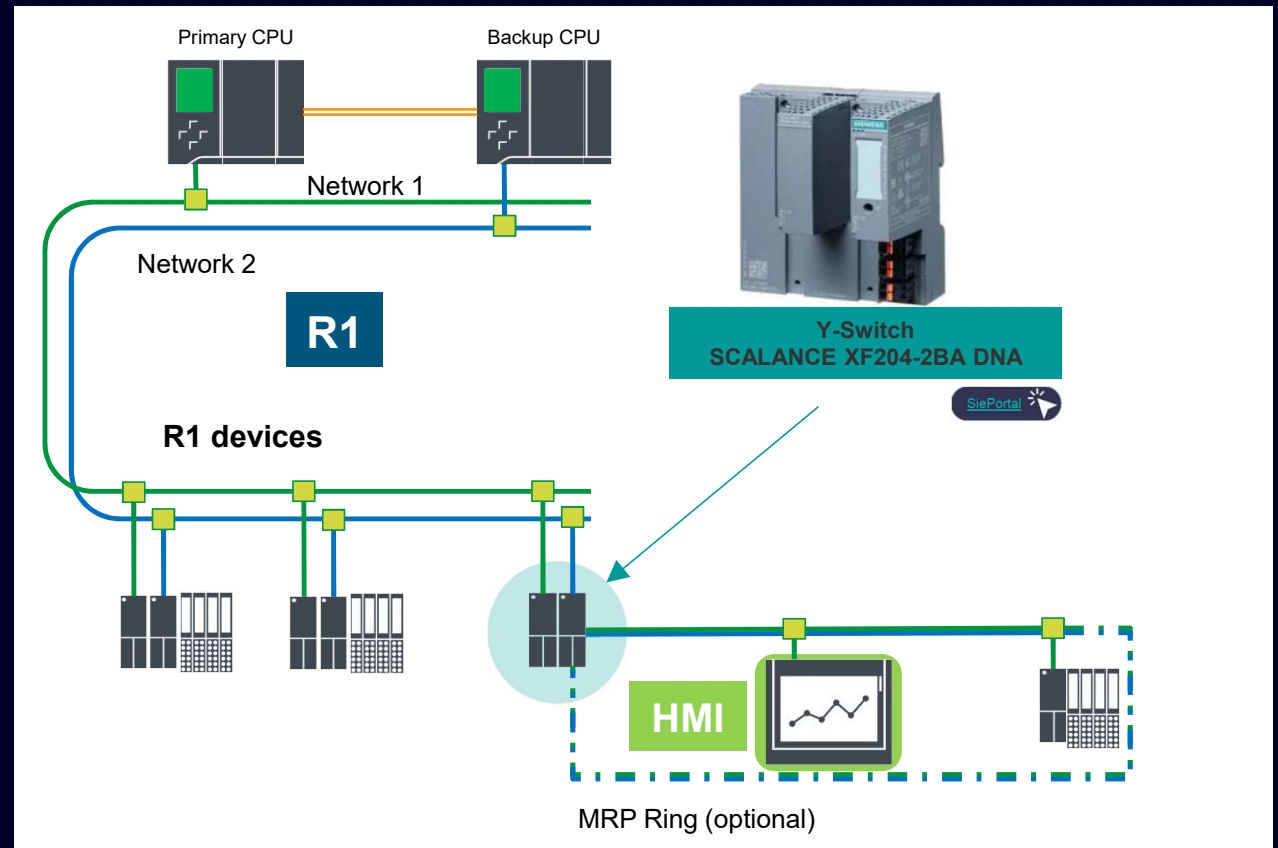
Behavior in case of line interruption



HMI Connection with R1 Redundancy Networks

Combination of R1 and HMI devices

R1 configurations can be extended with HMI devices using a Y-switch.



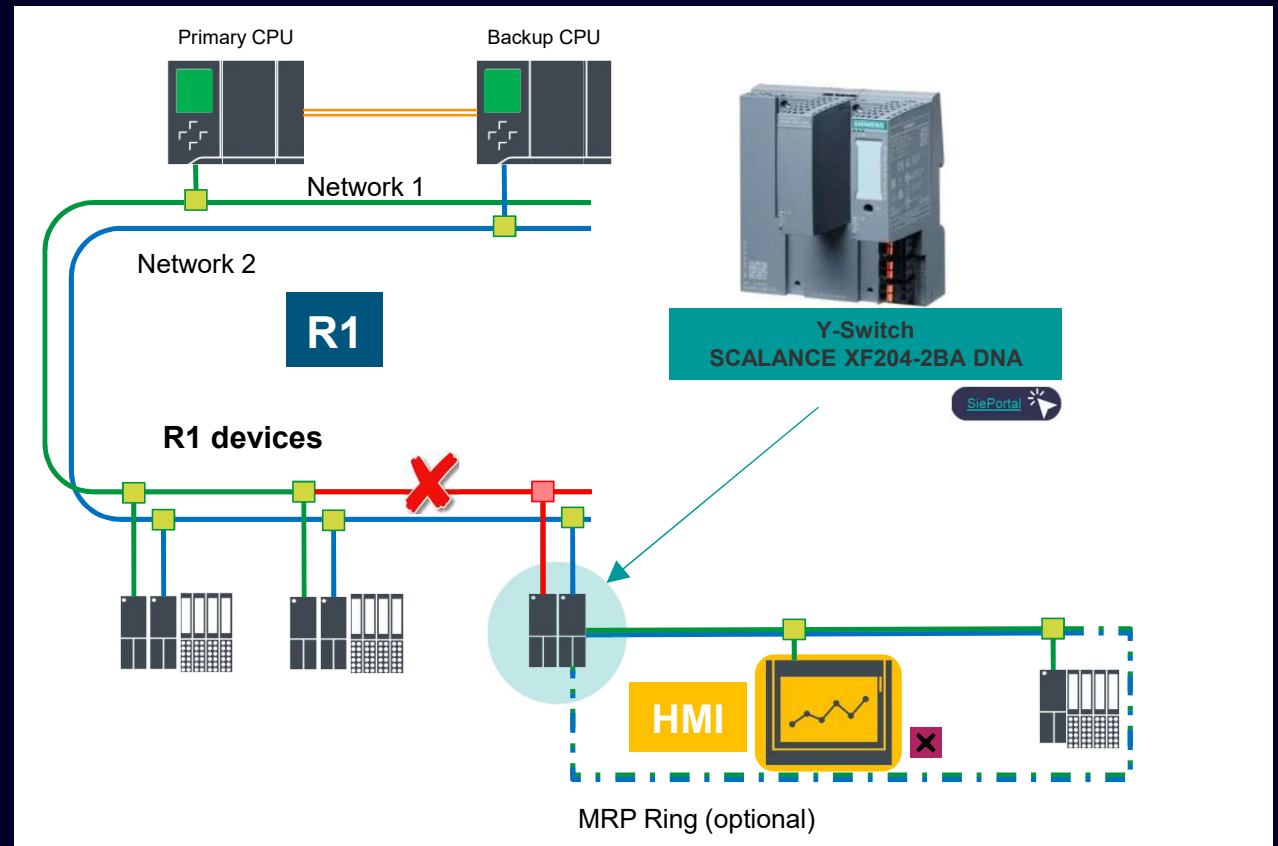
HMI Connection with R1 Redundancy Networks

Combination of R1 and HMI devices

If the connection to the Y-Switch fails, the connected HMI device also fails in line topology.

Solution:

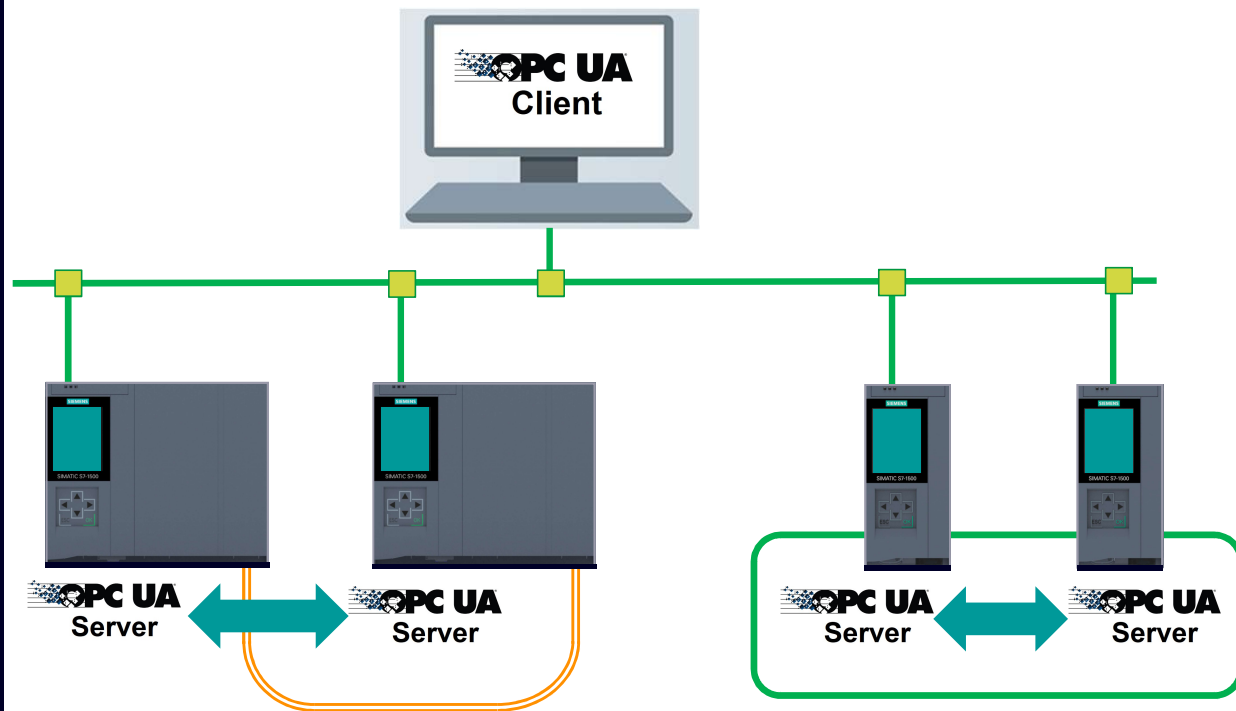
Use device IP addresses of the CPUs or build a ring structure



OPC UA Communication and Web-API

OPC UA Communication for R/H Systems

Support of OPC UA Server



As of firmware V3.1, the OPC UA communication standard is supported as a server.

The redundancy operation is implemented in accordance with the OPC UA specification.

OPC UA Communication for R/H Systems

Feature comparison to standard CPU

OPC UA works in the same way as the standard CPUs. However, there are differences in which functions are offered.

	Standard CPU	R/H CPU
Basic Functionality	Client + Server	Server
Data Access via	Configured Server Interface + Standard Server Interface	Configured Server Interface
OPC UA Security	Yes	Yes
Data Access	Yes	Yes
UA Methods	Yes	Yes
Alarms and Conditions	Yes	No
UA GDS	Yes	Yes
UA Server Redundancy	No	Yes

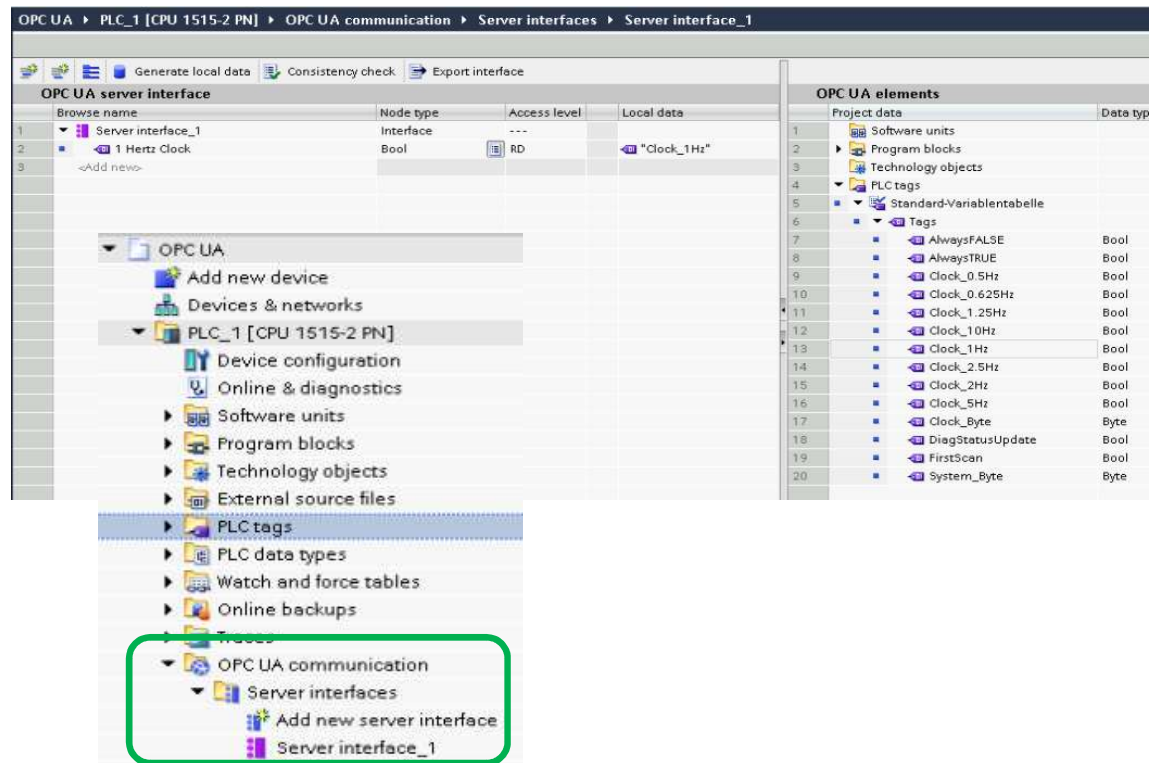
OPC UA Communication for R/H Systems

Access via configured server interface

The UA Server interface can be configured via the TIA Portal or the SIME Editor.

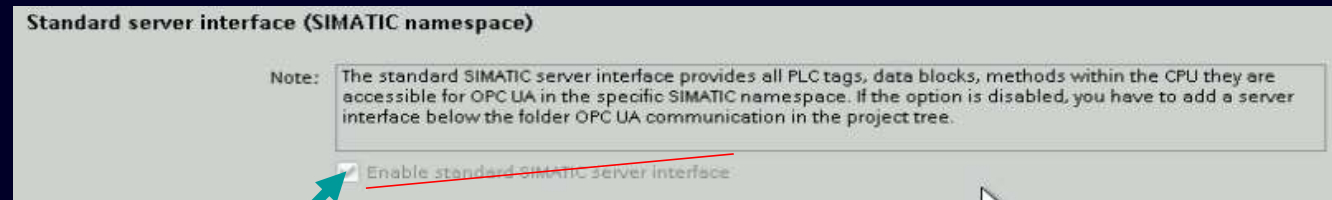
The interface can also be generated automatically with an AddIn

See [GitHub](#)



OPC UA Communication for R/H Systems

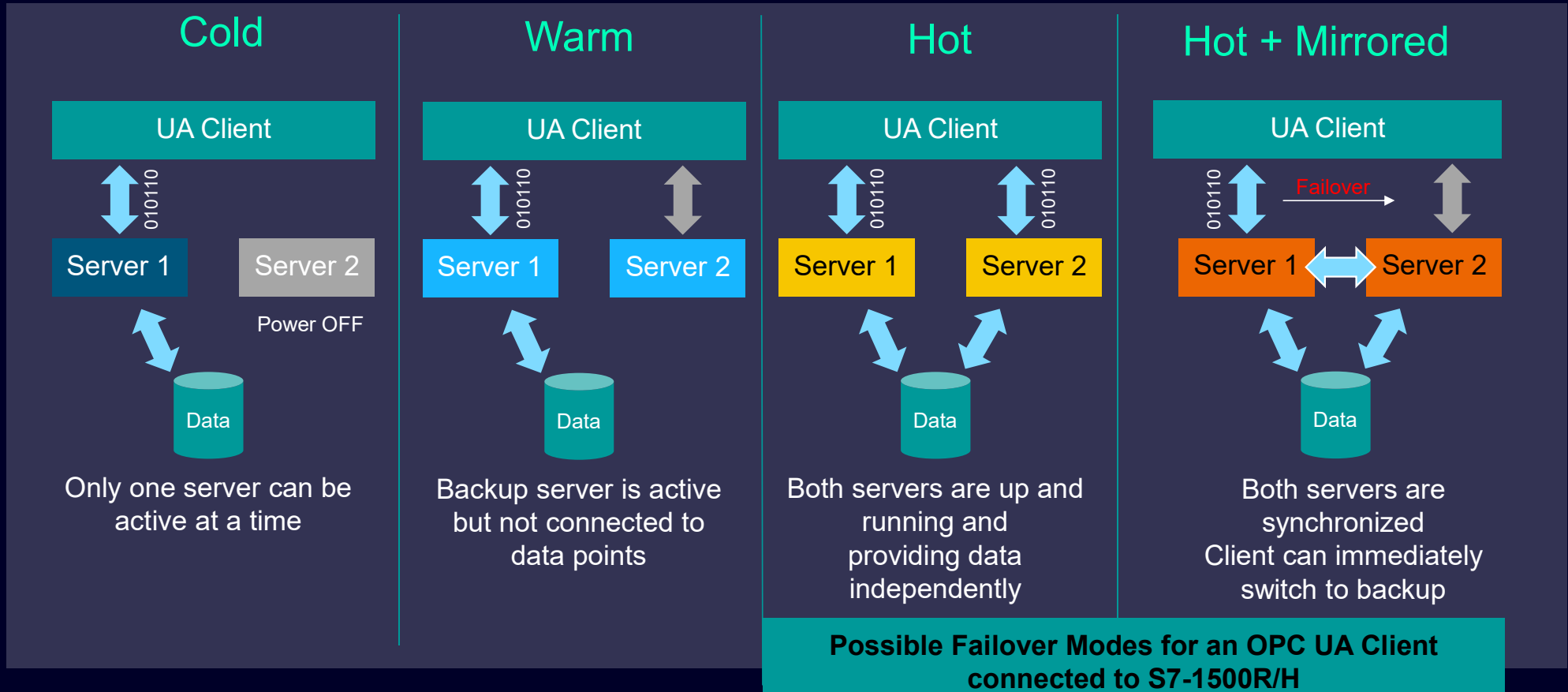
No access via standard server interface



Cannot be enabled for R/H Controller!

OPC UA Communication for R/H Systems

Server Failover Modes according OPC UA Specification

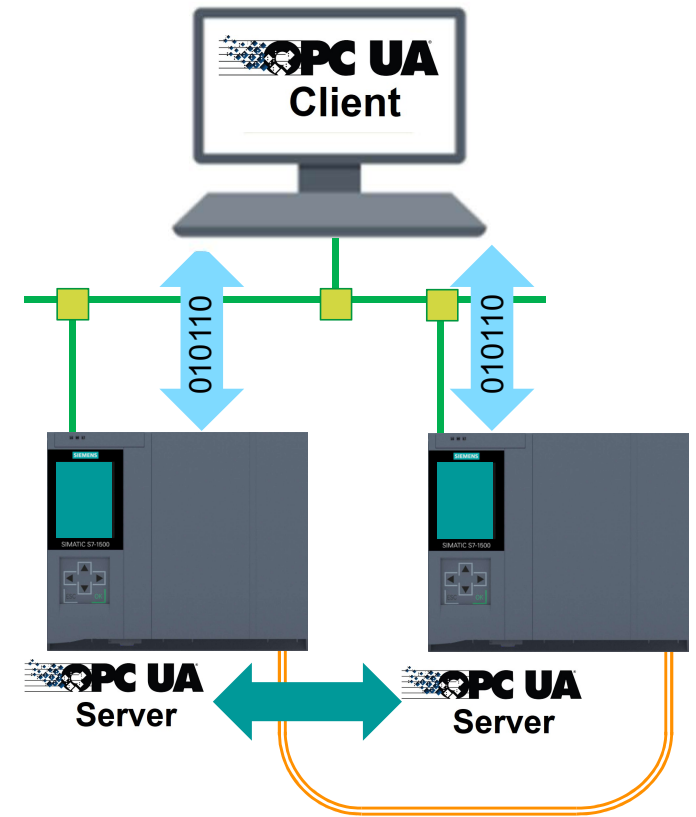


OPC UA Communication for R/H Systems Hot Mode

Hot redundancy mode is initiated by the OPC UA client

In this mode both controller are sending the same data to an OPC UA client. UA client decides which data is used

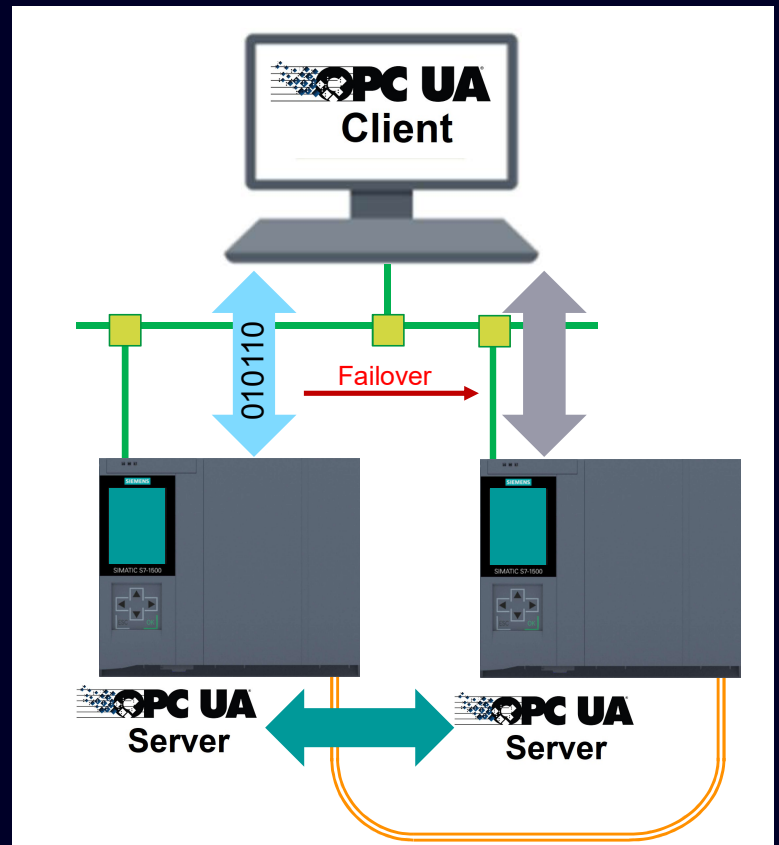
- + This mode allows a bumpless switch over in case of a controller failover since clients gets data twice
- Double network load
- Half quantity structure for OPC UA items



OPC UA Communication for R/H Systems Hot+Mirrored Mode

Hot+Mirrored redundancy mode is initiated by the OPC UA client. In this mode, the sessions of both servers are initiated but only one server sends data.

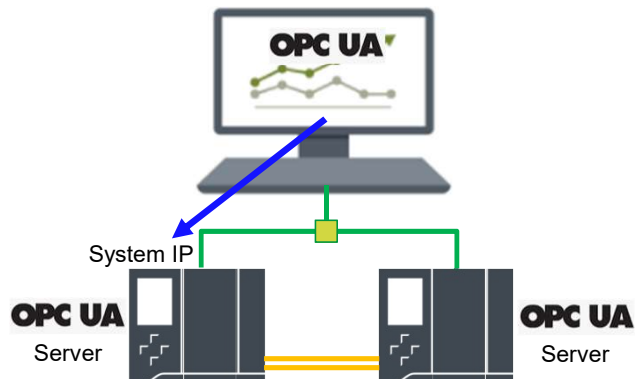
- + Optimal utilization of network capacity
- + Full quantity structure for OPC UA items can be used
- + Fast switch over
- Not bumpless



OPC UA Communication for R/H Systems

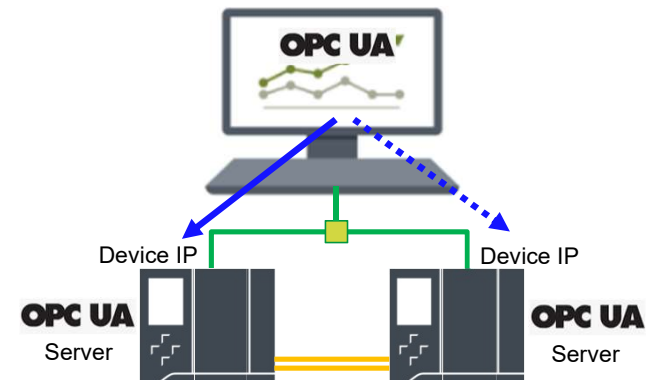
Redundancy modes

Transparent Mode



For “standard” Clients without redundancy support.
UA Client always connects to System IP Address

Non-Transparent Mode



For clients which are aware of a redundant server pair →
The client selects the appropriate OPC UA server using the Device IP Address of the CPU.
This mode is supported by WinCC OA

OPC UA Communication for R/H Systems Required Licenses

- To use the OPC UA Server, appropriate licenses must be purchased.
- **One license is required per CPU.**
i.e., 2 licenses are required for 1 R/H system.
- These are the same licenses as the non-redundant controllers.

CPU Type	Required License	MLFB	
CPU 1513R	SIMATIC OPC UA S7-1500 Small	6ES7823-0BA00-1BA0 6ES7823-0BE00-1BA0	
CPU 1515R	SIMATIC OPC UA S7-1500 Medium	6ES7823-0BA00-1CA0 6ES7823-0BE00-1CA0	
CPU 1517H	SIMATIC OPC UA S7-1500 Large	6ES7823-0BA00-1DA0 6ES7823-0BE00-1DA0	
CPU 1518HF	SIMATIC OPC UA S7-1500 Large	6ES7823-0BA00-1DA0 6ES7823-0BE00-1DA0	

| Web server

SIMATIC S7-1500 Redundant Systems

System Web Pages

V21

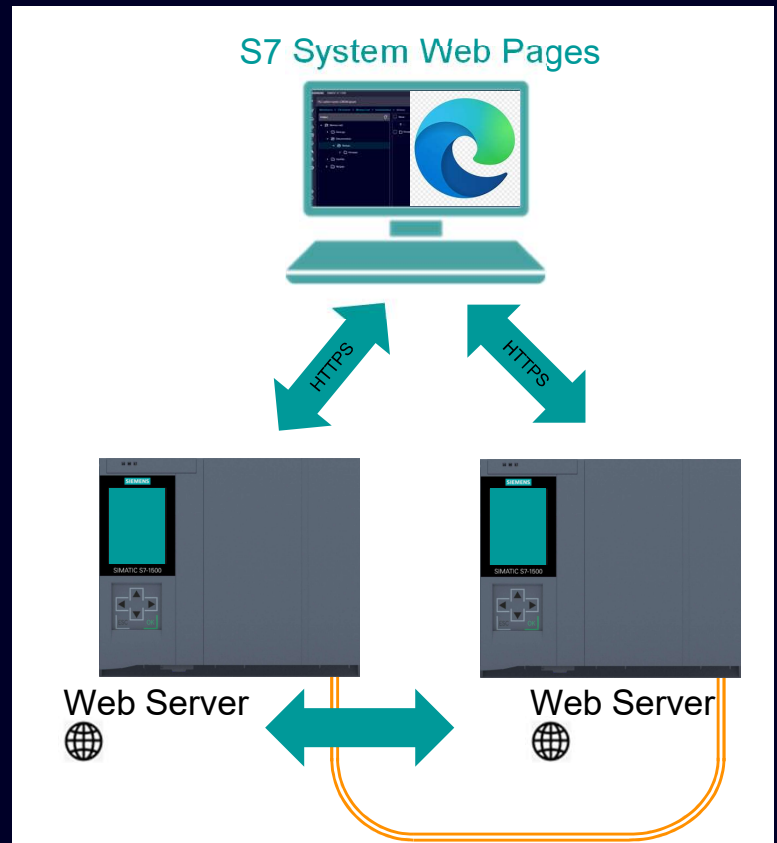
The web server offers a simple way of commissioning, diagnosing, operating and monitoring as well as supporting service and maintenance work – All you need is a Web Browser

Available diagnostic and configuration information:

- Alarms, Diagnostic buffer
- File Management for easy access to Data Logs, User Files and Recipes
- Safety administration

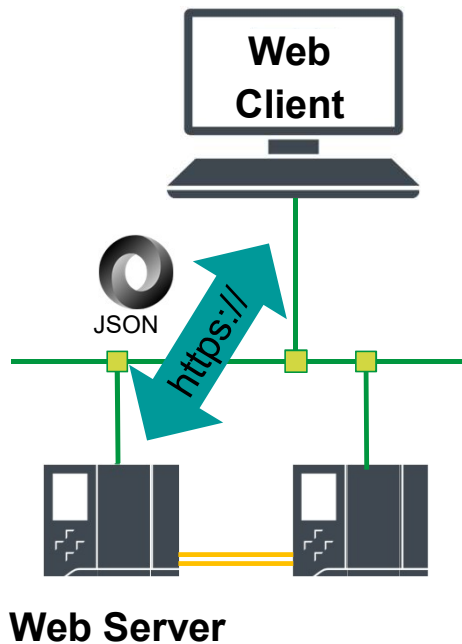
Control possibilities:

- Monitoring and change of operating mode and system state
- Password change



SIMATIC S7-1500 Redundant Systems

IT link access via Web API



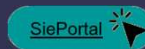
The Web API enables to:

- Implement web applications at the latest state-of-the-art technology
- Communicate with the Web server of the CPU via script and programming languages

Supported development platforms

- | | | |
|--------------|---------------|---------------|
| ▪ Windows | ▪ Python | ▪ Laptop |
| ▪ Linux | ▪ C# | ▪ Tablet |
| ▪ Mac | ▪ Java | ▪ Smart phone |
| ▪ iPhone/iOS | ▪ Power Shell | ▪ Edge |
| ▪ Android | ▪ others... | ▪ Cloud |

A detailed description of the API methods can be found in the Function Manual Webserver:



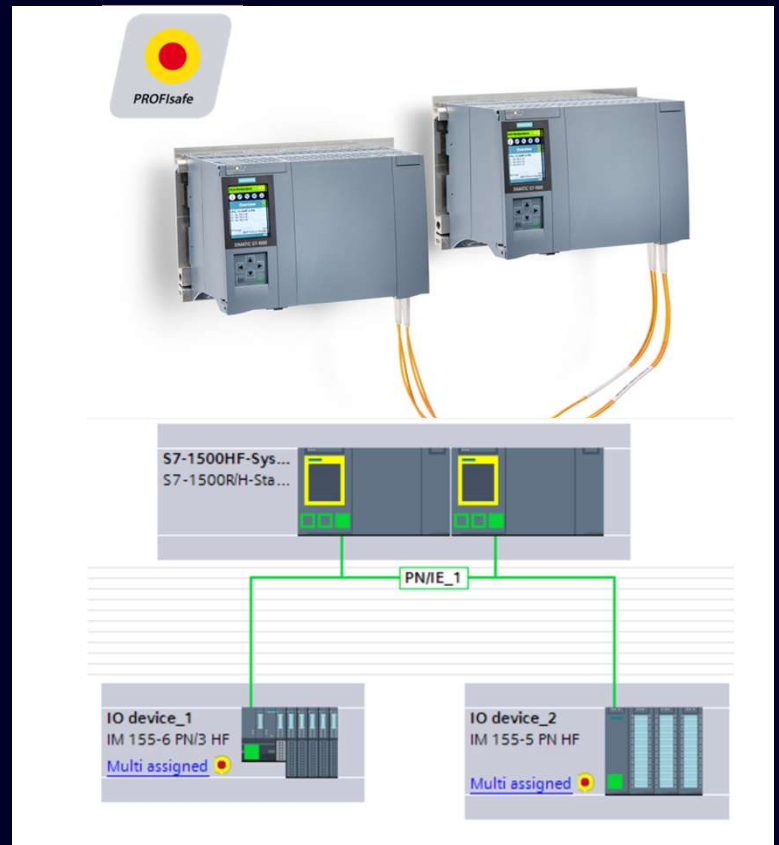
Safety with Redundant Systems

Safety for Redundant Systems

Failsafe applications with redundant Controller

High Availability + Failsafe = CPU 1518HF

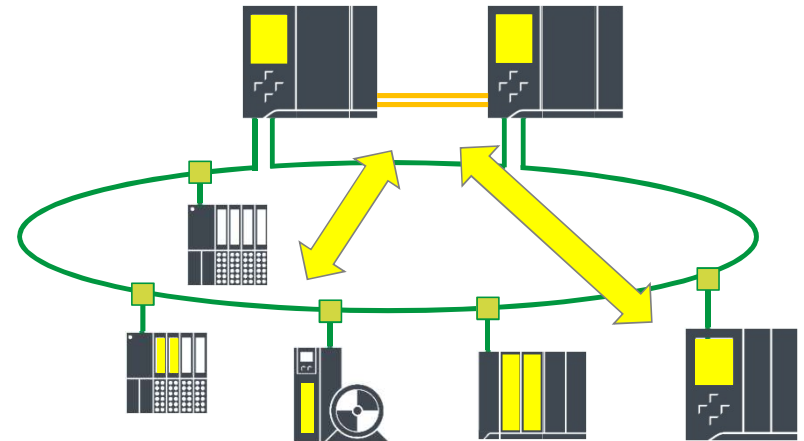
- Engineering with STEP 7 Professional und STEP 7 Safety,
- **Safety programming as for a non-redundant fail-safe CPU,**
- Supported PROFIsafe Communication,
- Supports flexible F-Link (safe controller/controller communication),
- Fail-over scenario without stop the safety program,
- Fast-commissioning mode as all S7-1500F supported,
- S2 and R1 System Redundancy can be used with Failsafe modules.



Safety for Redundant Systems

Direct integration of fail-safe devices with SIMATIC CPU 1518HF

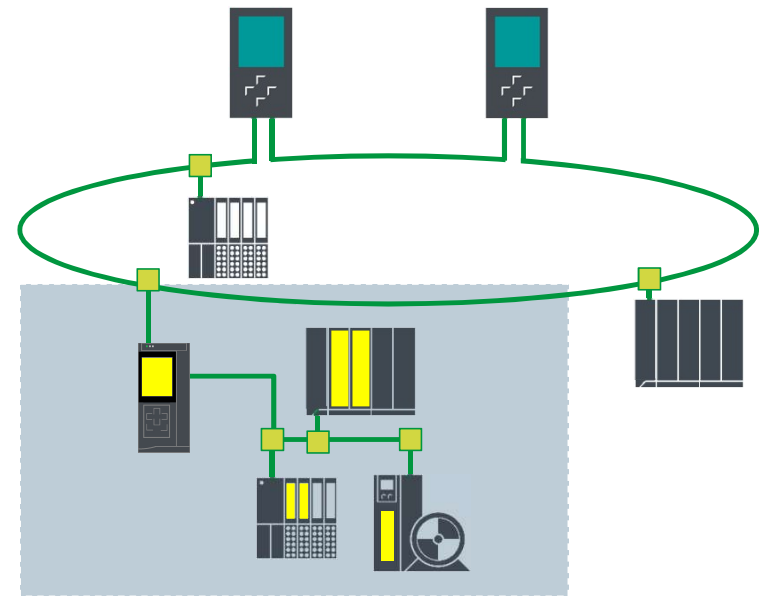
- Safety Application runs on CPU 1518HF,
- Direct connection of PROFIsafe IO-devices,
- Safe CPU-CPU communication via Flexible F-Link.



Safety for Redundant Systems

Connection of fail-safe devices via a subordinate F-controller

- R-System communicates to a subordinate F-controller via iDevice or Open User Communication (non-Safe).
- Safety Application runs in a separate F-Controller.



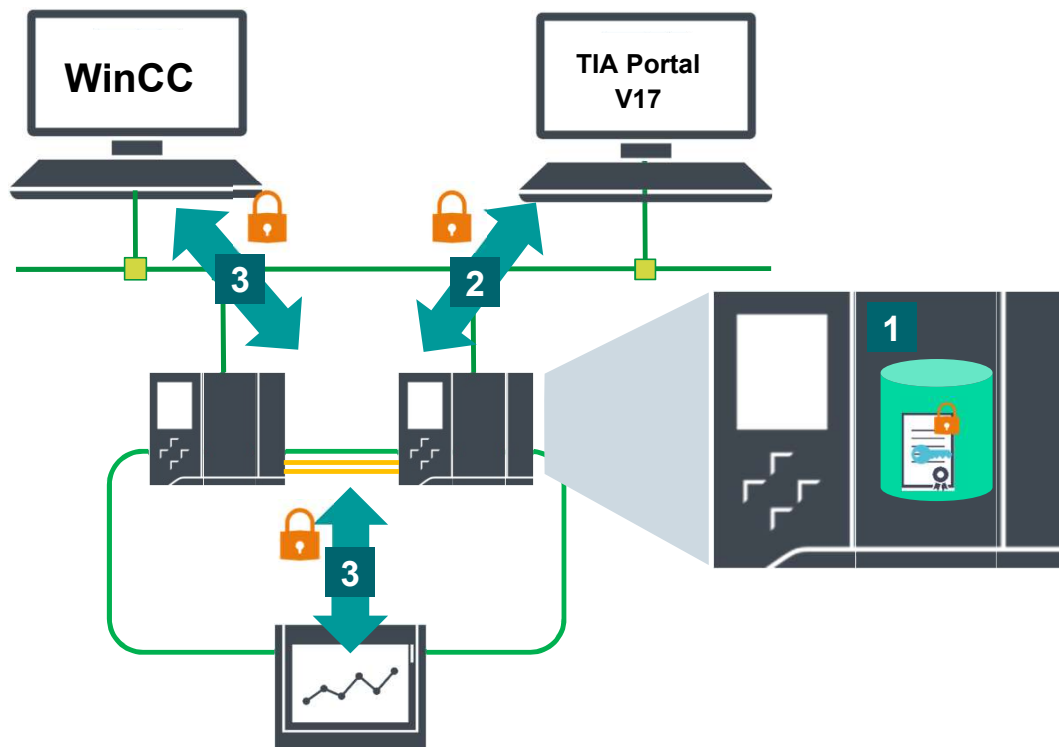
Connection of fail-safe devices by a subordinate F-controller with PN/PN coupler

-
- The diagram illustrates an MRP Ring network topology. A central green ring connects several nodes. Inside a light blue shaded area labeled "MRP Ring", there are various components: a server rack, a terminal, a server with a CD-ROM, and a server with a monitor. Outside this area, there are two more server racks and two terminal-like devices. The ring connects these components in a closed loop, ensuring redundancy and continuous communication.

Security

Security Enhancements

Protection of configuration data / Secure communication to HMI and TIA Portal

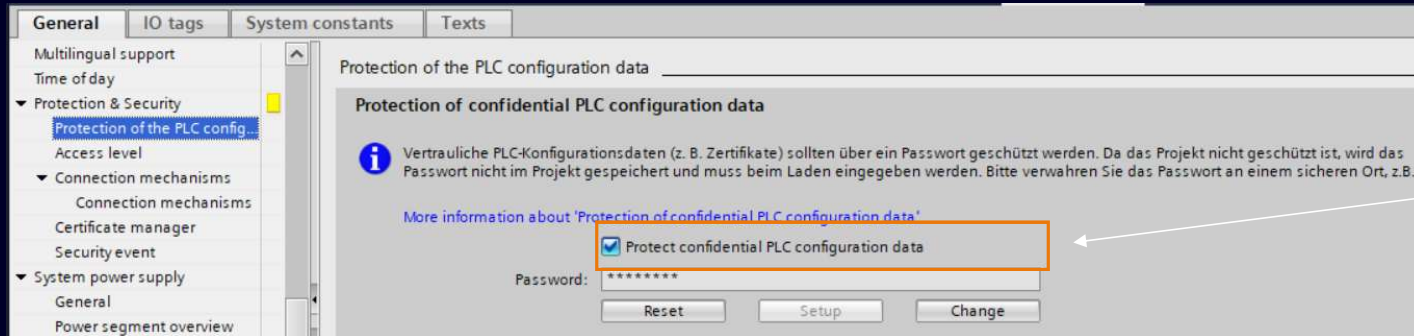


- 1** Protection of configuration data
- 2** Secure communication between controllers and TIA Portal
- 3** Secure communication between controllers and HMI

Protection of configuration data

Properties

Configuration data protection can be activated/deactivated via the TIA Portal.

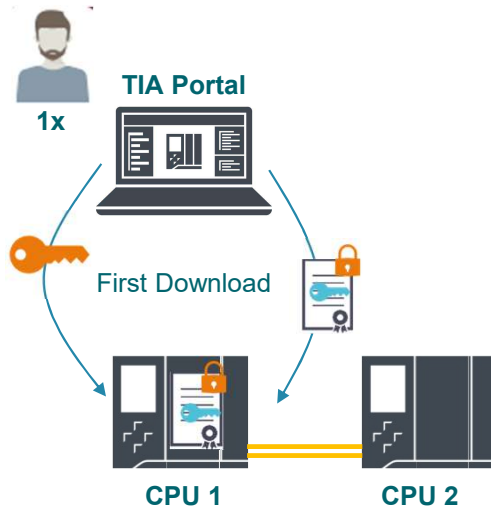


- The protection password is not stored in the project but stored independently of the project data in the CPU. It is not readable and must therefore be stored in a safe place
- If protection of the configuration data is activated, special features must therefore be taken into account during the initial commissioning and replacement of a controller. → see next slides

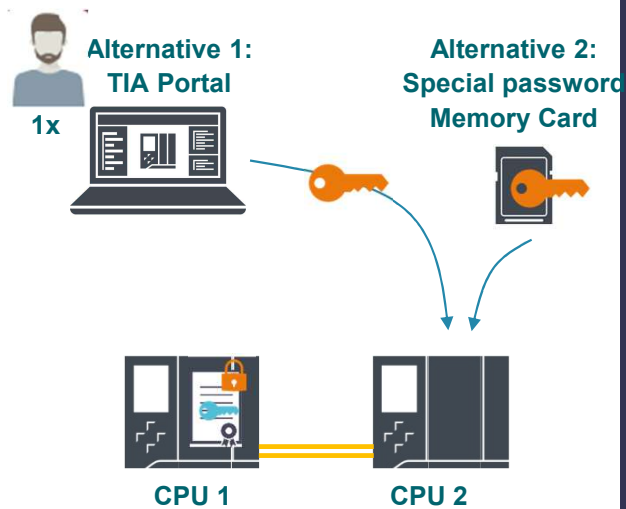
Special features when activating password protection of the configuration data

Initial commissioning of an RH system

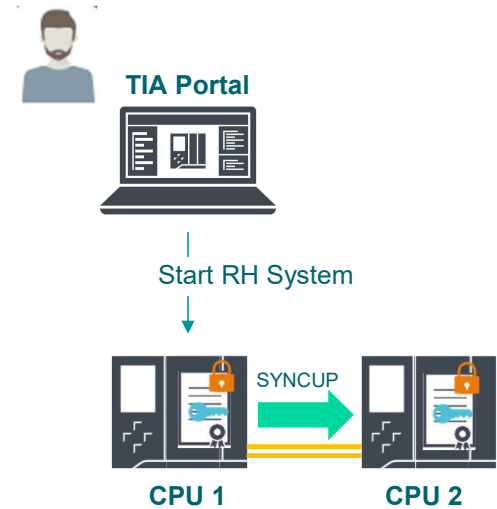
Step 1) Load the configuration and set the password



Step 2) Set password in CPU 2



Step 3) Start RH System → Synchronizing data



Without this step, the SYNCUP process will fail!



Certificate with private key for PG/HMI communication



Password to protect configuration data

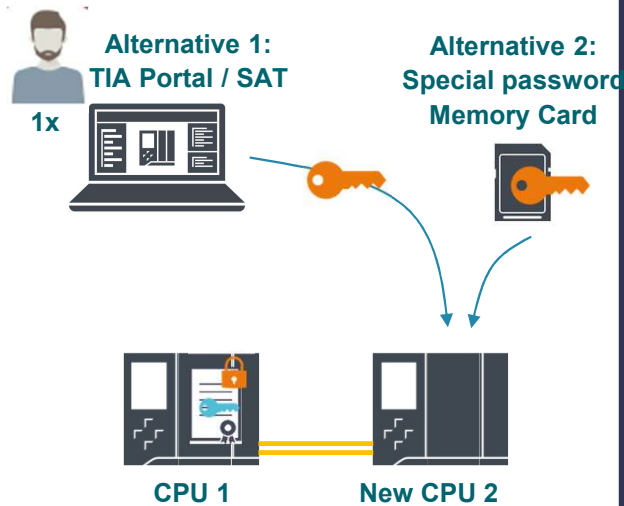
Special features when activating password protection of the configuration data

Replacing an R/H CPU

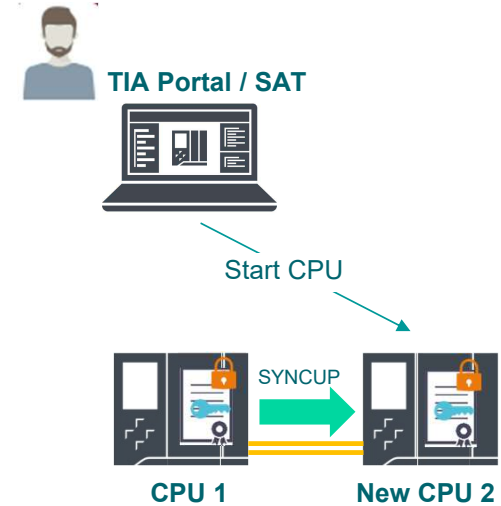
Event: A CPU fails and needs to be replaced



Step 1) Change CPU and reset password



3) Start the new CPU → Synchronizing data



Without this step, the SYNCUP process will fail!



Certificate with private key for PG/HMI communication

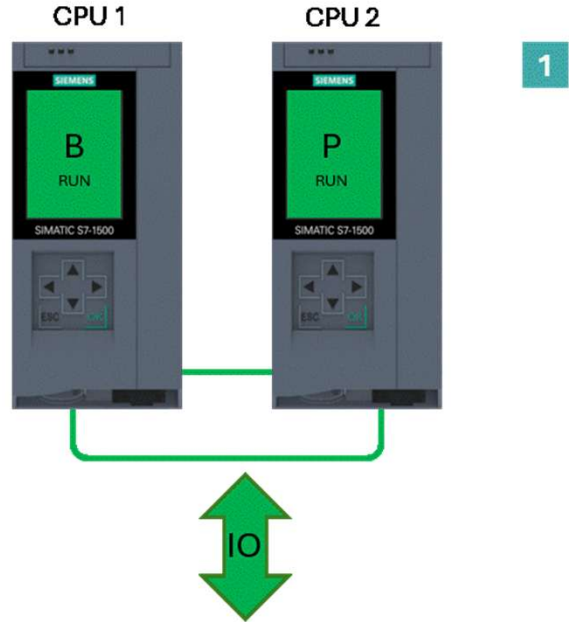


Password to protect configuration data

Loading a Project

Load program in S7-1500R/H

Procedure without changes to the hardware configuration



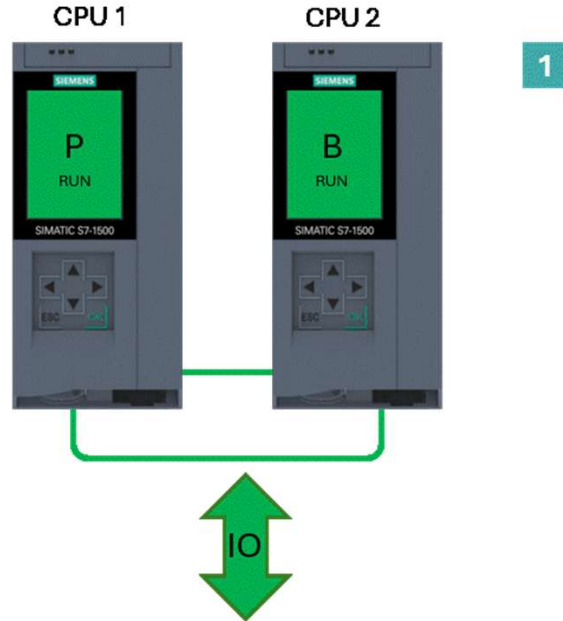
Loading program only

If only the program of the CPU has been changed, this can be loaded in the RUN-Redundant state

- 1 System in State „Run-Redundant“
- 2 Load new program in CPU 2
- 3 System continues with new program

Load project in S7-1500R/H

Procedure with changes to the hardware configuration



Loading hardware configuration

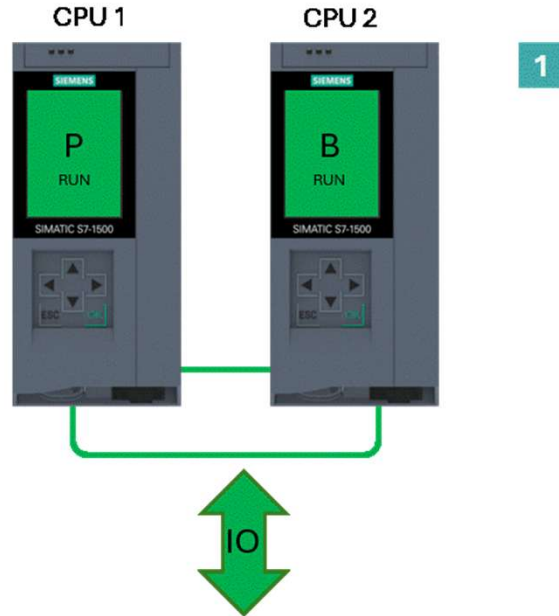
Changes to the hardware configuration can be loaded with the help of the [CiR function](#) in the RUN-Redundant state.

If changes have been made that cannot be loaded via CiR, the following procedure can be used to minimize downtime

- 1 System in State „Run-Redundant“
- 2 Stop CPU 2 (Backup CPU)
- 3 Load Project into CPU 2
- 4 Stop CPU 1 → Process is stopped
- 5 Start CPU 2 → New project is active
- 6 Start CPU 1 → Project is synchronized
- 7 System in State „Run-Redundant“

Load project in S7-1500R/H

Rollback to previous project status



Rollback procedure

With a similar approach, a quick "rollback" to the original status of a project planning can also be realized. This can be useful if a program change is not successful.

- 1 System in State „Run-Redundant“
- 2 Stop CPU 2 (Backup CPU)
- 3 Load Project into CPU 2
- 4 Stop CPU 1 → Process is stopped
- 5 Start CPU 2 → New project is active but not working correctly
- 6 Stop CPU 2 → Process is stopped
- 7 Start CPU 1 → Previous project is active

Programming Recommendations

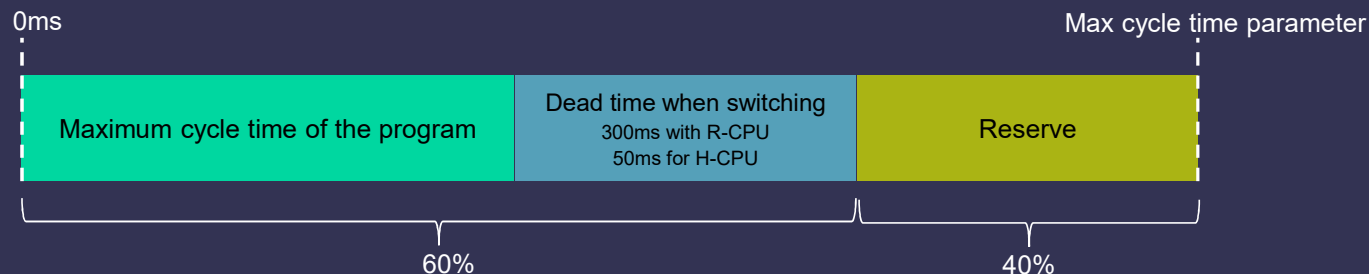
Programming recommendations SIMATIC S7-1500R/H (I)

1 Set maximum cycle time

The maximum cycle time should be set as large as the process allows.

This can shorten the time for the SYNCUP, as the transition to RUN-redundant only takes place when the actual cycle time is <80% of the set maximum cycle time.

Recommendation:



2 Set minimum cycle time as high as possible

By increasing the minimum cycle time to the minimum value required for the process, the system load due to synchronization is reduced. This also shortens the SYNCUP phase and leads to higher performance in communication.

Programming recommendations SIMATIC S7-1500R/H (II)

3 Avoid direct peripheral access

Each direct peripheral access is synchronized in the system state RUN-Redundant and leads to a higher cycle time.
Recommendation: In the RH system, the inputs and outputs of the IO-devices should only be accessed via the process image or the subprocess images.

4 Reduce communication during the SYNCUP phase

If possible, the communication load should be kept low during a SYNCUP phase. This will speed up the SYNCUP process. This can be done programmatically via the SFC "GET_DIAG".

```
#RetVal := GET_DIAG(MODE := 1, LADDR := "Local1~RHSys", CNT_DIAG => #CNT_DIAG_temp, DIAG := #DIS);  
IF #DIS.OperatingState <> 38 (* 38 = "SYNCUP system state" *) THEN  
  // not in syncup  
  "Load";  
END_IF;
```

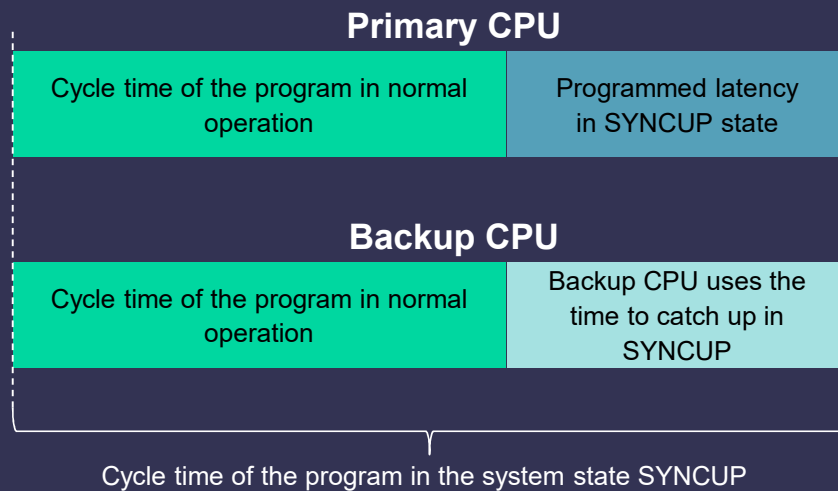
5 Reduce cyclic interrupts during the SYNCUP phase

If cyclic interrupts are used intensively in the program and it is tolerable to reduce the call frequency during the SYNCUP phase, then the sync load can be reduced by calling the system function "SFC SET_CINT". This measure also helps to shorten the SYNCUP time. In normal state (RUN-Solo or RUN-Redundant) the alarm times can then be changed back to the original state.

Programming recommendations SIMATIC S7-1500R/H (III)

6 Use of the WAIT function in the system state SYNCUP

If a longer cycle time is tolerable in the SYNCUP system state, the SYNCUP process can be accelerated by calling the system function "WAIT" at the end of OB1. This is possible because the WAIT function on the Backup CPU is used to catch up.



```
IF #DIS.OperatingState = 38 THEN
  // if in syncup
  WAIT(30000);
END_IF;
```

Specific Functions for R/H Controller

RH_CTRL Instruction

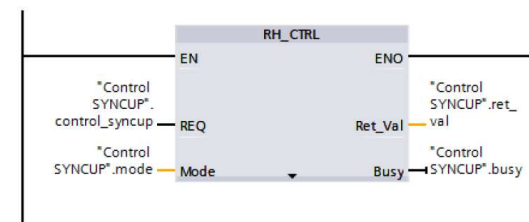
General Information

The RH_CTRL instruction is used to influence sequences in R/H systems.

The functions are selected by the “Mode” parameter.

RH_CTRL instruction works with all TIA Portal versions. Functionality only depends on the used firmware version of the R/H CPU.

Extended instructions		
Name	Description	Version
Alarming		V1.9
Diagnostics		V1.10
Recipe and data logging		V1.3
Data block control		V1.4
Addressing		V1.3
File handling		V1.1
Cryptography		V1.0
R/H system		V1.1
RH_CTRL	Influence processes in ...	V1.0
RH_GetPrimaryID	Specify redundancy ID ...	V1.0
Safety extensions		V1.0



RH_CTRL Instruction

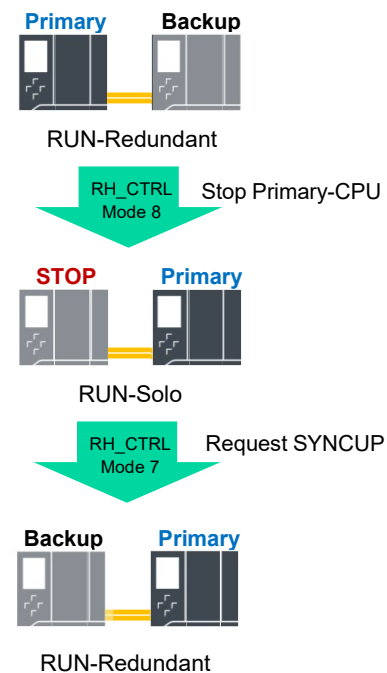
General Information

Mode	Function	Description	FW Version
3	Lock SYNCUP	Disables the system state "SYNCUP". The lock is in force until you revoke it with the "RH_CTRL" instruction or until the R/H system goes into "STOP" mode	V2.6
4	Enable SYNCUP	Re-enables the system state "SYNCUP" after "Lock SYNCUP" is called.	V2.6
7	Request SYNCUP	If the system is in the RUN-Solo state, the SYNCUP can be restarted by calling this function to bring the system back into the redundant state.	V2.9
8	Stop Primary-CPU	When running redundantly, the Primary CPU is stopped, and the Backup CPU runs in RUN solo mode. If the SYNCUP is then requested again (mode 7), the system works with swapped roles (primary/backup) redundant.	V2.9
9	Stop Backup CPU	When running redundantly, the Backup CPU is stopped, and the Primary CPU continues to run in RUN solo mode.	V2.9
10	SYNCUP is locked?	Query whether the system state SYNCUP is currently locked	V3.0
11	Stop Double Primary	Request STOP of an R/H-CPU and assign the role of the backup CPU to it	V3.1
14	Start SYNCUP with open MRP Ring	Switch off the intended MRP check in "SYNCUP" system state	V3.1
15	Move System IP	Allows to move the System IP between Primary- and Backup-Controller	V4.0
16	Query System IP status	Reads the status of the System IP of an interface (activated/deactivated)	V4.0

RH_CTRL Instruction – Mode 7 / 8

Use case: Switch CPU roles programmatically

To change the roles (Primary /Backup) of the two controllers, a sequence of RH_CTRL instructions can be used.



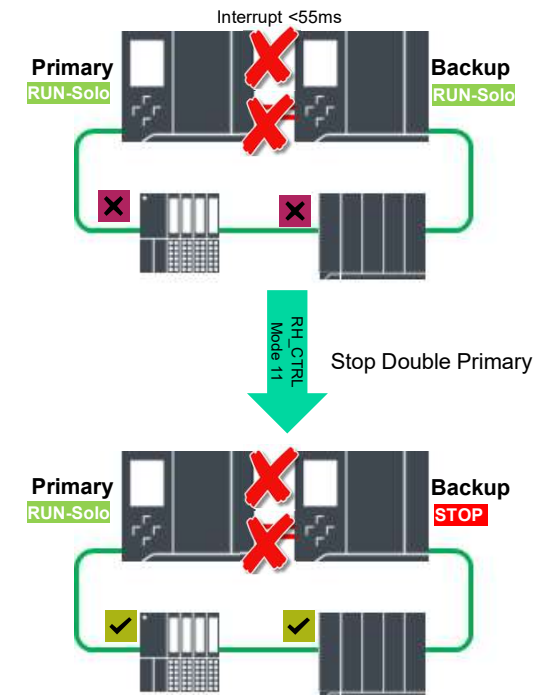
RH_CTRL Instruction – Mode 11

Stop Double Primary

Behavior of RH_CTRL Mode 11: Stop Double Primary

- In the case, that both controllers are operating in RUN-Solo mode, call RH_CTRL instruction in Mode 11
- After a successful execution of "RH_CTRL" the R/H system is in a defined state again: There is a primary CPU which is in the "RUN" operating state and a backup CPU which is in the "STOP" operating state. There is no pairing
- Ensure that "RH_CTRL" is only performed on one R/H-CPU. If "RH_CTRL" is performed on both R/H-CPU, the system state goes to "STOP".

To ensure that "RH_CTRL" is only performed on one CPU, you can, for example, determine the redundancy ID of the primary CPU using the instruction "RH_GetPrimaryID" and then stop the R/H-CPU with the redundancy ID 1



RH_CTRL Instruction – Mode 14

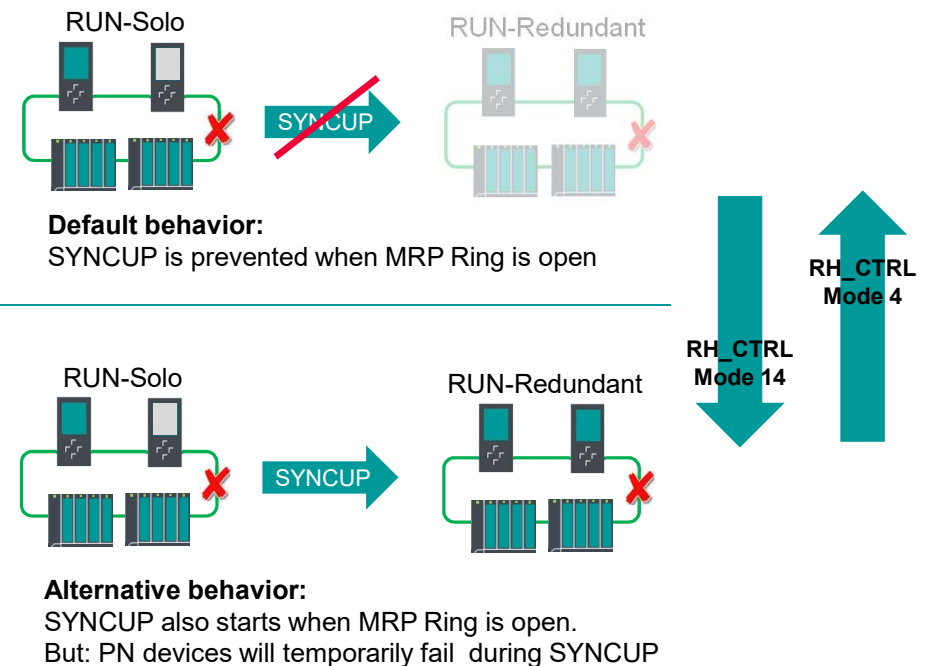
Start SYNCUP with Open MRP Ring (I)

Current SYNCUP behavior

- R/H System cannot be switched to RUN-Redundant when MRP Ring is not closed
- Reason: During SYNCUP the Backup-CPU makes a reboot and opens the internal switch between X1 P1 and X1 P2. As a result, communication to PN devices would be stopped

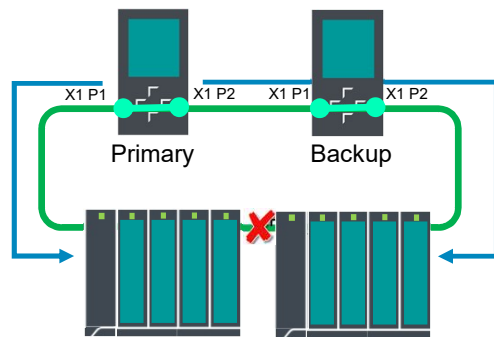
Alternative SYNCUP behavior

- In commissioning phase, the new Mode 14 of RH_CTRL allows to disable the MRP Check
- RUN Redundant operation state can also be reached if MRP Ring is open



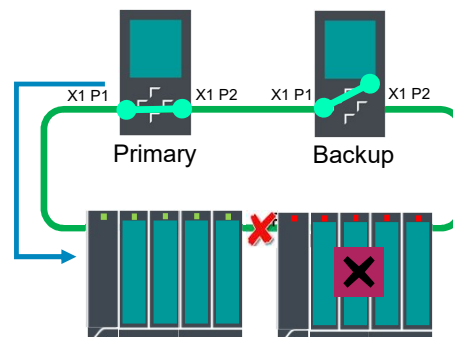
RH_CTRL Instruction – Mode 14

Start SYNCUP with Open MRP Ring (II)



RUN-Redundant / Solo Mode

- Switch between X1 P1 and X1 P2 is working on both CPUs
- Primary Controller can connect both PN Devices



SYNCUP

- Switch between X1 P1 and X1 P2 is open during reboot of Backup CPU
- → Ring is open on 2 locations
- PN devices on Backup CPU would be disconnected

Side effect of alternative mode

- In Mode 14, IO-Devices may be temporarily fail
- Reason: During SYNCUP the Backup-CPU makes a reboot and opens the internal switch between X1 P1 and X1 P2. As a result, communication to PN devices on Backup CPU are stopped

RH_CTRL Instruction – Mode 15

Move System IP Address between RH Controller

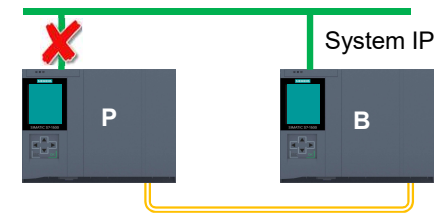
Previous behavior of the system IP address (up to firmware V3.1)

- System IP address is always assigned to the Primary CPU.
- If there is a communication interrupt to Primary CPU, the H-System cannot be reached via System IP.

New behavior of the system IP address (from firmware V4.0)

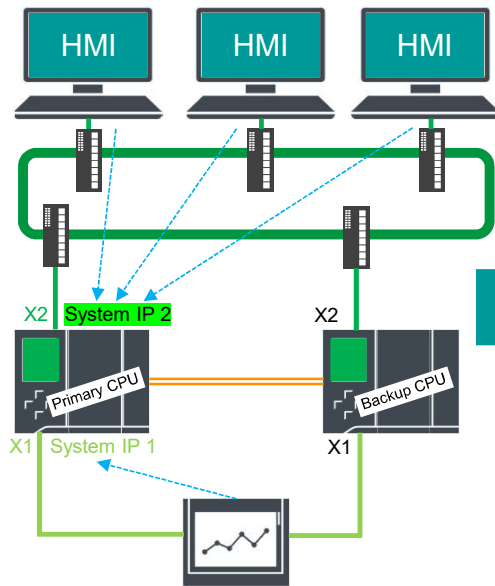
- Default target for the System IP address is the Primary CPU.
- Via user program, the System IP address can be moved to Backup CPU and vice versa.
- System IP address movement is done via RH_CTRL instruction.
- It can be performed individually for each interface (X1..X3)

Application example: See 



SIMATIC S7-1500 Redundant Systems

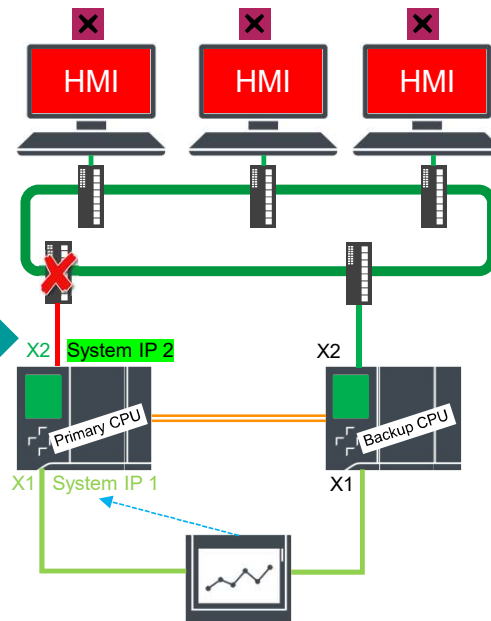
Move System IP Address between RH Controller



Starting Point:

- HMI Devices are connected to System IP on X2
- Panel is connected to System IP on X1

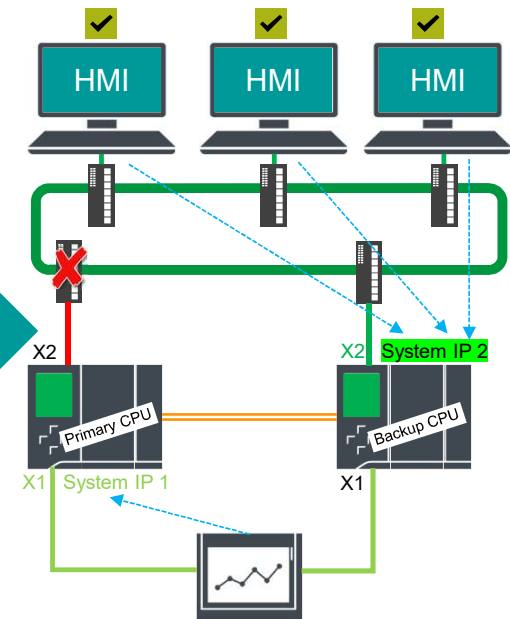
Fault happens



Failure Scenario:

- Switch fails → No more communication possible between HMI and Primary CPU
- Result: HMI devices on X2 cannot control or monitor the process!

RH_CTRL Mode 15



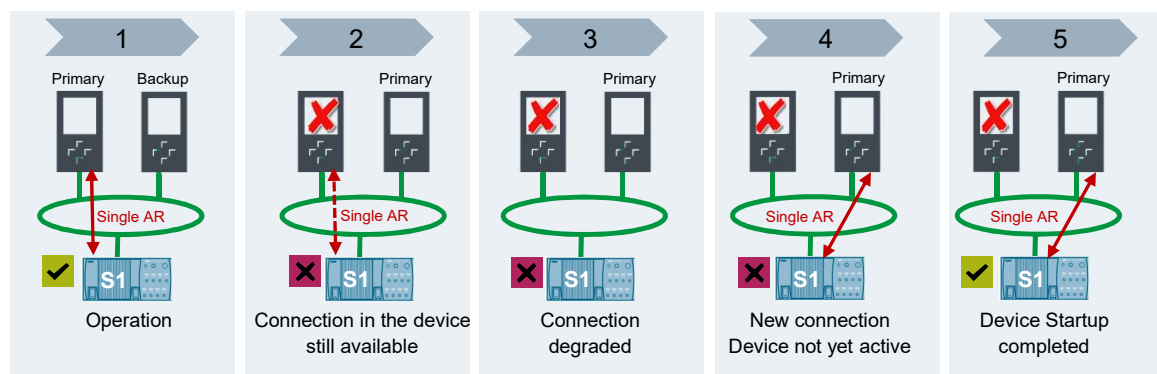
Result after System IP Movement:

- System IP for X2 is on Backup CPU
- Communication is re-established
- HMIs are connected to Backup CPU
- No role exchange of CPUs
- Panel on X1 remains on Primary CPU

Switched S1: Connection of PN Devices without System Redundancy Support

Adjustable switching time for the "Switched S1" function

- 1 System in redundant operation, IO data is exchanged
- 2 After fail-over: IO Device holds the existing connection until the expiration of the Watchdog-Time. As long as no further connection is possible.
- 3 IO device is ready for a new connection to the controller
- 4 The new connection is established, the IO device start-up begins
- 5 IO data is exchanged again



Switching process for S1 devices

AR = Application Relation (connection between controller and device)

Switched S1: Connection of PN Devices without System Redundancy Support

Adjustable switching time for the "Switched S1" function

The switching time can be optimized by changing the watchdog time setting of the S1 device

The optimal length of the watchdog time results from the time it takes for an IO device to be able to build a new AR again.



The screenshot shows the configuration interface for the S1 device. It has two main sections: 'Update time' and 'Watchdog time'. In the 'Update time' section, 'Calculate update time automatically' is selected, and the 'Update time' is set to 2.000 ms. In the 'Watchdog time' section, 'Accepted update cycles without IO data' is set to 112, and the 'Watchdog time' is set to 224.000 ms.

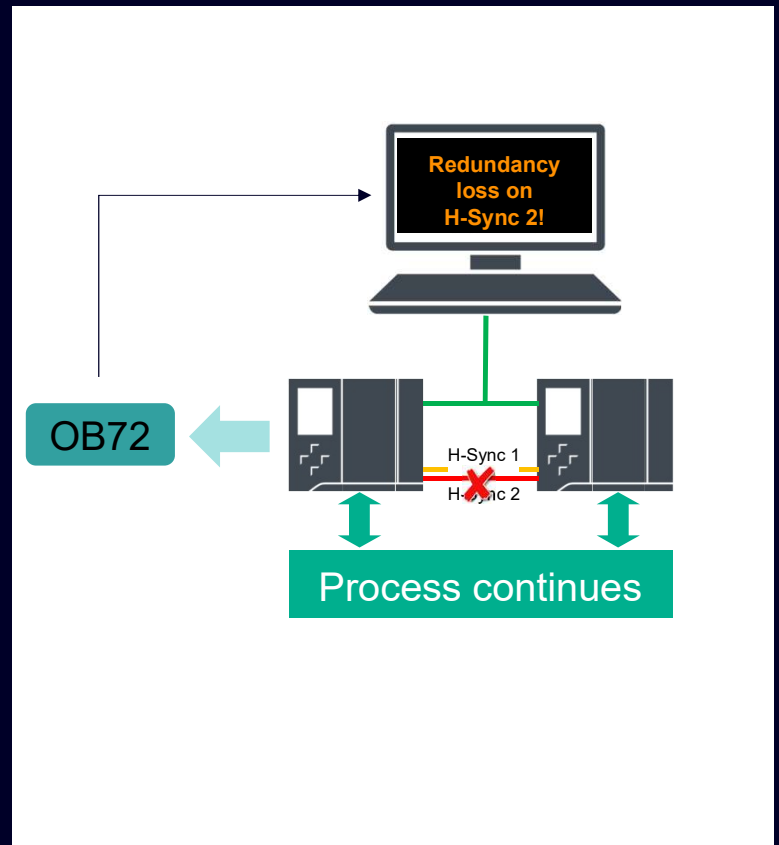
Section	Parameter	Value	Unit
Update time	Update time	2.000	ms
	Adapt update time when send clock changes	☐	
Watchdog time	Accepted update cycles without IO data	112	
	Watchdog time	224.000	ms

AR = Application Relation (connection between controller and device)

Reporting loss of redundancy

OB72 reports loss of redundancy during synchronization

- If a sync line fails, the maintenance LED on the controller lights up and the H-System continues to work in redundant mode.
- The failure of a sync line is reported from firmware V2.9 by calling the OB72 (redundancy loss).
- This can be used to report a request to restore the sync line.



Reporting of peripheral redundancy errors

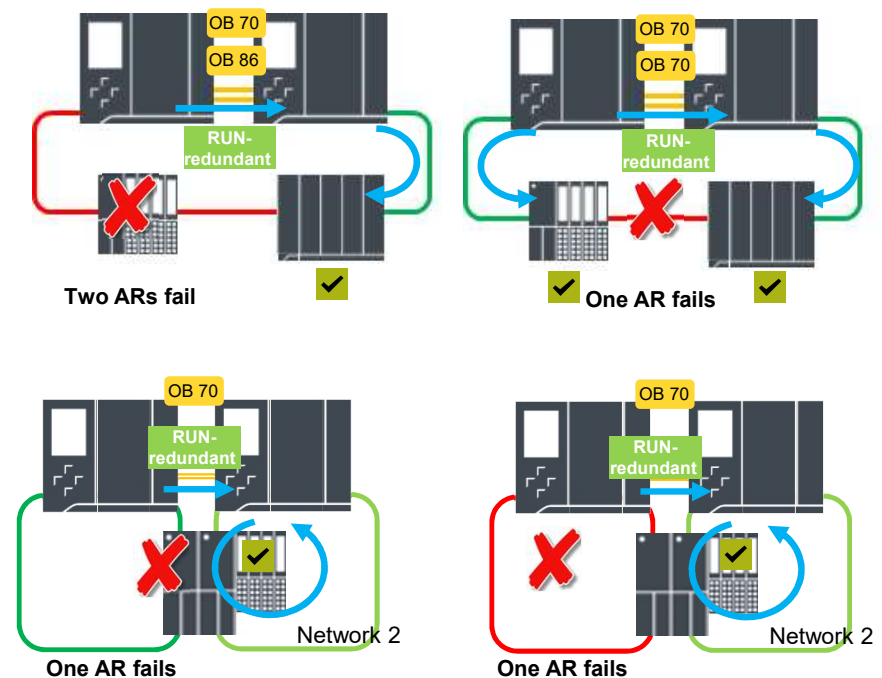
OB70 reports loss of peripheral redundancy

In the H system, the OB 70 is called if in the system state RUN-Redundant on a R1/S2 device redundancy loss or a redundancy return event.

Redundancy Loss: One AR of two ARs of an R1/S2 device fails

Redundancy Return: The second AR of an R1/S2 device is rebuilt

Example scenarios for the OB 70 call → see figure



SIMATIC S7-1500 Redundant Systems

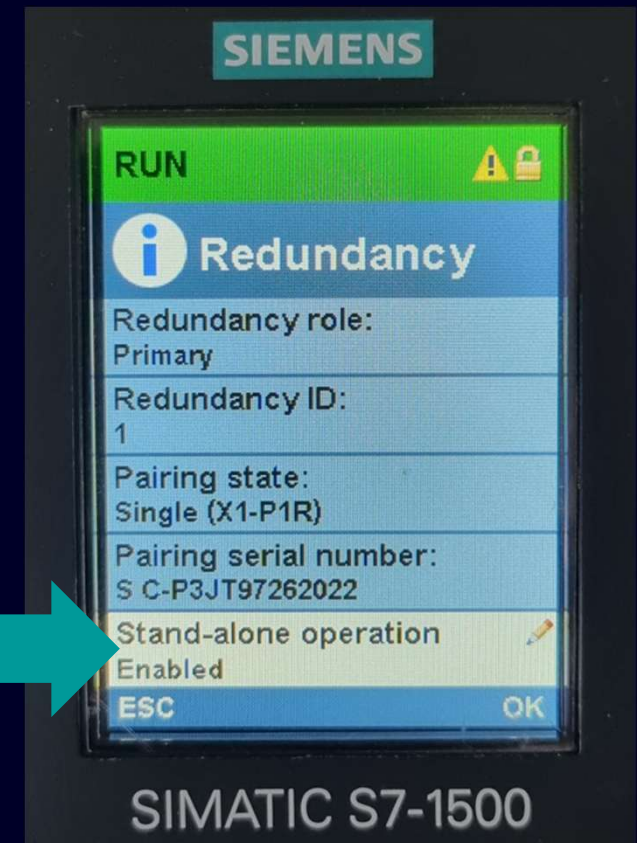
Single Mode for R-CPU

Single Mode for R-CPU

- New mode can be activated via CPU display
- In this mode, a single R-CPU can be operated without warnings
- Pairing of R-CPU is deactivated

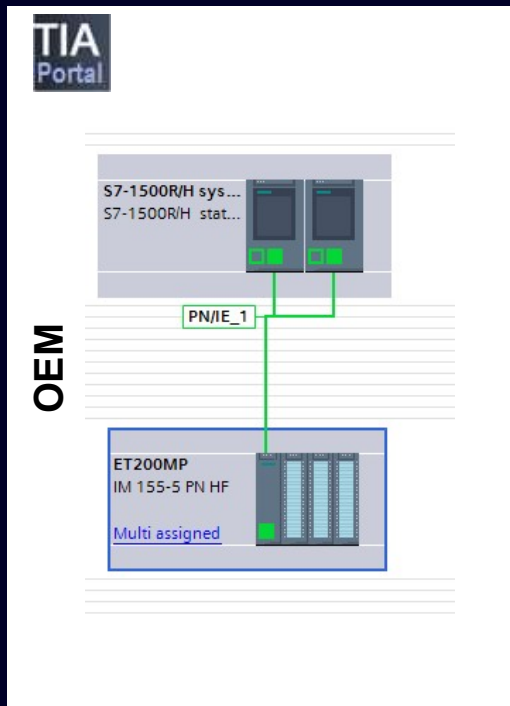
How to come to a redundant setup

- A system with a single R-CPU can be extended to a redundant system without project download from TIA Portal:
Just install second R-CPU and deactivate Single Mode on first CPU

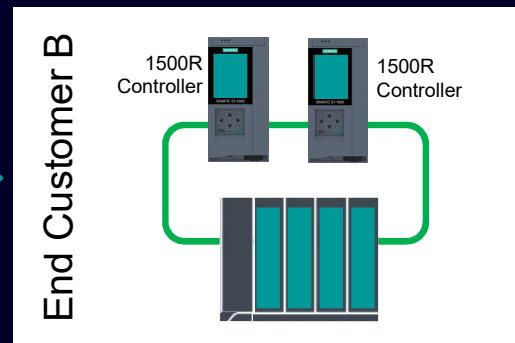
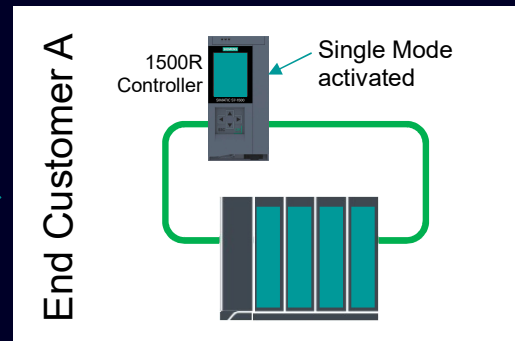


SIMATIC S7-1500 Redundant Systems

Use case scenario for single mode



OEM always configures a Redundant System



Depending on the end customer's requirement, 1 or 2 R-CPU's are installed

Benefits for OEM

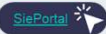
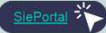
- Reduce engineering effort since the same TIA Project is used.
- Ability to offer a single or redundant set up with the same project/documentation to the end customers.

Benefits for end customer

- Easy way to increase the availability of the system when the requirements would change, OEM action is not required.
- Reduce spare parts since the same product is used. Optimize maintenance costs.

Feature Comparison / Restrictions

Functional comparison with S7-1500 and S7-400H

	S7-1500R	S7-1500H	S7-1500	S7-400H
Communications	Open User Communication; S7 communication as server		Open User Communication; S7 communication; Configured connections	Open device communication; S7 communication; Configured connections
OPC UA	Yes (UA Server)		Yes	No
Webserver	Yes		Yes	No
Redundant communication	Via LComRed Library			Yes
Redundant IO	Via LRedIO Library			
System IP address	Yes		Not required	No
DHCP with DNS	No		Yes	No
Technology Objects	Counting, measuring, PID, BasicPos, Ident		Yes	No
IRT, Isochrone Mode, MRPD	No		Yes	No
Hardware configuration in RUN	Yes Add devices and change parameter settings		--	With switching (H-CiR)
Firmware Update in RUN	No		No	Yes

Configuration Restrictions for S7-1500R/H

	S7-1500R	S7-1500H	S7-1500	S7-400H
Central connection of Modules IO, CM/CP, System Power Supply	Yes: CP 1543-1 (Ethernet), System-PS	Yes: CP 1543-1 (Ethernet), System-PS	Yes	Yes
PROFINET Network Structure	MRP Ring	any	any	any
PROFINET System Redundancy	Switched S1 S2	Switched S1 S2 R1	No	S2
Operation of the RH system as Shared Device or iDevice	No		Yes	No
Operation of PROFIBUS devices	Yes, via IE/PB LINK HA		Yes	Yes
Long Term Tracing	Yes		Yes	No
SYSLOG Messaging	Yes		Yes	No
Profiling	Yes		Yes	No

Add Ons

TIA Portal Add-In

Calculation of the watchdog time over a AddIn

Tool to Set/Reset the correct watchdog factor of PN IO-Devices connected to R/H systems

☐ Select All

Set correct watchdog factor for selected PN IO-Devices

Reset watchdog factor of the selected PN IO-Devices to default

Selected	R/H System	Device number	Device name	Update time in ms	Watchdog factor	Watchdog time in ms
☒	1517H System4 GroupLevel2	109 device(s)				
☒	1517H System3 Group1Level1	109 device(s)				
☒	1515R System6 Group2Level3	4 device(s)				
☒	1515R System6 Group2Level3_1	4 device(s)				
☐	1515R System6 G	1	io device, 327	2	112	224
☐	1515R System6 G	2	io device, 326	2	3	6
☐	1515R System6 G	3	io device, 325	2	3	6
☐	1515R System6 G	4	io device, 324	2	112	224
☒	1515R System5 Group2Level1	4 device(s)				
☒	1517H System2 PNV Level	111 device(s)				
☒	S7-1500R/H-System_1	5 device(s)				
☒	S7-1500R/H-System_2	4 device(s)				
☒	S7-1500R/H-System_3	6 device(s)				

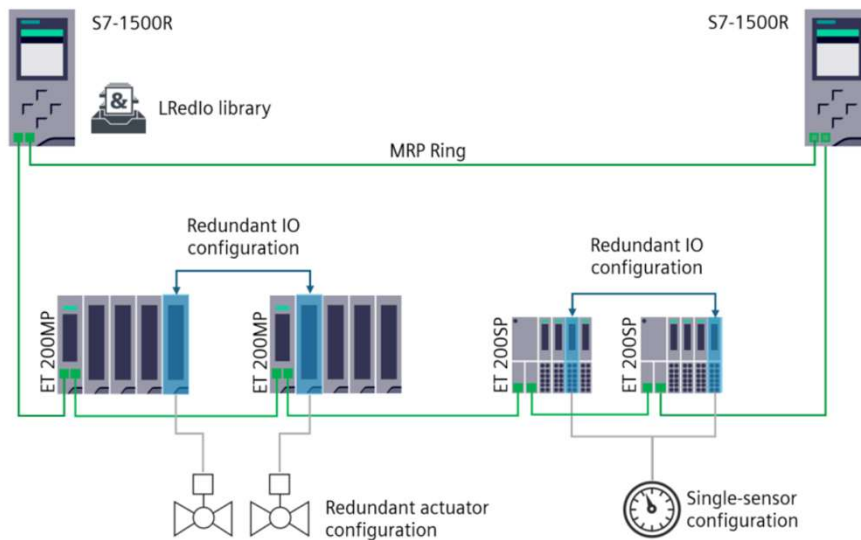
Save & Close

Siemens TIA Portal

To connect a PROFINET device to a redundant S7-1500R/H system, it is necessary to set the correct watchdog time for each device. The SIMATIC S7-1500R/H AddIn determines the correct factor and updates it in the settings

Application Example

Connection of redundant I/O in the user program

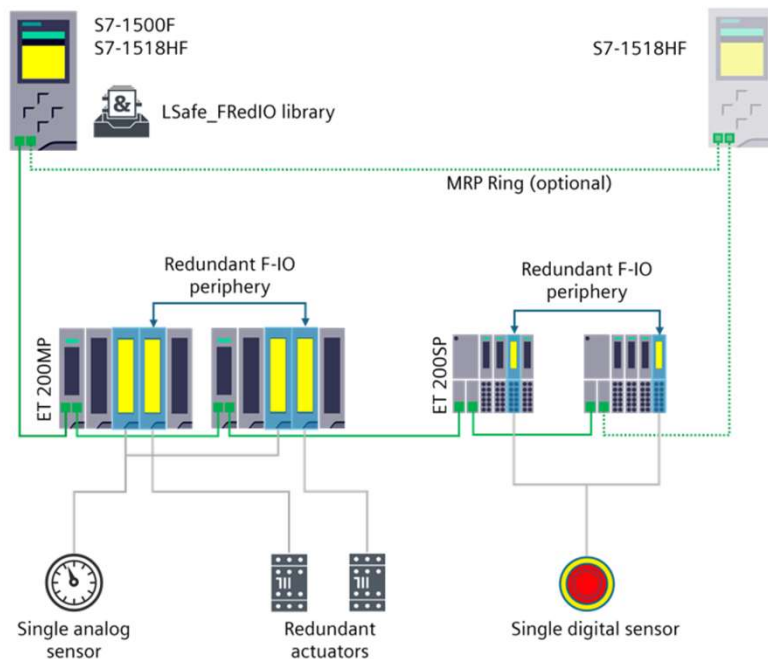


In this application example, a possibility is shown how I/O can be redundantly connected to a SIMATIC S7-1500 controller with blocks of the TIA Portal.

Function block	Function
LRedIO_RedDI	Redundancy function for two digital inputs
LRedIO_RedDQ	Redundancy function for two digital outputs
LRedIO_RedAI	Redundancy function for two analog inputs
LRedIO_RedAQ	Redundancy function for two analog outputs

Application Example

Connection of redundant F-I/O in the user program

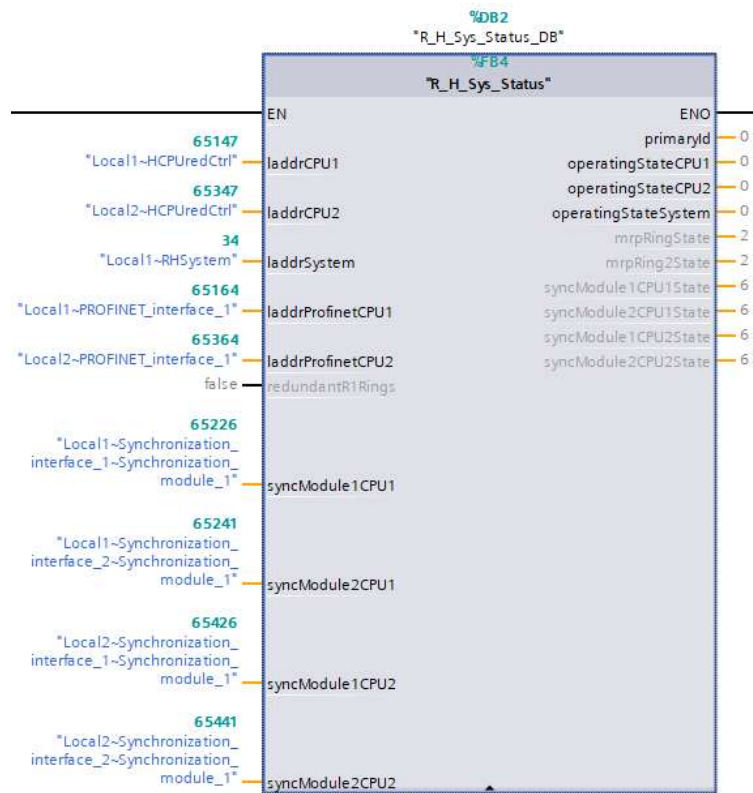


If the availability of sensor and/or actuator circuits of a safety system needs to be enhanced, this application offers effective solutions by incorporating redundancy of the F-I/O-modules connected to single or redundant sensors/actuators.

Function block	Function
LSafe_FRedDI	Receiving and comparing of two redundant F-DI modules
LSafe_FRedDQ	Receiving and comparing of two redundant F-DQ modules
LSafe_FRedAI	Receiving and comparing of two redundant F-AI modules

Application Example

Diagnosis of the operating status of a SIMATIC S7-1500R/H system using FB



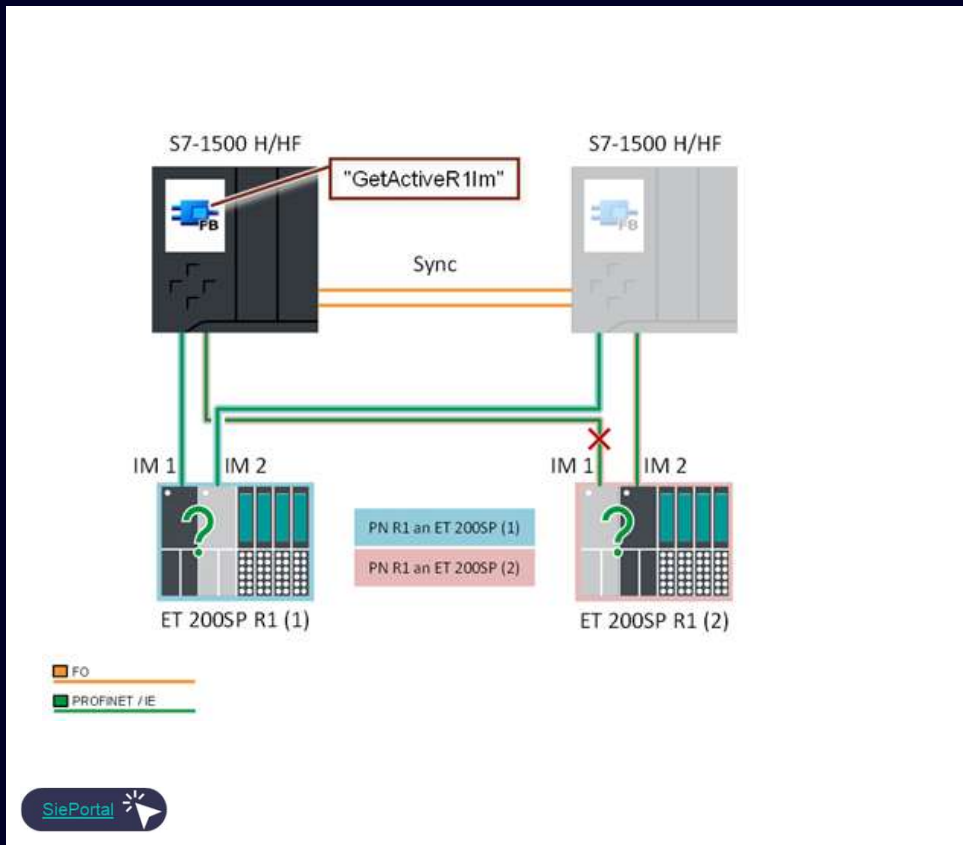
A function block "**R_H_Sys_Status**" is available for extended diagnostic information of the SIMATIC S7-1500R/H system.

Advantages

- Ready-to-use diagnostic block,
- Simple connection of different hardware addresses,
- Integrated self-diagnosis function (in addition to the standard diagnostic functions) of the S7-1500R/H system for early detection and reporting of faults before they affect the process.

Application Example

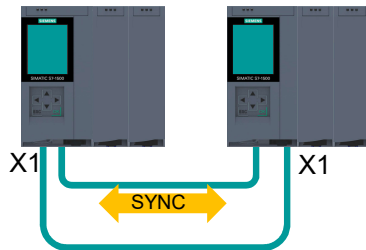
Determining the Active ET 200SP Interface Module for R1 Configuration



This application example provides you with the **"GetActiveR1Im"** function block so that you are always informed about the status of the redundant connection and the R1 IM used by the ET 200SP in your user program.

Application Example

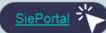
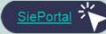
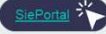
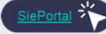
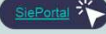
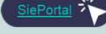
Automatic SYNCUP of S7-1500R/H controllers



The function block detects autonomously when a SYNCUP is required between primary and Backup CPU – and thus eliminates the need for manual intervention by operators. As a result, synchronization between redundant controllers is ensured at the earliest possible time, thus increasing the system reliability and reducing the downtimes.

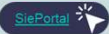
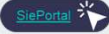
Communication Functions

Telecontrol and substation control technology with SIMATIC S7-1500R/H

Component	Version	Protocols	Additional Information
TIM 1531 IRC  	From V2.1	• SINAUT ST7 • DNP3 • IEC 60870-5 / 101, 104	Device redundancy with the TIM 1531 IRC 
SIPLUS RIC Library for SIMATIC S7-1500 	From V1.7	IEC 60870-5 101,102,103, 104	How do you implement the IEC 60870-5 protocols with SIMATIC S7-1500? 
IEC 61850 MMS Client AS LIBRARY 		IEC 61850 MMS	Setting up an IEC 61850 MMS/OPC UA Gateway with TIA Portal and Station Gateway for TIA 

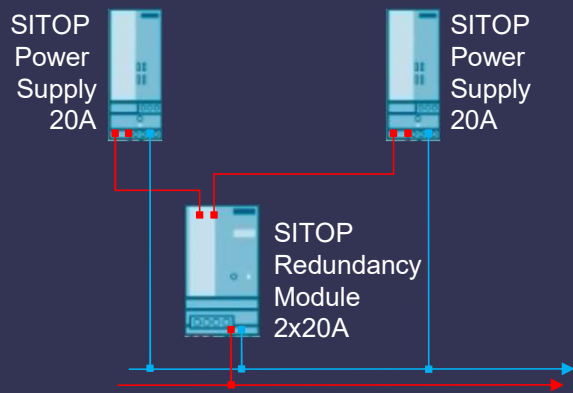
Communication Functions

Redundant communication

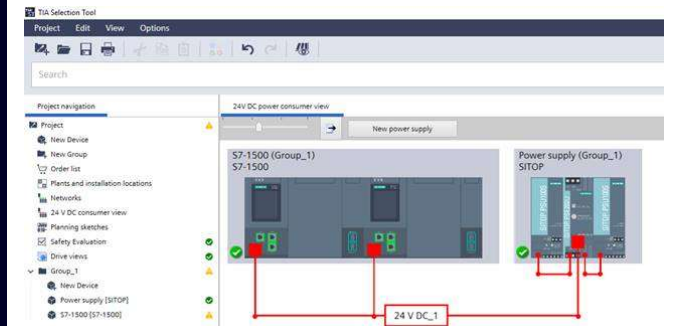
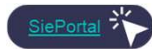
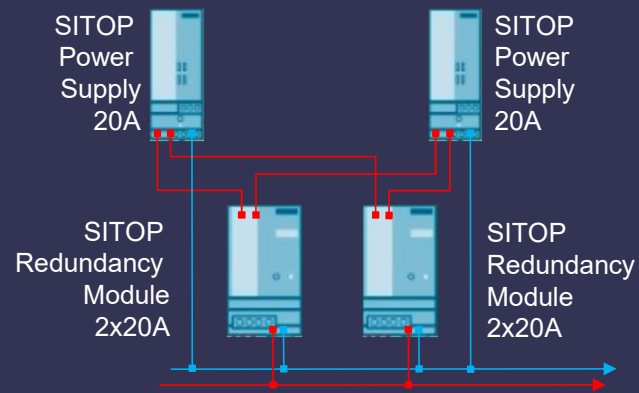
Component	Protocols	Link
SIMATIC Modbus/TCP Red S7-1200/S7-1500 	Modbus/TCP	
Redundant Open User Communication	Various	Redundant Open User Communication 

Redundant power supply SITOP Redundancy Module RED1200

Redundancy of the
SITOP Power Supply
with
single “Redundancy module”



Redundancy of the
SITOP Power Supply
with
redundant “Redundancy module”



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